

# Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant  $c_3 = \frac{1}{8\pi}$  and a five-slot carrier — build  $E_8$   
and read off the Standard Model: status assessment and reading guide

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## In one sentence

**Topological Fixed-Point Theory (TFPT)** constructs, from exactly **two inputs** — a boundary constant  $c_3 = \frac{1}{8\pi}$  (P1) and a five-slot carrier  $g_{\text{car}} = 5$  (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with  $E_8$  ( $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ( $\alpha^{-1} = 137.0359992$ ); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ), not as free outputs. The honest one-line claim is on the status card below.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading the introduction — the entry point and reading guide.**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed (**[E]**); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status                  |
|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive <b>[O]</b>    |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | <b>[E]</b> / <b>[C]</b> |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | <b>[C]</b>              |

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**[E]** exact / machine-proven**[C]** conditional**[O]** open / axiom**[X]** kill test

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**Reading the markers.** Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g.  $240 = 16 \cdot 5 \cdot 3$ ), a Lie/lattice theorem (e.g.  $D_5 \oplus A_3 + \mu_4 = E_8$ ), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Physical — under hypotheses        | RG masses, QG    QG.AMB.01                   |
| Numerical — fit-free fixed point   | $\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01 |
| Lattice — Lie / glue / root system | $E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01   |
| Formal — Lean / exact script       | P2 algebra    FORM.P2.02                     |
| Axiom — declared inputs            | P1 $c_3$ , P2 $g_{\text{car}}$ AX.P1.01      |

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[E]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function  $e_k(a)$  of the anchor  $a = (1, 1, 2)$ ; (ii) a power sum  $p_n(a)$  or difference  $p_{n+1} - p_n$ ; (iii) a determinant, spectrum, Smith normal form or trace of  $(R, Q, K, L)$ ; (iv) an anchor quadratic form  $u^\top M v$  with  $u, v \in \{1, a\}$ ; (v) a Plücker coordinate of the anchor plane  $\langle 1, a \rangle$ ; (vi) an  $E_8$  Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

**Reviewer path** (read in this order; the rest is support):

1. the architecture: the two axioms  $\{c_3, g_{\text{car}}\}$  and the two-engine picture (§1, the figures);
2. the  $E_8$  glue  $E_8 = (D_5 \oplus A_3) + \mu_4$  — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the  $\alpha^{-1}$  fixed point plus ablation (`v3_em_alpha.py`);

4. the **single status ledger verification/status\_ledger.csv** — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (**tfpt\_research\_contracts**: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

## 1 What this is about: the jump in construction principle

### In plain words

**What we found.** TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number  $\frac{1}{8\pi}$  and a five-slot carrier) — builds the exceptional symmetry  $E_8$ , and  $E_8$  is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

**Why this is simple.** There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number  $\pi$ .

### Sharper still: one anchor $a = (1, 1, 2)$ , and $E_8$ as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity / splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums*  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the big Lie data directly:  $p_1=4=|\mu_4|$ ,  $p_2=6=|R^+(A_3)|$ ,  $p_3=10=\mathcal{A}_\Lambda$ , and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder  $p_{n+1} - p_n = 2^n$ . So the inputs collapse from  $\{c_3, g_{\text{car}}\}$  to  $\boxed{a = (1, 1, 2) \text{ plus } \pi}$ .  
[verification/v23\_anchor\_generator.py]

And  $a$  is itself half-forced ([E]/[C]): on  $\mathbb{P}^1$  every bundle splits (Birkhoff–Grothendieck), so once  $|\mu_4| = 4$ ,  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  are set, the three positive anti-degrees are positive integers summing to 4, and  $4 = 2 + 1 + 1$  is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why  $a = (1, 1, 2)$ ” but “why the carrier  $g_{\text{car}} = 5$  /  $(|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with  $c_3$ ) is the [O] input layer.

*Two readings, one source (no contradiction).* P1 and P2 are the *operational compiler inputs*; their anchor provenance is  $a = (1, 1, 2)$  plus  $\pi$ . The *single structural bedrock postulate* is the keystone SEAM.EQUIV.01, whose conformal-deck face QGEO.SYM.01 (the carrier  $\mu_4$  clock is the seam’s conformal deck, see **tfpt\_research\_contracts**) forces the  $\mu_4$  deck, hence  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  and the tripartition  $4 = 2+1+1$ , of which  $c_3$  and  $g_{\text{car}}$  are the two elementary readouts.

“Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication are now Lean-formalised (FORM.QGEO.02/FORM.QGEO.01); the postulate itself stays [O].

*One source, four readings (the cleanest origin).* The four-point seam divisor  $\mu_4$  on  $\mathbb{P}^1$  generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates  $D_5$  itself ([verification/v197\_rr\_carrier\_clifford\_d5.py]):  $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor. So  $\mu_4$  is not merely glue: it is the *common source* of  $A_3$ ,  $D_5$ ,  $c_3$  and  $E_8$  ([verification/v189\_riemann\_roch\_carrier.py]) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

**What  $E_8$  is (and is not).** In TFPT  $E_8$  is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct  $E_8$  gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal  $E_8$  world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain  $E_8$  as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull.*

In one line:

■ The anchor generates the syntax; the boundary generates the scale;  $E_8$  checks the consistency.

i.e. the discrete content (carrier  $D_5 \oplus A_3 + \mu_4$ , the operators  $R, Q, K, L$ ) flows from  $a = (1, 1, 2)$ ; the continuous content ( $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0$ ,  $\alpha^{-1}$ , the exponential scales) flows from  $\pi$ ; and  $E_8$  is the unimodular *checksum container*, not the world group.

**The anchor ratio triad** ([verification/v119\_review\_validation\_2.py]). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly  $(2, 5) = (e_3, e_2)$  with inflection  $\frac{7}{2} = \frac{e_3 + e_2}{2}$ : the fractions  $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$  are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]:  $\Omega_{\text{adm}} = 2p_1p_2 = 48$ ,  $10b_1 = 1 + p_1p_3 = 41$ ,  $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$ ,  $h(E_8) = e_3p_0e_2 = 30$ ,  $\text{rank } E_8 = p_0 + e_2$ .

$$a = (1, 1, 2) \quad [E]$$

#### elementary $e_k(a)$

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

#### power sums $p_n = 2 + 2^n$

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

#### derived motifs

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

## The companion documents

The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

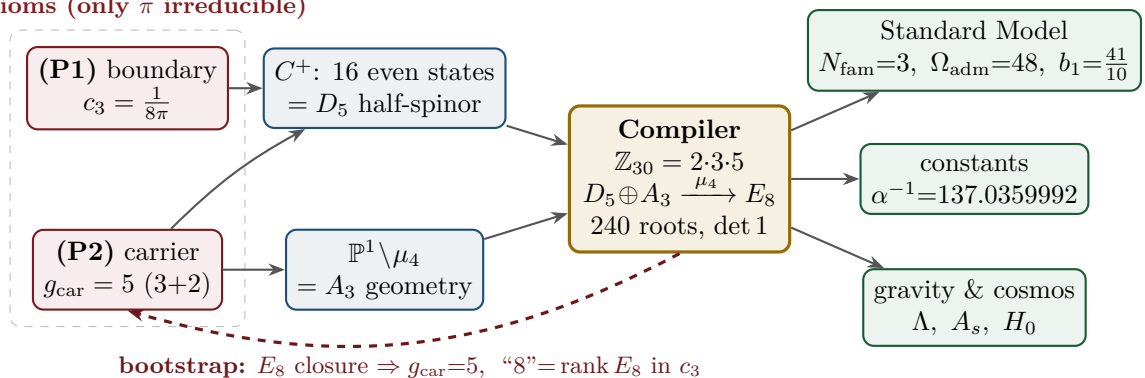
| Document                          | What it contains (former notes now as parts)                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tfpt_1_architecture_e8            | <b>Core.</b> The two axioms $\{c_3, g_{\text{car}}\}$ , the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$ , the explicit $D_5\oplus A_3+\mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).                                                                        |
| tfpt_2_standard_model             | <b>Standard Model.</b> The full SM in one $\varphi_0^{\text{ret}}$ -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ $n_s, r$ , Starobinsky $M$ , derived $\theta_{12}$ , quark $c$ , the H2 splitting type, $\Lambda$ -typing (Part III).                                                |
| tfpt_3_e8_audit_bootstrap         | <b><math>E_8</math> audit &amp; bootstrap.</b> The seven $E_8$ slices as an audit raster ( $E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III).                                                                                     |
| tfpt_4_frontier                   | <b>Frontier.</b> The honest status of $\eta_B$ , $m_p/m_e$ , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.                                                                                                                                                                                                                                                 |
| tfpt_5_redteam                    | <b>Red Team.</b> The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.                                                                                                                                                                                                                                              |
| tfpt_research_contracts           | <b>Research contracts.</b> The open gates as numbered contracts with lemma chains and closing theorems: ( $U_{\text{wall}}$ ) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).                |
| tfpt_horizon_readouts<br>(App. H) | <b>Appendix H — horizon unit system (reframe, not new physics).</b> One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ( $1920= W(D_5) $ , $ \mu_4 =4$ ); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138). |
| origin_theory                     | <b>Origin Theory.</b> Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and $\pi$ (the dimensionful $v_{\text{geo}}$ and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.                 |

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure  $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$  in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the  $\theta_{12}$  texture and the quark  $c$  in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a  $U_{\text{point}}$  anchor;  $\theta_{12}$  the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

## 2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only  $\pi$  irreducible)



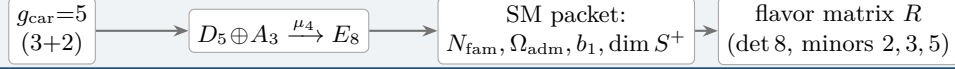
**Key statement (the loop closes):** previously  $D_5$ ,  $A_3$  and  $E_8$  sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the  $E_8$  closure feeds back and *fixes the inputs*.  $g_{\text{car}} = 5$  is the unique rank-fill/Coxeter/Pascal solution ( $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ), and the integer 8 in  $c_3 = \frac{1}{8\pi}$  is rank  $E_8 = h(D_5) = \det R$ . **Inputs and output form an overdetermined fixed point** — the structure does



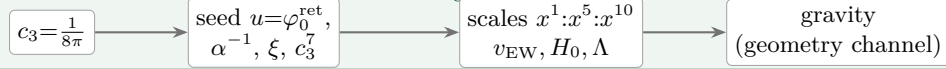
not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous  $\pi$  (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

### Engine 1 — discrete closure (from $g_{\text{car}} = 5$ )

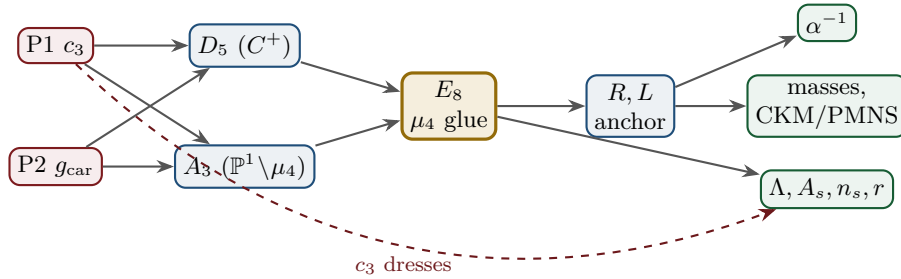


### Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$ )



## 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



| Node           | Definition                                               | Inputs      | St.     | Output                              | Failure mode                |
|----------------|----------------------------------------------------------|-------------|---------|-------------------------------------|-----------------------------|
| P1             | RP boundary kernel, $c_3=1/(8\pi)$                       | — (axiom)   | [O]     | $c_3$                               | no reflection-positive seam |
| P2             | 5-slot carrier $g_{\text{car}}=5$                        | — (axiom)   | [O]     | carrier 3+2                         | wrong family/charge lattice |
| $D_5$          | $C^+$ even-Hamming = half-spinor 16                      | P2          | [E]     | $\dim S^+=16, Y$                    | group/matter mismatch       |
| $A_3$          | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.       | P1          | [E]     | $N_{\text{fam}}=3, \mathbb{Z}_6$    | wrong multiplicity          |
| $E_8$          | $\mu_4$ glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$ | $D_5, A_3$  | [E]     | 240, 248                            | not even-unimodular         |
| $R, L$         | residue + winding matrix, anchor (1, 1, 2)               | $A_3$       | [E]     | $\det 8, \text{ minors } (2, 3, 5)$ | wrong $D_6$ branch          |
| $\alpha^{-1}$  | root of $F_{U(1)}(\alpha)=0$                             | $c_3, M=41$ | [E]     | 137.0359992                         | no/second root (excl.)      |
| masses         | $\varphi_0^{\text{ret}}$ -ladder $\lambda_V^L \Lambda$   | $R, L, c_3$ | [E]/[C] | 9 masses, mixings                   | hierarchy mismatch          |
| $\theta_{12}$  | TBM + seam $\varepsilon=c_3$                             | $R, L$      | [E]/[C] | 0.307                               | seam-misalign. lemma fails  |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron                               | $c_3$       | [C]     | $\Lambda, A_s, n_s, r$              | ambient RP fails            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

## The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.2 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the  $R + R^2$  spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the  $R$ -bridge amplitude normalisation  $U_{\text{point}}$  *reduces to*  $v_{\text{geo}}$  (Gate 1 closed, **v75** — the same anchor as  $1/G$ ), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, **v76**) — which is in turn the rigorously-constructed  $(E_8)_1$  lattice net (**v77**), so  $G_{\text{metric}}$  is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale  $v_{\text{geo}}$  — the *dimensional-analysis floor* that no theory escapes (**v78**), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces ( $F_{\text{frontier}}$ ).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).** (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ ), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([**E**] for the attractor, [`verification/v56_unique_attractor.py`]); the *physical identification* of the transport operator stays [**C**], reinforcing the static bootstrap web ( $g_{\text{car}} = 5$  forced three ways *given* the  $E_8$  closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). (ii) *Newton’s constant is not an independent input:* fixing the physical  $\Lambda$  branch ( $((8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2})$  pins  $\bar{M}_{\text{Pl}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([**E**]/[**C**], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just  $\pi$ , the discrete  $E_8$ -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale  $v_{\text{geo}}$  — and that single scale is shared by *both* the flavor sector ( $U_{\text{point}} \rightarrow v_{\text{geo}}$ , **v75**) and gravity ( $1/G$ , **v68**): the two [**A**] anchors are *one* (dimensional analysis; no metre from pure mathematics).

## The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels  $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$  are kept only for ledger continuity). In current terms:  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$  — the flavor ratios are *closed* (Plücker/Readout Rigidity, **v49/v71/v75**), what remains is *one dimensionful unit*, not a flavor gap;  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$  — reduced to a single boundary-net theorem, “the seam–Calderón net is the index-4  $\mu_4$  simple-current extension of the carrier net” (holomorphy then follows, **v89/GATE.METRIC.06**); since the QBL chain (**v108–v113**) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (**v110**), the certified channel counts the code identically (**v112**), pair transport is minimally complete (**v111**), and the one quasi-free kernel — rank =  $c$ : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (**v113**), so closing  $G_{\text{net}}$  also closes the carrier selection [**E**];  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$  — named QFT/cosmology *transfer* interfaces (Koide source→pole,  $\eta_B$  Boltzmann, axion relic,  $m_p/m_e$ ), never compiler powers; the four are one typed functor with a four-axiom contract (`CONTRACT.F.01`, **v213**) and *one shape* — each is a gapped



relaxation to a unique attractor (the main-branch discrete→dynamic motif), with  $F\_pole$  at the *seam* rate  $(2/3)^6$  and the other three external (**v303**). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion  $G_{net}$  is the index-4  $\mu_4$  simple-current extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$  (Simple-Current Extension Theorem, **v154**, exact: index 4,  $c=8$ ,  $\mu=1 \Rightarrow$  holomorphic  $\Rightarrow E_8$ ); net existence and full-cone reflection positivity are discharged to **[E]** (**v175**); and the **No-Unit Theorem** (**v153**) makes  $v_{geo}$  an irreducible metrology primitive ( $U_{point} \sim v_{geo}$ ,  $1/G \sim v_{geo}^2$ ,  $m/\mu = e^{3/4}$  — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). What stays open is the *one seam-side identification*, now packaged as the single named theorem **SEAM.EQUIV.01** — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — whose conformal-deck premise **QGE0.SYM.01** is its *downstream* definitional face (per **v323**; the residual is Lean-pinned to named cited steps, **v308**). The sprint-by-sprint reduction that produced it is recorded in the changelog and in **tfpt\_research\_contracts**, not replayed here. The status card:

| Item                                  | Status         | Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter-freeness (self-consistency) | [E]            | gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)                                                                                                                                                                                                                                                                                                                                                                              |
| Newton constant $G_N$                 | [E]/[C]        | not <i>independent</i> : branch pins $\bar{M}_{\text{PI}}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor                                                                                                                                                                                                                                                                                                                                                                                   |
| ratio $c_u/c_d = \frac{55}{117}$      | [E]            | $g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$ ; rigid on the derived stratum (v49/v71), no solve                                                                                                                                                                                                                                                                                                                                                                                       |
| flavor absolute scale                 | [E]/[O]        | ratios <i>closed</i> (v49/v71); only the absolute scale remains $= v_{\text{geo}}$ (Gate 1, v75), the same anchor as $1/G$                                                                                                                                                                                                                                                                                                                                                                                   |
| $R + R^2$ gravity                     | [E]/[C]        | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| physical QG sector                    | [E]/[C]        | gap-decoupled admissible measure ( $\Delta_{\text{eff}}=1.648>0$ )                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ambient QG measure                    | [O](reduced)   | <b>Gate 2 tier B</b> : holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1 \otimes (A_3)_1 \times \langle (1,1) \rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$ ) and factors into the already-open $A_2$ net-existence + the keystone <b>SEAM.EQUIV.01</b> — zero new gates (v160–v165) |
| $v_{\text{geo}}$ metrology unit       | [O](primitive) | No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu$ a ratio — one unit, irreducible                                                                                                                                                                                                                                                                                                                     |
| Koide                                 | [C]            | $Q_{\text{src}} + \varphi_0/24$ as source $\rightarrow$ pole transfer conjecture                                                                                                                                                                                                                                                                                                                                                                                                                             |
| axion DM                              | [C]/[O]        | $f_a = M_{\text{scal}}/128$ as a scenario                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\eta_B$                              | [C]            | readout closed; leptogenesis Boltzmann solve missing                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $m_p/m_e$                             | [O]            | cross-sector QCD/EW ratio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $N_*$                                 | [C]            | reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $a = (1, 1, 2)$                       | [E]/[C]        | forced by $4 = 2 + 1 + 1$ , conditional on the carrier                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| carrier selection $g=5$ (QBL)         | [E]/[C]        | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$ ); residue = the $G_{\text{net}}$ premise itself — gate closes $\Rightarrow$ carrier [E]                                                                                                                                                                                                                                                                                             |
| selector establishment R4'            | [E]/[C]        | $(d, n) \Rightarrow R$ columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment                                                                                                                                                                                                                                                                                                                                          |
| sheet/parity (Q-geometry derived)     | [E]/[C]        | $H^1$ characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius $D_4$ matched parity by parity (v146); residue = the module identification, folded into <b>SEAM.EQUIV.01</b>                                                                                                                                                                                                                                                                          |
| seam quantum clock R1                 | [E]/[C]        | typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born <sup>2</sup> Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference                                                                                                                                                                                                                                                                              |
| $E_8$ Lean                            | [E]            | hardening, not a blocker (script-certified)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

The structural open core is the single named keystone **SEAM.EQUIV.01** (the  $G_{\text{net}}$  inclusion: the raw RP seam =  $(E_8)_1$  at  $\tau=i$ ) together with the separate ambient-measure summit **QG.AMB.01**;  $v_{\text{geo}}$  is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes.  $G_{\text{net}}$  is now heat-kernel grounded (G2: the spectral action gives  $R + R^2$  structurally) and *gap-decoupled* (G5:  $\Delta_{\text{eff}}=1.648>0$  isolates the closed admissible sector from the un-built ambient), so the ambient projective limit **QG.AMB.01** is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion **tfpt\_research\_contracts**, with priority on the  $v_{\text{geo}}$  scale anchor

and the  $G_{\text{net}}/\text{SEAM.EQUIV.01}$  inclusion theorem. [\[verification/v36\\_spectral\\_action\\_g2.py\]](#)

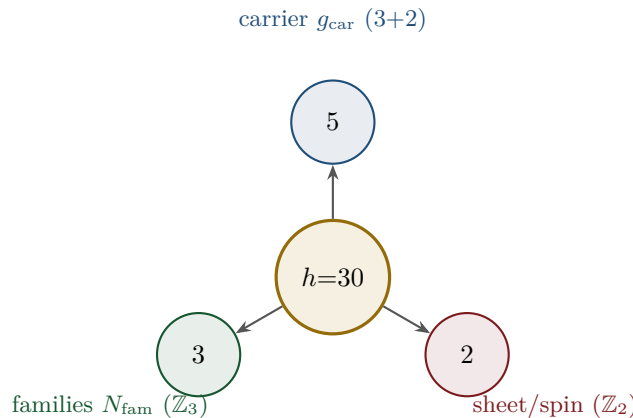
**Sharper still: the structural residual is *one* condition.**  $G_{\text{metric}}$ , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “the seam carries no nontrivial abelian sector” — holomorphy ( $\mu$ -index 1), a homology-sphere seam link ( $\Gamma$  perfect  $\Leftrightarrow 2I$ , [\[verification/v232\\_e8\\_kleinian\\_seam.py\]](#)), exactly one 1-dim irrep ([\[verification/v219\\_icosahedral\\_mckay.py\]](#)) — all of which force  $E_8$  ( $\#(\text{mark-1}) = |H_1| = 1$  only for  $E_8$ ; holomorphic  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic  $\Leftrightarrow \det K = 1$ , the extension tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  being anyon condensation (the Kitaev  $E_8$  state); the one open analytic step is then “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” = the single named keystone **SEAM.EQUIV.01** (with **QGE0.SYM.01** its downstream conformal-deck face). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, **v261**): the finite Dirac is a covariance induction of the seam KMS state (**v258**), the spectral-action cutoff *is* that KMS weight ( $f_2/f_0 = 1$ , **v259**), the seam, the carrier-16 and  $E_8$  live on one Kummer/K3 surface (**v260**), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone **SEAM.EQUIV.01** and adds *no new open item*, while the ambient quantum-gravity measure stays separate by design. [\[verification/v234\\_seam\\_holomorphy\\_selection.py\]](#) [\[verification/v235\\_seam\\_chern\\_simons.py\]](#)

#### Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ( $N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$ , the flavor matrix, the  $E_8$  glue, the scale grammar) is a *consequence*. And even those two *tighten*: the bootstrap note shows  $g_{\text{car}} = 5$  and the integer 8 in  $c_3$  are *overdetermined*  $E_8$ -closure *fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5;  $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$ ), leaving *one* genuine continuous primitive:  $\pi$  from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence:  $E_8$  is the exceptional top of the tower of finite  $SU(2)$  subgroups, so its branch data is the icosahedron and the three atoms (2, 3, 5) are its rotation-axis orders — the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees equal to the affine- $E_8$  Kac marks,  $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$  ([\[verification/v219\\_icosahedral\\_mckay.py\]](#)). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [\[verification/v6\\_bootstrap.py\]](#) [\[verification/v14\\_carrier\\_uniqueness.py\]](#) [\[verification/v23\\_anchor\\_generator.py\]](#) [\[verification/v219\\_icosahedral\\_mckay.py\]](#)

## 4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of  $E_8$  is  $h = 30 = 2 \cdot 3 \cdot 5$ . Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

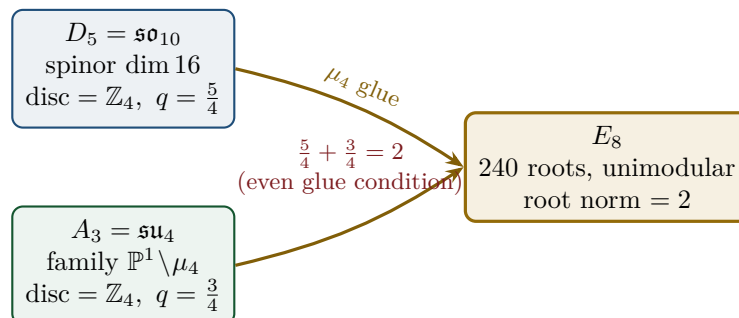


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$  (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$  appears in family, action ladder and the  $E_8$  root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$  and  $\dim E_8 = 240 + (5 + 3) = 248$ . [E]
- the  $\mathbb{Z}_{30}$  Coxeter element  $\leftrightarrow$  the cyclotomic polynomial  $\Phi_{30}$ ; the corner rotation  $z \mapsto iz$  on  $\mathbb{P}^1 \setminus \mu_4$  is the  $A_3$  Coxeter element. [E]
- **why these three primes:**  $E_8$  is the icosahedral top of the McKay tower ( $2I$ , order 120, irrep degrees = affine- $E_8$  marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the  $E_8$  exponents  $(\mathbb{Z}/30)^\times$  carry an order-4 totative character  $\langle 7 \rangle$  — the same  $\mu_4$  seam clock. [E] [verification/v219\_icosahedral\_mckay.py] [verification/v223\_coxeter\_totative\_clock.py]
- **a network reading (computational substrate).** On the same McKay  $(2, 3, 5)$  graph the affine- $E_8$  marks are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4 \cdot m$  (it converges from *any* positive start to  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ ); the spherical tower terminates at  $E_8$  and the cascade  $E_8 \supset E_7 \supset E_6 \dots$  is nested-subgraph coarse-graining. So  $E_8$  reads as the *universal attractor of a minimal  $(2, 3, 5)$  network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing the *full* structure (lattice + recovery + readouts) stays open, and  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction. [O] [verification/v298\_e8\_network\_attractor.py]
- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate  $Am \leq 2m$ , Collatz–Wielandt = Smith's  $\rho \leq 2$ ; the Perron eigenvector enters only as an audit) *reaches*  $E_8$  as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary  $\hat{E}_8$ ) — but *only* from a 3-arm  $(2, 3, \cdot)$  seed (the controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$ , so neither family ever reaches  $E_8$ ). So  $P_2$  is exactly the irreducible seed on which the rule reaches  $E_8$  as the last finite point — a second network reading of the  $(2, 3, 5)$  hull, not a derivation of  $P_2$ . [O] [verification/v299\_hypergraph\_0d\_autonomous.py]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ( $Am=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5)$  = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate  $(2/3)^6$  is *not* any eigenvalue of the adjacency and  $\varphi_0^{\text{ret}}$  is not a rational edge fraction. So a substrate reproducing the *full* structure must inject exactly two non-graph-spectral data — the cusp weight  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  (the gap  $6 \ln(3/2)$ ) and the analytic seed  $\varphi_0^{\text{ret}}$  — a precise delimiting of the open question. [O] [verification/v312\_hypergraph\_substrate.py]

## 5 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step:  $D_5$  and  $A_3$  have the *same* discriminant group  $\mathbb{Z}_4$ , and their discriminant-form norms are exactly two TFPT constants which add up to the  $E_8$  root norm.



**Glue theorem (proved)**

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue index  $|\mu_4| = 4$ , and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 = \text{the } E_8 \text{ root norm}$ . Hence  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1\_e8\_glue.py]

**6 How derivation now works: the derivation map**

What can one concretely *compute* with  $\{c_3, g_{\text{car}}\}$ ? The following map summarises what is now a consequence rather than an assumption.

| Sector                       | How it now arises                                                                   | Status  |
|------------------------------|-------------------------------------------------------------------------------------|---------|
| gauge group / families       | carrier $3+2$ , $\dim S^+ = 16$ , $N_{\text{fam}} = 3$ , $\Omega_{\text{adm}} = 48$ | [E]     |
| hypercharge                  | roots of the charge polynomial $bsY^2 + (s-b)Y - 1$                                 | [E]     |
| $U(1)$ index $b_1$           | $b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$          | [E]     |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha) = 0$                                               | [E]     |
| flavor matrix                | transport theorem (H1 proved, H2/H3 force the matrix)                               | [E]/[C] |
| electroweak scale            | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides $\alpha^{-1}$ by 5)        | [E]     |
| cosmological constant        | $\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)                | [E]/[C] |
| Hubble scale                 | $H_0 \sim \sqrt{\Lambda}$                                                           | [C]     |
| inflation amplitude          | $A_s = \frac{N^2}{24\pi^2} c_3^7$ in the $R^2$ branch                               | [C]     |
| $E_8$                        | $D_5 \oplus A_3 + \mu_4\text{-glue}$                                                | [E]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**Notable:** the scale grammar is *one* exponential engine. The same number  $\alpha^{-1} \approx 137$  generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

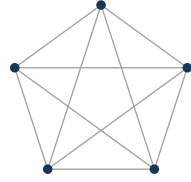
**Typing note on  $\Lambda$  (a deliberate danger zone).** “ $\Lambda$ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent  $e^{-2\alpha^{-1}}$  (an exact [E] readout), (ii) the dimensionless density quotient  $\rho_\Lambda / \bar{M}_{\text{Pl}}^4$  (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

**Cosmology is the Pascal action grammar of the carrier**

With  $x = v_{\text{geo}} / (5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$  the cosmological readouts are a *power tower* on the five-slot carrier graph  $K_5$ :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit  $x^1$ , Hubble is the product over the  $\binom{5}{1} = 5$  vertices,  $\Lambda$  is the product over the  $\binom{5}{2} = 10$  edges — the *same* Pascal triple that reads one generation, the action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . “Hubble = slot product,  $\Lambda$  = pair product.” [E]

 $K_5$ 

- $\binom{5}{1} = 5$  vertices — action  $x^1$  (Hubble slot)
- $\binom{5}{2} = 10$  edges —  $\Lambda$  density pair  $x^{10}$
- $\binom{5}{3} = 10$  triangles — dual sector
- $\binom{5}{4} = 5$  complements — inverse sector
- $\binom{5}{5} = 1$  determinant — closure

$K_5$  on the five carrier slots; the action powers  $x^1, x^5, x^{10}$  are its  $\binom{5}{k}$  faces (mnemonic; the physical scale map stays  $[C]$ ).

### The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths  $L_{f,j}$  are the *fixed residue matrix of the compiler*, whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers ( $\text{tr} = 9 = N_{\text{fam}}^2$ ,  $\det = 8 = h(D_5)$ , principal 2-minors 2,3,5 with product  $30 = h(E_8)$ ). Through the  $\varphi_0^{\text{ret}}$ -ladder  $\hat{m} = \frac{v_{\text{geo}}}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$  all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark mass ratios* are closed too (integer Plücker readouts on the derived selector stratum, [v49/v71](#)), with only the *absolute* amplitude scale reducing to the  $(U_{\text{wall}})$  anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional;  $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$  are RG scheme-layer ( $m_H$  rests on the closed UV quartic  $\frac{1}{16\pi^2}$ ); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; see below). (*Full derivation: document 2, [tfpt\\_2\\_standard\\_model](#).*) [[verification/v18\\_quark\\_yukawa.py](#)] [[verification/v20\\_lepton\\_c\\_derivation.py](#)] [[verification/v46\\_grand\\_mass\\_volume.py](#)]

## 7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*,  $\alpha^{-1}$  is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the  $C^+$  spinor and the puncture geometry  $\mathbb{P}^1 \setminus \mu_4$  produce  $D_5$  and  $A_3$ , glued to  $E_8$ ; *topologically*, the common discriminant  $\mathbb{Z}_4$  (the  $\mu_4$  winding) forces the glue and  $\Phi_{30}$ /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence.

**Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions  $\Rightarrow$  low overfitting risk), (ii) *rigid* (the glue condition  $5/4 + 3/4 = 2$  and the discriminant  $\mathbb{Z}_4$  leave little freedom), and (iii) *checkable* (machine-checkable  $E_8$  appendix, explicit fixed-point equation for  $\alpha^{-1}$ ). More explained from less — exactly the criterion that makes a theory more probable.

## 8 What role $E_8$ now plays exactly

$E_8$  is no longer the starting point but the *output format*:

- **Bookkeeping:**  $E_8$  is the unimodular container in which carrier ( $D_5$ ) and family ( $A_3$ ) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ( $\det 1$ ) is a hard condition; it closes only because the discriminant-form norms  $5/4$  and  $3/4$  (both already present in TFPT) add up to 2.  $E_8$  thus tests the internal coherence of the two atoms.
- **Not an end in itself:**  $E_8$  does not explain “everything”; it is the smallest even unimodular hull of the datum  $D_5 \oplus A_3$ . The physics sits in the *two inputs*, not in  $E_8$ .



**New — the secondary atlas.** Beyond that,  $E_8$  is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that  $E_6 \times A_2$  reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2 = h(D_5)$ . The discipline rule is: *every load-bearing number must appear in at least one  $E_8$  projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

## 9 What this means for the theory-of-everything question

### The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic  $\leftrightarrow$  unitary representation); the algebraic selector  $R$ , the  $Q$  spectrum and the splitting type are fixed (now *derived*:  $\det R = 8$  lattice,  $\text{Spec}(Q_+)$  from  $D_4$ , `v69/v70`; since `v136/v139` the dual pair  $(d, n)$  forces  $R$  *columnwise*,  $d$  is pure anchor data, and  $\text{Spec}(Q_+) = (1, 2, 3)$  is the  $A_3$ -exponent grading on the seam curve’s cohomology, `v137`), and the charged-lepton amplitudes are closed. The quark mass *ratios* ( $c_u/c_d = \frac{55}{117}$  etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, `v49/v71`) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ( $U_{\text{point}}$ , an anchor; Research Contract 1). [\[E\]](#)/[\[O\]](#)
2. **Gravity — exact in the  $R^2$  regime.** Spectral action  $\rightarrow R+R^2$ , BRST/Hodge kills the gauge directions, FRW reduction  $\rightarrow$  Starobinsky; the only remainder is the *controlled* low-curvature truncation ( $R^{n \geq 3} \rightarrow R^2$ , valid for curvature  $\ll M^2$ ). [\[E\]](#) in the regime
3. **Strong CP — decoupled.**  $\theta_{\text{eff}}=0$  follows from three structural facts ( $\gamma_5$ -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [\[E\]](#) (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap  $6 \log \frac{3}{2}$  of the  $C_6$  transfer operator (eigenvalues  $(k/3)^6$ ), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (`v258`), the spectral-action cutoff *is* that KMS weight (`v259`), the seam/carrier/ $E_8$  live on one Kummer/K3 (`v260`), all certified together (`v261`) — closed modulo the single keystone **SEAM.EQUIV.01**. *What this does not close*: the full measured pole masses (the absolute scale  $v_{\text{geo}}$  + standard RG), the absolute QCD scales ( $m_p/m_e$ , [\[O\]](#)), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [\[C\]](#)/[\[O\]](#)), and the real 4D interacting QFT / ambient measure ([\[O\]](#)). [\[E\]](#) (transfer model + boundary relative object) / [\[C\]](#) (continuum) / [\[O\]](#) (real 4D)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (3357 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [\[E\]](#)/[\[O\]](#)

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.*)

### The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

| TOE piece                                                                          | Status  | If not complete: what closes it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gauge group, $N_{\text{fam}}$ , hypercharge, $b_1$                                 | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $E_8 = (D_5 \oplus A_3) + \mu_4$ , group closure                                   | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| bootstrap $(g, \mu) = (5, 4)$ , “8” in $c_3$                                       | [E]     | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| fine structure $\alpha^{-1}$                                                       | [E]     | — <b>existence+uniqueness proved</b> ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| fermion masses ( $\varphi_0^{\text{ret}}$ -ladder)                                 | [E]/[O] | leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)                      | [E]/[C] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| flavour matrix / H2                                                                | [E]     | dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); $R$ forced columnwise by $(d, n)$ (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map                                                                                                                                                                                                                                                                                                                                             |
| seam quantum clock (R1)                                                            | [E]/[C] | typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)                                                                                                                                                                                                                                                                                                                                                                                                |
| inflation $A_s, n_s, r$ , scalaron $M$                                             | [C]     | scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; $n_s, r$ are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| boundary QFT (Modular Spectral Closure)                                            | [E]/[C] | <b>one relative object</b> TFPT <sub>QFT</sub> : $D_F$ = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ $E_8$ on one K3 (v260), certified (v261) — closed modulo SEAM.EQUIV.01                                                                                                                                                                                                                                                                                                                                                               |
| perturbative 4D S-matrix ( $S_{\text{pert}}$ )                                     | [E]/[C] | <b>Epstein–Glaser-constructible</b> (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278) — matter+gauge; gravity $R^2/\text{Weyl}^2$ non-unitary (Stelle ghost)                                                                                                                                                                                                                                                                               |
| <i>genuinely open — the actual TOE remainder (what the closure does not cover)</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| QFT measure (admissible sector)                                                    | [E]     | <b>closed: RP from cusp spectrum, OS reconstruction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| full measured pole masses                                                          | [C]     | ladder $\times v_{\text{geo}}$ , dressed by standard RG (no new physics); $v_{\text{geo}}$ over-determined (v274)                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| absolute QCD scales ( $m_p/m_e$ , $\Lambda_{\text{QCD}}$ )                         | [O]     | QCD matching contract (FR.MPME.01); not a compiler number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| full PMNS dynamics (mass scale, CP)                                                | [C]/[O] | angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute $\nu$ -scale = one seesaw ratio (v272)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (1) SEAM.EQUIV.01 — one named theorem ( <i>raw seam</i> = $(E_8)_1$ at $\tau=i$ )  | [O]     | the single arrow unifying the flat- $\tau=i$ geometry and the det $K=1$ holomorphy (v282/v286, import firewall). After the closing arc (v300–v302) its residual is a composition of <i>cited</i> theorems over <i>established</i> TFPT facts: Flat-Away hardened to a discrete obstruction + its pin derived from $(E_8)_1$ (v300); Route A’s invertibility discharged by the free-fermion classification (v301); the last input = the derived Recovery gap $6 \ln(3/2) > 0$ (v302). No undischarged TFPT-internal assumption remains (stays [O]: not machine-proved end-to-end) |
| (2) QG.AMB.01 — nonperturbative ambient measure                                    | [O]     | the genuinely separate strict-TOE summit: the perturbative $R^2/\text{Weyl}^2$ gravity ghost (Stelle) is the frontier; Tier-B’s holomorphy bit (det $K=1$ , v277) is absorbed into SEAM.EQUIV.01                                                                                                                                                                                                                                                                                                                                                                                 |
| analytic boundary origin of $\pi$ ( $c_3 = \frac{1}{8\pi}$ )                       | [C]     | <b>hardened</b> : $c_3 = 1/( \mathbb{Z}_2  \oint_{S^2} K) = 1/(8\pi)$ ( <b>Gauss–Bonnet</b> ); “8” = rank $E_8$ ; $\pi$ stays the one primitive                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\eta_B, m_p/m_e$ , Koide, dark matter                                             | [O]     | frontier handles only (see document 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one  $\varphi_0^{\text{ret}}$ -ladder (lepton rationals exact, quark  $O(1)$  digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and

on top of it the whole boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}}$  (the *Modular Spectral Closure*, v258–v261:  $D_F$  a covariance induction, the cutoff the seam KMS weight, seam/carrier/ $E_8$  on one Kummer/K3) — closed modulo the single keystone **SEAM.EQUIV.01**. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder  $\times v_{\text{geo}}$  + standard RG, [C]), the absolute QCD scales ( $m_p/m_e$ , [O]), the complete PMNS dynamics ([C]/[O]), and the nonperturbative *ambient/metric-sector* measure, i.e. *full quantum gravity*. The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278); only the nonperturbative ambient measure stays [O] — its  $R^2/\text{Weyl}^2$  gravity sector is renormalizable but non-unitary (the Stelle ghost), the lone perturbatively-irreducible piece, with Tier-B reduced to the single holomorphy bit  $\det K=1$  (v277) and the whole bedrock reduced to one geometric postulate, *the raw seam is the flat  $\tau=i$  pillowcase* (v276). The single open lemma is the named theorem **SEAM.EQUIV.01** — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$* . (The flat  $\tau=i$  geometry and the  $E_8$  holomorphy are *two faces of one object*: the  $(E_8)_1$  character is  $\chi_{E_8}=j^{1/3}$ , so  $\chi_{E_8}(i)=1728^{1/3}=12$  and  $\tau=i$  is at once the order-4 CM point and the  $(E_8)_1$  partition-function modulus, v282.) It stays [O] (not machine-proved end-to-end), but its whole residual is now a composition of *standard cited theorems* (Osterwalder–Schrader/quasi-free clustering, the Kitaev free-fermion classification, the Route-A AQFT stack) over *established* TFPT facts, with *no* undischarged TFPT-internal assumption left — Lean-pinned to exactly those named steps plus the derived Recovery gap  $\Delta = 6 \ln(3/2) \approx 2.43 > 0$  (v308); the conformal-deck premise **QGE0.SYM.01** is its *downstream* definitional face (v323). This is by design the *one* irreducible structural postulate of the theory — the role the *constancy* of  $c$  plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; “closing the theory” is therefore the concrete task of proving (or refuting) exactly this one arrow, not an open-ended search. The full sprint-by-sprint reduction that pinned **SEAM.EQUIV.01** is recorded in the changelog, not replayed here. The seam seed is *hardened*:  $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$  is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank  $E_8$  and only the Gauss–Bonnet  $\pi$  left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and  $\pi$  is the lone continuous primitive. *4D-QFT fork policy (frozen, v265)*: the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

## 10 Does this make the theory more probable?

**Plausibility balance.** *Pro*: (a) two inputs instead of many; (b)  $\alpha^{-1}$  to  $2.9 \times 10^{-10}$  (CODATA-2022,  $1.9\sigma$ ) as a *fixed point*, not a fit; (c)  $E_8$  constructed, not postulated; (d) one compiler generates several sectors  $\Rightarrow$  fewer degrees of freedom, more predictive power.

*Contra/open*: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ( $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

## 11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

| Observable                          | TFPT value & derivation                                                                                                                                                                       | Experiment (running / upcoming)                                                                                                      | St.     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------|
| solar<br>$\sin^2 \theta_{12}$       | angle<br>prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.) | <b>JUNO</b> : first 59.1 d run $0.3092 \pm 0.0087$ (compatible), precision phase pending                                             | [E]/[C] |
| reactor<br>$\sin^2 \theta_{13}$     | angle<br>$\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ ; seed $\times$ carrier-trace $e^{-\gamma}$ , $\gamma = \frac{5}{6}$                                                                      | Daya Bay / RENO / JUNO (PDG 0.0220)                                                                                                  | [E]     |
| atmospheric<br>$\sin^2 \theta_{23}$ | $\approx \frac{1}{2}$ ( $\mu\tau$ -symmetric limit; octant not selected)                                                                                                                      | NOvA / T2K / <b>DUNE</b> (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)                                                             | [C]     |
| fine structure $\alpha^{-1}$        | 137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)                                                                                                             | CODATA-2022 (data through 2022)<br>137.035999177(21): dev. $2.9 \times 10^{-10}$ ( $1.9\sigma$ of its uncertainty)                   | [E]     |
| tensor ratio $r$                    | $r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$ ); scalaron-reheating point 0.0045 ( $N_*=51.4$ , v86)                                                           | now $r < 0.036$ (BICEP/Keck BK18); <b>CMB-S4</b> ( $\sigma_r \leq 5 \times 10^{-4}$ , obs. $\sim 2033$ )                             | [C]     |
| tilt $n_s$                          | $n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ( $N_*=51.4$ , v86)                                                                                 | Planck ( $0.9649 \pm 0.0042$ ); CMB-S4 sharpens                                                                                      | [C]     |
| amplitude $A_s$                     | $\frac{N_*^2}{24\pi^2} c_3^7$ ; $2.0 \times 10^{-9}$ at $N_*=55$ (the measured $A_s$ prefers fast reheating, v86)                                                                             | Planck ( $2.1 \times 10^{-9}$ )                                                                                                      | [C]     |
| neutron EDM                         | $\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)                                                                                                                   | <b>PSI nEDM</b> / SNS (running)                                                                                                      | [E]     |
| CP phase $\delta_{\text{CKM}}$      | $\delta = \frac{\pi}{3} + 3\lambda^2 = 68.65^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$                                                               | <b>LHCb</b> / Belle II $\gamma$ combination; decision at $\sigma_\gamma \leq 0.96^\circ$                                             | [E]     |
| cosmic<br>birefringence $\beta$     | $\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)                                                                                                                   | ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); <b>Simons Obs.</b> / LiteBIRD | [C]     |
| $0\nu\beta\beta$ / ordering         | Majorana branch; normal-ordering preference                                                                                                                                                   | <b>LEGEND</b> / <b>nEXO</b> , DUNE, KATRIN                                                                                           | [C]     |
| flavor invariants                   | $\det R = h(D_5) = 8$ , principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)                                                                                                   | any future CKM/PMNS global fit must satisfy these                                                                                    | [E]     |
| $H_0$ vs. $\Lambda$                 | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ , $\Lambda \sim e^{-2\alpha^{-1}}$ , $H_0 \sim \sqrt{\Lambda}$ : one engine                                                                           | SH0ES / <b>DESI</b> / Planck (Hubble tension)                                                                                        | [C]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

### The two decisive near-term tests

(1) **JUNO and  $\theta_{12}$ .** JUNO is *taking data since August 2025* and targets sub-percent  $\sin^2 \theta_{12}$ . TFPT commits to the single prediction of record  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$  (misalignment  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ ). The first 59.1-day JUNO run reports  $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$  — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from  $\sim 0.307$  at high significance falsifies the seam mechanism.

(2)  **$r$ .** The  $R^2$  scalaron with  $M$  fixed by  $c_3^7$  predicts a small  $r = 12/N_\star^2$ : the frozen band  $r \in [0.0033, 0.0048]$  over  $N_\star \in [50, 60]$ , with the scalaron-reheating point  $r = 0.0045$  ( $N_\star=51.4$ , **v86**) — already below the current bound  $r < 0.036$  (BICEP/Keck BK18) and within reach of CMB-S4 ( $\sigma_r \leq 5 \times 10^{-4}$ , first observations  $\sim 2033$ –34). A detection of  $r \gtrsim 0.01$  would be incompatible with the  $R^2$  branch on which  $M_{\text{Pl}}$  and  $A_s$  rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable  $E_8$  appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate  $E_8$ ; and the discipline rule “*every load-bearing number appears in at least one  $E_8$  projection*” (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix **[E]** [[verification/v100\\_numerology\\_null\\_mc.py](#)].

### Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry [verification/freeze\\_file.csv](#). The decisive entries:

- **solar**  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$  — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor**  $r \approx 0.0040$  — any robust  $r \gtrsim 0.01$  kills the  $R^2$  branch carrying  $M_{\text{Pl}}, A_s$ ;
- **ordering** normal, small  $m_{\beta\beta}$  — inverted ordering or large  $m_{\beta\beta}$  kills the Majorana branch;
- **strong CP**  $\theta_{\text{eff}} = 0$  — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$  — a robust  $w \neq -1$  kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: [verification/predictions\\_frozen.json](#) registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [[verification/v84\\_frozen\\_registry.py](#)] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by [manifest.sha256](#); ledger REG.FREEZE.01). In particular the registry admits exactly *one*  $\theta_{12}$  prediction of record (the seed  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards.  $r$  and  $n_s$  enter only as *bands* over the declared external  $N_\star \in [50, 60]$ , never as point predictions.

### What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, **v49/v69/v71**); the only residual is the *absolute* amplitude scale, an anchor ( $U_{\text{point}}$ ) of the same nature as the one dimensional scale — no new physics;
- further constants as carrier traces (analogous to  $b_1 = 41/10$ ), since the compiler systematically produces traces over  $S^+$ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ $R^2$  branch is lifted from **[C]** to **[E]**.



## Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator  $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$  builds, from two inputs  $\{c_3, g_{\text{car}}\}$  via  $D_5 \oplus A_3 + \mu_4$ , the  $E_8$  container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms  $\{c_3, g_{\text{car}}\}$  alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

| Script                                          | What it machine-checks                                                                                                                                                               |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>v1_e8_glue.py</code>                      | glue theorem $E_8 = (D_5 \oplus A_3) + \mu_4$ : $\text{disc} = \mathbb{Z}_4$ , $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ , $240 = 16 \cdot 5 \cdot 3$ , $248 = 240 + 8$      |
| <code>v2_carrier_pascal.py</code>               | $g_{\text{car}} = 5$ unique Pascal solution; $16 = 1 + 5 + 10$ , $N_{\text{fam}} = 3$ , $\text{rank } E_8 = 8$ , $\Omega_{\text{adm}} = 48$ , $b_1 = \frac{41}{10}$                  |
| <code>v3_em_alpha.py</code>                     | EM closure: unique root of $F_{U(1)}(\alpha) = 0$ , $\alpha^{-1} = 137.0359992168$ , existence/uniqueness, inverse test, ablation                                                    |
| <code>v4_flavor_matrix.py</code>                | residue matrix $R$ : $\det = 8$ , minors $\{2, 3, 5\}$ , $\chi_R = t^3 - 9t^2 + 10t - 8$ , SNF $\text{diag}(1, 1, 8)$ , $\ R\ _F^2 = 78$ , $\sum L = 40$                             |
| <code>v5_e8_cascade.py</code>                   | cascade $D_n = 60 - 2n$ : endpoints, exponent rungs $\rightarrow 240$ , IR tail 46080, variance 6552                                                                                 |
| <code>v6_bootstrap.py</code>                    | reverse glue $\mu^2 - 5\mu + 4 = 0$ ; $g_{\text{car}} = 5$ three ways; $8 = \text{rank } E_8 = h(D_5) = \varphi(30)$                                                                 |
| <code>v7_gravity_cosmo.py</code>                | scalaron exponent 7, $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^7$ , $A_s, n_s, r$ , $\Omega_b$ , $\sin^2 \theta_{12}$                                                         |
| <code>v8_horizon.py</code>                      | horizon code: $\frac{1}{2\pi} = 4c_3$ , $1920 =  W(D_5) $ , $S_{dS}$ , Page, $\beta_{\text{rad}} = 0.2424^\circ$                                                                     |
| <code>v9_neutrino_texture.py</code>             | solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23} = 45^\circ$ , $\theta_{13} = 0$    |
| <code>v10_projection_involution.py</code>       | $Q, \Sigma$ algebra: $K = R + Q\Sigma$ , $L = R + Q(I + \Sigma)$ ; $\chi_Q, \chi_K$ ; det ladder $3 \cdot 4 \cdot 8 \cdot 20 = 1920 =  W(D_5) $ ; $a^\top K a = 41$                  |
| <code>v11_unique_KQ.py</code>                   | $K, Q$ are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)                                                               |
| <code>v12_mass_generation_polynomials.py</code> | sector/generation polynomials of $K$ and the anchor-block det ladder $(9, 10, 16, 40)$                                                                                               |
| <code>v13_open_gates.py</code>                  | gate closures: $M = 41 = 10b_1 = \sum L + N_\Phi$ ; $Q_+ = A_3$ exponents, $Q_-^2 = N_{\text{fam}}$                                                                                  |
| <code>v14_carrier_uniqueness.py</code>          | $g_{\text{car}} = 5$ unique ( $N_{\text{fam}} \in \mathbb{Z}^+$ , $\text{rank } E_8 = 8$ ); split $(3, 2)$ from $b + s = 5$ , $b^2 + s^2 = 13$ ; $\text{Tr } Y = \text{Tr } Y^3 = 0$ |
| <code>v15_bootstrap_classification.py</code>    | $D_5 \oplus A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$                                                                                 |
| <code>v16_solar_dual_anchor.py</code>           | $a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ ; $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; full PMNS                                |

*continued on next page*



| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v17_hexagonal_resolvent.py         | finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark $c$ backbone)                                                                                                                                                                                              |
| v18_quark_yukawa.py                | quark source ratios $\frac{m_u}{m_d}=0.470085$ , $\frac{m_c}{m_s}=13.61$ , $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ ; full hierarchy                                                                                                            |
| v19_monodromy_moduli.py            | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ( $3^7$ ); $D_4$ $SU(3)$ monodromy constructible but $D_4$ -fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$                                                                                                     |
| v20_lepton_c_derivation.py         | lepton $c$ 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7}, \frac{4}{3}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9}\Rightarrow\frac{7}{6}$                                                       |
| v21_solar_product_quark.py         | solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$ ), $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$ ; quark law fails ( $\frac{1}{7}$ vs 0.724)                                                   |
| v22_open_gates_audit.py            | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)                                                                                                                                                      |
| v23_anchor_generator.py            | anchor $a=(1,1,2)$ : $e_k=(4,5,2)$ , $c_3=\frac{1}{2e_1\pi}$ ; $p_n=2+2^n\Rightarrow 240, 248$ , binary ladder; inputs $\rightarrow\{a,\pi\}$                                                                                                                                             |
| v24_quark_ratio_closure.py         | quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $<0.03\%$ ; rational $c$ gauge, $\Lambda_q$ in $O(1)$                                                                                                                                      |
| v25_frontier_conjectures.py        | conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$ ; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\mu\text{eV}$                                                                                                                          |
| v26_flavor_frontier_unification.py | the “11” is not uniquely forced ( $\Rightarrow$ [C]); flavor frontier unifies to one $(U)$ gate; cusp config on the parabolic stability wall                                                                                                                                              |
| v27_wall_representative.py         | explicit balanced wall rep $W_{\text{wall}}$ (cols = perm(0,1,2), rows $w=(2,1,1)$ , pardeg 0); selector $\det R=8$ , $\text{Spec}(Q_+)=\{1,2,3\}$                                                                                                                                        |
| v28_gravity_fR.py                  | $R+R^2$ closed: $f(R)$ , $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_\star=57\rightarrow n_s, r, A_s$ ; full metric measure open; $248c_3^2<6\log\frac{3}{2}$                                                                                                                           |
| v29_research_contract_certs.py     | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick $W_{\text{wall}}$ ) and $G5$ gap-dominance $2\ V\ <\Delta$                                                                                      |
| v30_d4_character_variety.py        | $C_U^{(2)}$ : full $D_4$ -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)                                                                           |
| v31_R_dictionary.py                | $R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); $R$ integer but $\rho$ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)                                                                      |
| v32_rh_splitting.py                | RH route: $\sum_k U^k A_0 U^{-k}=4\text{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow\text{diag}(A_0)=(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k=I+R$ -bridge stay open ( $c_u/c_d$ not forced)                  |
| v33_explicit_flat_bundle.py        | explicit valid $(U_{\text{wall}})$ flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2 + \ M_\infty-I\ \sim 10^{-9}$ ( $\prod M_k=I$ ) + irreducible (case A)                                                                                                          |
| v34_h2_bridge_attempt.py           | H2 bridge: explicit per-puncture $M_k$ (cusp, $\prod M_k=I$ ); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\text{min}}$ dictionary is the remaining input ( $c_u/c_d$ not obtained)                                                                       |
| v35_quark_qcd_boundary.py          | the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)                                                                                                           |
| v36_spectral_action_g2.py          | G2: spectral action $\rightarrow R+R^2$ ( $a_2\sim-R/3$ , $a_4 _{R^2}\sim R^2/72$ ); $M^2/\bar{M}_{\text{Pl}}^2=6(4\pi)^2/f_0$ , $c_3^7$ fixes $f_0$ ; gap-decoupling $\Delta_{\text{eff}}=1.648>0\Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target |
| v37_plucker_anchor.py              | anchor-plane Plücker apparatus: $\ Pl(K)\ _1=11$ , $\ Pl_R(K)\ _1=26$ , $\sum Pl_R(L)=60$ ; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$ ; $K_e-K_u=a$ , $L_e-L_u=\text{Exp}(A_3)$ ; lepton ring $c_e c_\tau= \mathbb{Z}_2 c_\mu$ ; Koide $u\rightarrow\frac{53}{54}u$ ([C])                        |

*continued on next page*

| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v38_uwall_killswitch.py       | $(U_{\text{wall}})$ U2 kill-switch hardened: $D_4$ -cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq 0.35$ , reducible locus measure-zero $\Rightarrow$ case A generic, $(U_{\text{wall}})$ survives; amplitude dictionary still research-level                                                                                                                 |
| v39_uwall_selectors.py        | $(U_{\text{wall}})$ selector side closed on the explicit point: splitting type = anchor $a=(1,1,2)$ , pardeg=0; $\det R=8=n \cdot a$ , $\text{Spec}(Q_+)=\{1,2,3\}=3\alpha+1$ read off the bundle; only the amplitudes $c_u/c_d$ ( $\Gamma^{\min}$ Hitchin) remain                                                                                                                        |
| v40_harmonic_metric.py        | $(U_{\text{wall}})$ harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$ ): unique invariant form $H>0$ , unitarised $\widetilde{M}_k$ cusp-class with $ \widetilde{M}_k  \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ ; the feared Hitchin PDE collapses to unitarisation + combinatorial $R$                                                    |
| v41_leg_assignment.py         | $(U_{\text{wall}})$ final leg test: amplitudes $=\delta=\frac{1}{2}$ resolvent (closed); $\Lambda_\mu>1$ rules out holonomy moduli; harmonic holonomy $ \text{diag} =(0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta=\frac{1}{2}$ (suggestive); honest negative on literal $c_u/c_d$ reproduction                                                                                         |
| v42_exterior_leg.py           | $(U_{\text{wall}})$ Exterior Leg Lemma: $u, d$ share $(r, w)=(1,1)$ so any scalar leg gives $c_u/c_d=1$ ; the $u/d$ split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$ , $\ \cdot\ _1=11$ generated $\Rightarrow \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]                                                                                          |
| v43_exterior_bridge.py        | $(U_{\text{wall}})$ $\Lambda^2 F$ bridge: $\Lambda^2(\widetilde{M})=\widetilde{M}$ (exterior leg $=\bar{3}$ ) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ( $\det R=8$ , $\text{Spec}(Q_+)$ confirmed), not continuous                                                                                                     |
| v44_carrier_exterior.py       | $(U_{\text{wall}})$ Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$ ; $u^c \in \Lambda^2(5)$ , $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2,4)$ , sum=6= $ R^+(A_3) $ , diff=2= $ \mathbb{Z}_2 $ ; the exterior leg is the carrier's own $\Lambda^2$ grading (no special math)                                                                                       |
| v45_family_exterior.py        | $(U_{\text{wall}})$ the “11” is Pascal: $ \text{Pl}(K) =(1,4,6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$ ; $\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of $\mu_4$ up to $\deg u^c=2$ ); $15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$                                                                                        |
| v46_grand_mass_volume.py      | absolute mass-scaling is [E], not a product rule: sector dets $(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \Lambda_\Lambda \Rightarrow \det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ ; $Q$ -rows $(4,5,6)$ sum 15= $\dim A_3$ ; $25+15=40=\sum L$ |
| v47_selection_theorem.py      | Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$ ; 16-spinor $\rightarrow D_5$ ; $q(D_5)+q(A_3)=2$ , glue 4= $ \mu_4 $ ; only $D_5 \oplus A_3$ meets all four conditions ( $D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]                                                                                                                           |
| v48_em_ward.py                | EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^2 \alpha^2 - \text{transport } 8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ; $8b_1=\frac{4}{5}M=\frac{164}{5}$ , $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]                                                                                                                                   |
| v49_readout_rigidity.py       | Readout Rigidity (Thm U2): $c_u/c_d=\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}})=U_{\text{unitary}}+U_{\text{H2}}+U_{\Lambda^2}+U_{\text{point}}$ , only $U_{\text{point}}$ open                                                                                                           |
| v50_q_geometry.py             | Q geometry (Thm Q): $Q_+ = \text{diag}(3,2,1)$ -block $\chi=(t-1)(t-2)(t-3)$ , $Q_- \chi=t(t^2-3)$ ; $Q$ unique under rows(4,5,6)/cols(9,5,1)/spectra; geom. origin [C]                                                                                                                                                                                                                   |
| v51_boundary_half_step.py     | glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $ : $q(D_5)=\frac{5}{4}$ , $q(A_3)=\frac{3}{4}$ ; four ops forced $\rightarrow \{\text{sum}=2$ root norm, diff= $\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$ , prod= $\frac{15}{16}$ , ratio= $\frac{5}{3}\}$ . So $\delta=\frac{1}{2}$ is the carrier-family count over the glue index, not chosen [E]                               |
| v52_pencil_endpoints.py       | $K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $ , $P(0)=4= \mu_4 $ , $P(1)=20=\det L$ , $P(2)=68=2p_5(a)$ ; $\det(K-Q)=2$ , $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]                                                                                                                                                                                                          |
| v53_compiler_core.py          | Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5,3)$ — rank $E_8=8$ , $ \mathbb{Z}_2 =2$ , Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2  \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]                                                                                |
| v54_seam_horizon_keystones.py | The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned $(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1, (2/3)^6, (1/3)^6\}$ , $\lambda_2=(2/3)^6$ ) governs both SM flavor gap and horizon Page recovery [E]                                                                                       |

continued on next page

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v55_coxeter_cycle.py           | Computed $E_8$ Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$ , eigenphases = exponents = totatives of 30, so $\text{rank } E_8=8=\varphi(30)$ = live phases of the order-30 cycle; one $\alpha^{-1}$ sets EM, $S_{dS}$ and $\Lambda$ ( $S_{dS}\rho_\Lambda=32\pi^4$ ) [E]                                                      |
| v56_unique_attractor.py        | Gapped boundary transport (gap $6\log\frac{3}{2}>0$ ) $\Rightarrow$ <i>unique</i> attracting fixed point, geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes; entropy reset = rank-1 projector [E]                                                                                                               |
| v57_horizon_crosslinks.py      | Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ( $1/4=1/ \mu_4 $ entropy density); Hod QNM $\omega_R/T_H=\ln 3=\ln N_{\text{fam}}$ ; Hawking power denom $1920= W(D_5) $ ; one $\alpha^{-1}$ for EM/ $S_{dS}$ / $\Lambda$ /Hubble [E] (+ [C] identification)                                                                    |
| v58_seam_horizon_chain.py      | Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided $S^2$ Gauss–Bonnet $c_3=1/(2\cdot 4\pi)=\frac{1}{8\pi}$ ; KMS $4c_3\kappa$ ; $S_{BH}=M^2/2c_3$ ; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open                                                                             |
| v59_area_law_evidence.py       | Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3}\log L$ ), by the <i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2  \mu_4 $ . The $c_3$ coefficient stays [O] open                                                             |
| v60_lambda_metrology_branch.py | $\Lambda$ branch selection: $(8\pi)^2\delta_{\text{top}}=\frac{3}{4\pi^2}$ , mis-scale $2c_3/\delta_{\text{top}}=\frac{64\pi^3}{3}=661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215^\circ\pm 0.074^\circ$ vs $0.2424^\circ$ ( $0.37\sigma$ ) [E]                                                              |
| v61_cft_bridge.py              | Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$ , $c(D_5)=5$ , $c(A_3)=3$ = (rank $E_8, g_{\text{car}}, N_{\text{fam}}$ ); conformal embedding $c_{\text{coset}}=0$ ; $SU(3)\leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$ , $4\rightarrow 3+1$ (one geometry $\mathbb{P}^1\setminus\mu_4$ , two sides) [E]                       |
| v62_data_scorecard.py          | TFPT vs 2024/25 data: $\alpha^{-1}$ , $\sin^2\theta_{12}$ , $\beta_{\text{rad}}$ match; $\sin^2\theta_{13}$ ( $2.0\sigma$ ) and Starobinsky $n_s$ vs DESI ( $\sim 2\sigma$ ) mild tensions; $\theta_{23}$ open; $r\approx 0.004$ future falsifier [E]                                                                                                           |
| v63_seam_engineering_index.py  | Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2\log(3/2)}\approx 0.323$ ( $2\ V\ =\frac{31}{4\pi^2}$ , $248=\dim E_8$ , $31=1+h^\vee$ ), $\Delta_{\text{eff}}\approx 1.648$ ; four-class possibility taxonomy [E]/[O]                                                                                                                         |
| v64_causal_boundary_nogo.py    | Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)                                                                                     |
| v65_falsification_layer.py     | Prediction layer with explicit failure thresholds: 3 core ( $v_{\text{GW}}=c$ , no 4th gen, seam=8), 4 observational ( $\beta, r, n_s, \theta_{12}$ ), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]                                                                                                                                     |
| v66_e8_casimir_degrees.py      | Organizing simplification: the compiler atoms <i>are</i> the $E_8$ Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ( $2= \mathbb{Z}_2 $ , $8=\text{rank}$ , $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $30=h(E_8)$ ); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]                                                                       |
| v67_area_law_coefficient.py    | Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=2\pi c_3A=\frac{1}{4}A$ ( $2\pi c_3=\frac{1}{4}$ ); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$ . Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]                                               |
| v68_seeley_dewitt_residual.py  | Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ( $\propto d\Lambda^2 f_2$ ), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2/\text{scalaron}$ derived [E]                                     |
| v69_d4_q_geometry.py           | $D_4$ -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1\setminus\mu_4)=B_1\oplus E$ , $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$ ; $\Sigma$ -split $=D_4=\mathbb{Z}_4\rtimes\mathbb{Z}_2$ [E]                                                |
| v70_q_integer_lift.py          | Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action $R$ unimodular (eigenvalues $\{-1, i, -i\}$ ); $\det Q=3=N_{\text{fam}}$ , SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$ , not closable by $D_4$ alone [E]                                                                                           |
| v71_simple_r_bridge.py         | <i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ( $\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ( $\frac{55}{117}$ , no monodromy, v49 rigidity); residual $U_{\text{point}}=$ absolute norm. = an anchor [E]/[O] |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v72_q_det_from_cusp.py              | $\det Q = N_{\text{fam}}$ derived from the cusp class: $\det Q =  \text{coker } Q  = \text{cusp-class order} = \text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$ ); $\text{coker } Q = \mathbb{Z}/N_{\text{fam}} = \text{triality deck group} \Rightarrow \text{integer-lift residual closed}$ [E]                                                                                                                                                         |
| v73_k_c3_half.py                    | Central coefficient $k=c_3/2$ forced: $(1/2)$ variational $\times 1/( \mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ( $\chi(S^2)=2$ ), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$ ; only the absolute 4D Newton scale stays the anchor (v68) [E]                                                                                                                                                                                                                                         |
| v74_compiler_micro_lemmas.py        | Spine quotient ladder $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}) = \text{adjacent quotients of } (2, 3, 4, 5)$ ; pencil differences $(2, 16, 48) = ( \mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L = R + 61e_1^\top$ , $a^\top R^{-1}\mathbf{1} = 0$ (Sherman–Morrison rank-one invariance) [E] |
| v75_upoint_to_vgeo.py               | <b>Gate 1 complete:</b> $U_{\text{point}} \rightarrow v_{\text{geo}}$ . Ratios closed (v49/v71) + lepton $c$ exact (v20) + sector products = $\varphi_0^{\text{ret}}$ -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale $v_{\text{geo}}$ = the same anchor as $1/G$ (v68); two $[A] \rightarrow \text{one}$ [E]/[O]                                                                                                   |
| v76_gmetric_reduction.py            | <b>Gate 2:</b> Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , $\ V\  \leq 248c_3^2 = \frac{31}{8\pi^2}$ ); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]                                                                                                                                                                    |
| v77_e8_conformal_net.py             | <b>G6 route:</b> the seam boundary measure = the $(E_8)_1$ lattice net (rigorous, holomorphic $c=8$ ); level-1 $c(E_8) = \frac{248}{31} = 8$ , $c(D_5)=5$ , $c(A_3)=3$ , embedding $5+3=8$ (coset $c=0$ ); imports $G_6$ into RCFT/conformal-net rigor [E]/[C]                                                                                                                                                                                                                                                |
| v78_vgeo_floor.py                   | $v_{\text{geo}}$ <b>floor:</b> no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times \text{one } \bar{M}_{\text{Pl}}$ ; $S_{\text{dS}}\rho_\Lambda = 32\pi^4$ pins it from one measurement; theory at the minimum (one scale + $\pi$ ) [E]/[O]                                                                                                                                                                                                      |
| v79_review_identities.py            | Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n = (3^n - (-2)^n)/5 = 1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ( $\mathbf{1}^\top M^{-1}\mathbf{1} = 1/\text{atom}$ , $a^\top M^{-1}a = 1$ for $R, K, L$ ); MacWilliams $(1, 10, 5)$ ; $6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)$ . Rejects $m_p/m_e$ , $\eta_B$ near-formulas (firewall) [E]                                                                                                                  |
| v80_operator_pencil_geometry.py     | Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ} = (N_{\text{fam}}x +  \mathbb{Z}_2 )(N_{\text{fam}}x + g_{\text{car}}) = (3x+2)(3x+5)$ ( $\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3, 10, 16, 20)$ ; type checker $\det B_{Q,K,R,L} = (9, 10, 16, 40)$ ; audit: differential spectrum, $F_4 \times G_2$ shadow $248 = 52 + 14 + 26 \cdot 7$ [E]/[C]                                                                                                                |
| v81_singular_pencil_matrices.py     | Double cover $y^2 = \det B_{K+xQ} = (3x+2)(3x+5)$ : Koide $x = -\frac{2}{3}$ and carrier $x = -\frac{5}{3}$ are the two branch points (deck $2 =  \mathbb{Z}_2 $ ; disc $= 81 = N_{\text{fam}}^4$ ; separation $= 1$ transport period). Clearing matrices $C_{2/3} = 3K - 2Q$ ( $D_5 \oplus A_3$ glue, $B = (5, 7)^\top (9, 11)$ , $\sum = 240$ ), $C_{5/3} = 3K - 5Q$ (neutral, $\chi = (\lambda+2)(\lambda^2 + \lambda + 6)$ ); scalaron $7 = \text{mixed anchor disc}/N_{\text{fam}}$ [E]                  |
| v82_koide_attractor_splitting.py    | Koide attractor forced: the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6 = \lambda_2$ of the established transfer operator $\Rightarrow F'(2) < 1$ attractor, $F'(5) > 1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$ : the clean split is non-generic (anchor-first) [E]/[C]                                                                                                                                               |
| v83_e8net_holomorphic_uniqueness.py | Holomorphy pins $(E_8)_1$ : #primaries = det Cartan ( $D_8:4$ vs $E_8:1$ ), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $ ); carrier Pascal truncation $K = (g-1)/2 = \text{half-spinor split}$ [E]                                                                                                                                                                                                        |
| v84_frozen_registry.py              | Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> $\theta_{12}$ prediction of record; $r$ , $n_s$ as $N_\star$ bands; conditional entries carry their hypothesis by name [E]                                                                                                                                                                                                                                                 |
| v85_master_cover.py                 | Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $= N_{\text{fam}}^4 \det(G)^2$ ); the disc-81 family is its orbit ( $-\frac{8}{3} = -\text{rank } E_8/N_{\text{fam}} = \text{carrier} - \text{one period}$ ); $\mu_4$ is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]                                                                          |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v86_nstar_reheating.py           | $N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128$ GeV, $T_{\text{reh}}=9.6\times 10^9$ GeV, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$ , $r=0.0045$ (inside the frozen band); honest tensions: $n_s + 0.9\sigma$ , $A_s$ dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]                                                                                                                                 |
| v87_bulk_uniqueness_reduction.py | Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow$ Rep = Vect $\Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both $E_8$ pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]                                                                                                                                                                                              |
| v88_cp_phase_audit.py            | Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$ ; the $0.07^\circ$ -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [E]                                                                                                                                                                                                         |
| v89_carrier_index_lemma.py       | Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); $\mu$ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]                                                                                                                                                                                                         |
| v90_conical_defect_chain.py      | Fursaev–Solodukhin factor $4\pi$ <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]                                                                                                                                                                                                   |
| v91_spine_tetrahedron.py         | Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $ ; $240= \mu_4  E(K_4)  E(K_5) $ with $K_6$ negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]                                                                                                                                                                                                                              |
| v92_glue_uniqueness.py           | Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$ ; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ , nothing else [E]                                                                                                                                                         |
| v93_koide_relaxation_toy.py      | Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$ ; source at $\rho=-\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]                                                                                                                                                                   |
| v94_sheet_diamond.py             | Sheet diamond: all four flavor operators on <i>one</i> surface $M(s,t)=R+Q \text{ diag}(s,t,t)$ (pencil = the cut $(1+x, x-1)$ ; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all $s$ ; cofactor seam normal $(5, -9, 6)$ ; Plücker canonicalisation: 11 and 26 both live in $K$ ; spine-quotient firewall <i>rejected</i> [E]                                                                                                                               |
| v95_centered_diamond.py          | Centered cross: $Q=U+V$ , $R/L=C\mp U$ , $K/F=C\mp V$ (five objects $\rightarrow$ three); Spec $V=\{0,1,2\}=N_{\text{fam}}\cdot\text{cusp}$ ; $\det C=14$ , $\sum C=31=2^{g_{\text{car}}}-1$ ; $\text{Pl}_R(C)=7(2,3,1) \parallel \text{Pl}_R(L)=10(2,3,1)$ ; $G_2$ reading audit-typed [E]                                                                                                                                                                                               |
| v96_branch_kernel_selection.py   | Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$ : up = $-$ down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$ ); anchor-placement controls [E]                                                                                                                              |
| v97_sheet_conjugation_bridge.py  | Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer $T_A$ , $\det=-1$ ; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$ ; $\langle T_A, \Sigma \rangle = D_4$ ; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $ ); remaining [C]: $Q_+$ grading = $A_3$ discriminant grading [E]/[C]                                                         |
| v98_discriminant_dictionary.py   | Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A\Sigma$ acts integrally on the cusp basis ( $Ge_1=-e_1$ , $Ge_3=e_2$ , $Ge_2=-e_3 = B_1\oplus E$ ); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A GT_A^{-1}=G^{-1}$ ( $k \mapsto -k$ ); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$ , $q_{A_3}(2)=\frac{1}{2}$ ; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing $Q$ -geometry gate [E] |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v99_koide_flow_time.py         | Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}})\det B(q)$ , time-1 map = $F$ [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> ( $Q$ crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$ , $t$ passes every integer $\geq 3$ within $+0.5\sigma$ ) [E]; conditional steps: $n=2$ excluded ( $-2.9\sigma$ ), $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427\text{ MeV}$ ( $+0.14\sigma$ ), decision at $\sigma(m_\tau)\sim 0.01\text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)                                                                           |
| v100_numerology_null_mc.py     | Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$ , robust to budget/window variation; $2\times 2\cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$ , <i>conditional on the declared grammar</i> [E]                              |
| v101_horizon_anchor.py         | The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2) =$ traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$ ); interpolation $(x^2+1)/\Phi_3(x)$ ; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}] =$ the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$ , deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$ ; six atom landings, zero free parameters [E]/[C] (bulk reading)                                                         |
| v102_seam_orientation.py       | One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$ , $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate $\Delta$ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$ ; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]                                                                                               |
| v103_trisection_normal_form.py | Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — <i>one</i> cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=( \mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N=-\frac{8}{9}=-\text{rank } E_8/N_{\text{fam}}^2$ ; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$ , missing = one <i>clock</i> [E]/[C] |
| v104_nariai_clock.py           | The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the $dS_2$ static-patch equation with $m^2=-2\Lambda=- \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$ ; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$ ; entropy-deviation rate $2H= \mathbb{Z}_2 H$ ; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]                                            |
| v105_residual_inventory.py     | The residual inventory: <i>one constant</i> $(2/3= \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the $\Lambda$ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + $\pi$ ; a sixth gap would fail the script [E]/[C]                                         |
| v106_review_validation.py      | External-review validation: seed normal form $\varphi_0^{\text{ret}}=\frac{ \mu_4 }{N_{\text{fam}}}c_3+\Omega_{\text{adm}}c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2=120=5!$ computed from the charge table; factorial spine $5!= R^+(E_8) =\sum \text{exponents}$ , $240=2\cdot 120$ , $1920=2^4\cdot 5!$ ; degree-2 inventory (Pascal $K=2$ unique at $g=5$ , glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$ , $\binom{5}{2}=10$ ); named [C] hypothesis <i>Quadratic Boundary Locality</i> ( $\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240=16\times 15$ non-unique [E]/[C]      |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v107_quantum_clock_target.py | Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/M_{\text{Pl}}^2$ : $10^{-122}$ cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]                                                                  |
| v108_pascal_ladder.py        | Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- $g$ midpoint identity + even- $g$ straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]                                                                                                                                                                     |
| v109_sheet_pairing.py        | Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$ ; odd $g$ flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ( $256=1+45+210$ , zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms ( $2\Lambda^1 + 2\Lambda^3 + \Lambda^5$ ); tower tops at $\Lambda^{2K}$ , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]                                                                                            |
| v110_calderon_sheet.py       | Calderón-sheet selection theorem: a Calderón involution $\varepsilon$ on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2= \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ( $g=3, 5, 7$ ): sheet-oddness forces $K=(g-1)/2$ , <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ $\varepsilon$ sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees $\leq 2$ ) [E]/[C]                                                                             |
| v111_quadratic_transport.py  | Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$ , the $\Lambda^2$ term of $1+45+210$ ) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length 2 $\Rightarrow$ <i>transport degree exactly 2</i> (degree $\leq 1$ generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]                                                                                                                                   |
| v112_selfcounting_channel.py | Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels = $2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$ ; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]                                                                      |
| v113_quasifree_kernel.py     | One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 =  \mu_4 $ ); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$ , seam level $8=\text{rank } E_8$ ( $c = \text{rank of the kernel}$ ); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]                                                                         |
| v114_torsion_delta.py        | Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}}) \mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ( $\mu_4 = \text{order of the twisted cusp generator}$ ); on the involutive branch ( $T^2 = \mathbf{1}$ ) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C] |
| v115_anchor_residue.py       | The anchor pins the residue: $\mu_4$ -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $ ; off-weights uniquely $(8, 0, 5)/144$ ( $= (\text{rank } E_8, 0, g_{\text{car}})/( \mu_4  N_{\text{fam}})^2$ , total $13 = \Delta_Q$ ); <i>exact</i> residue $A_0^*$ with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ( $\ M_\infty - \mathbf{1}\  \sim 10^{-10}$ ); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]                                                                    |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v116_resonance_uniqueness.py      | Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted $\mu_4$ averages collapse to $B_1=4a_{12}E_{12}+4\bar{a}_{13}E_{31}$ ; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\bar{a}_{13} \Rightarrow M_\infty=\mathbf{1} \Leftrightarrow a_{13}=0$ ; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact $A_0^* - U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\  \sim 8.5 a_{13} $ [E]                                                                                   |
| v117_monodromy_weyl_a3.py         | The monodromy is $W(A_3)$ : the unitarised holonomy of $A_0^*$ is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$ ), $\text{diag } \widetilde{M}_0=(0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta=\frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$ , characters $(3, -1, 0, 1) = S_4=W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the $A_3$ glue factor as its Weyl group; ODE match $3.5 \times 10^{-10}$ [E]                                                                                             |
| v118_hexagon_family_dictionary.py | First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6=\mathbb{Z}_2 \times \mathbb{Z}_3$ ); $\det(\mathbf{1}-t\widetilde{M}_0)=1-t^3 \Rightarrow \det_{\mp}=(\frac{7}{8}, \frac{9}{8})$ ; lepton coefficients = resolvent determinants ( $c_e= \mu_4 /(2\det_-)$ , $c_\mu=N_{\text{fam}}/(2\det_+)$ , $c_\tau= \mu_4 \det_-/N_{\text{fam}}$ , product = $ \mu_4 /\det_+=\frac{32}{9}$ ); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48=\Omega_{\text{adm}}$ ; the address table stays open [E]/[C] |
| v119_review_validation_2.py       | Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0=(\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5)=(e_3, e_2))$ ; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a)=121=11^2=\ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$ , only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}}=2p_1p_2$ , $10b_1=1+p_1p_3$ , $\Delta_Y=e_2^2=p_0^2+2\text{rank}$ ; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]                                                                  |
| v120_address_table.py             | Address table (frozen v18/v20 integers): lepton words $(8, 5, 3)=(\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$ ; addresses = divmod by the hexagon ( $p_2=6$ ): $e \rightarrow (2, 1)$ , $\mu \rightarrow (5, 0)$ , $\tau \rightarrow (3, 0)$ , $\sum r=p_3$ ; sheet parity: $\tau$ sits at the real twisted eigenvalue $-1$ (= the product-rule site); quark sums: up = $p_3$ , down = $p_1+p_3$ , all quarks = $24= W(A_3) $ , all nine = $40=p_1p_3=a^T Ra$ ; top at the vacuum site $(0, 0)$ ; the assignment derivation stays open [E]/[C]                                                                 |
| v121_address_pinning.py           | Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator $R$ + the generation-1 winding, $R$ column 1 = the anchor $a$ ; margins = atoms (rows $(e_1, \text{rank}, p_3)$ , columns $(e_1, \Delta_Q, g_{\text{car}})$ ); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is $R$ (also unique with SNF $(1, 1, 8)$ ; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C]                                        |
| v122_margin_theorem.py            | Margin theorem: the <i>established</i> selectors — the $D_4$ annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$ , $\det K=4$ (the v71 quartet) — pin $R$ <i>uniquely</i> among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]                                                                                                |
| v123_inventory_update.py          | Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 $Q$ realisation — + $v_{\text{geo}}, \pi$ ; a sixth class fails the script [E]                                                                                                                                                                                                     |
| v124_resummed_clock.py            | Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$ , the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$ ; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]                                                                                                                                                          |
| v125_glue_qsystem.py              | Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ( $q(k(1, 1))=k^2 \in \mathbb{Z}$ , v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step, } 4=2 \times 2$ ; R2 sharpens from “find” to “identify” [E]/[C]                                                                                                                                                                        |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v126_clock_wall_bridge.py | Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly spec $A_0^*$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$ ; checkpoint 1 passed <i>classically</i> ( $p_2/N_{\text{fam}}=2= \mathbb{Z}_2 $ = the v104 entropy rate $2H$ ); $\det(\mathbf{1}-A_0^*)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]                                                                                                        |
| v127_ring_resummation.py  | The clock is a log determinant: rate = $-p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$ ; v107's "O(1) resummation" is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]                                                                      |
| v128_graded_hull.py       | Graded hull + zero-mode multiplicity: the 240 $E_8$ roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$ ; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [E]/[C]                                                                                                                                                      |
| v129_entropy_power_law.py | The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); rate = $p_2 \ln(S_{dS}/S)$ ; triple identity $\alpha=n/N_{\text{fam}}$ = MS weight = entropy-deficit share $\Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]                                                                   |
| v130_born_square.py       | Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability = $ A ^2=(S/S_{dS})^6 \Rightarrow h=n_{\text{zero}}/2=3=N_{\text{fam}}$ , $p_2=2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of $S^2$ ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C]                                                                              |
| v131_measure_is_area.py   | The measure is the area: $\ Y_{1m}\ ^2=r^2=A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude $S^3$ , Born $S^6$ — every exponent factor with a derivation origin; the $4\pi$ = the same Gauss–Bonnet primitive as in $c_3$ , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n/\det'_{dS}=1+O(\text{deficit})$ [E]/[C] ( <i>cancellation since derived within the SdS family, v144</i> )                                  |
| v132_detratio_anomaly.py  | Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) = $a_1-3=\frac{7}{3}-3=-\frac{2}{3}=- \mathbb{Z}_2 /N_{\text{fam}}$ exactly ( $a_1=\frac{1}{N_{\text{fam}}}+ \mathbb{Z}_2 $ ; numerical continuation $\sim 10^{-7}$ ) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL)=c^{\zeta(0)}\det'(L)$ ; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) [E]/[C]                                                            |
| v133_zeta_budget.py       | The $\zeta$ budget, both routes exact (Euclidean Nariai = $S^2 \times S^2$ ): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}=-e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2=\frac{161}{45}$ , $\zeta(0)=-\frac{109}{45}$ ( <i>no</i> atom; continuation $\sim 10^{-8}$ ); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C]                             |
| v134_dual_anchor.py       | Dual anchor lemma: $d:=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$ , $d \cdot \mathbf{1}=0$ , $d \cdot a=1$ , $(1, 1, -2)=-2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ( $\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1, 1, -1)$ ); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); $(d, n)$ = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge [E]/[C] |
| v135_det_surface.py       | Determinant surface: $\det M(s, t)=3s(t+1)(t+2)+t^2+5t+8$ — two $s$ -independent iso-volume walls $t=-2 \Rightarrow 2= \mathbb{Z}_2 $ , $t=-1 \Rightarrow 4= \mu_4 $ ( $\det K=4$ is a <i>layer</i> ); winding line $6s+8 \Rightarrow (8, 14, 20)$ , $\det F=32=2^{g_{\text{car}}}$ ; $\det B(s, 0)=2 \det M(s, 0)$ exactly, $B$ -wall only at $t=-2$ ; row-sum axes $C=(7, 11, 13)$ , $U=(3, 3, 3)$ , $V=(1, 2, 3)=\text{Spec}(Q_+)$ ; micro-identities $e_1^2+e_2^2=41=10b_1$ , $p_1^2+p_2^2=52$ = the carrier coset root count (v128) [E]/[C]                 |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v136_dual_normal_selector.py | Dual-normal selector: $(d, n)$ pins $R$ columnwise — $d \cdot c_j = a_j$ , $n \cdot c_j = 8\delta_{1j}$ , kernel = primitive $(6, 8, 7)$ , census: each column unique (1 of 216) $\Rightarrow$ $R4'$ shrinks from $6^9$ to three column problems; the $A_0^*$ zero mode carries $\sigma = (2, -9, 5) = ( \mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ( $\ v\ ^2 = \frac{16}{9}$ , $16 = \dim S^+$ ; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$ ; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element <b>[E]/[C]</b> |
| v137_qplus_cohomology.py     | The $Q_+$ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ( $\omega_k = z^{k-1}dz/(z^4-1)$ , character $i^k$ , residues $\zeta^k/4$ ); degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the $A_3$ exponents ( $h=4= \mu_4 $ ); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = $Q_+$ : one spectrum, four readouts; the QGEO residue = the naturality of the canonical map <b>[E]/[C]</b>                                                                                                                     |
| v138_vw_firewall.py          | Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); <b>edge match</b> : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget ( <b>v133</b> , the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall <b>[E]/[C]</b>                |
| v139_selector_triangle.py    | Selector triangle: $d = \frac{3}{2}a - 2 \cdot 1$ exactly (pure anchor data, no $R$ ) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(1, a, \sigma)$ has $\det = 11 = \ \text{Pl}(K)\ _1$ , and $n$ is the <i>unique</i> covector with atom pairings $(2, 8, 121) = ( \mathbb{Z}_2 , \text{rank } E_8, 11^2)$ ; on the constrained line the pairing is $11(11+t)$ , a square only at the true $n$ ; review lift exact (audit): $n = \text{reverse}(\sigma) +  \mu_4 e_3$ ; $R4'$ residue = three pairings <b>[E]/[C]</b> ( <i>since v142: two — the third pairing is integrality-forced</i> )               |
| v140_canonical_map.py        | The canonical map exists and is rigid: the plain deck $U$ (characters $(0, 1, 3)$ ) cannot match $H^1$ ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential $i^{Q+}$ $((1, 2, 3))$ both carry the $H^1$ character set $\{1, 2, 3\}$ (the <b>v118</b> sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); $i^{Q+}$ pulls the Euler degree back to $Q_+$ , $-U$ to $Q_+ \circ \rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice <b>[E]/[C]</b> ( <i>the choice since derived, v141</i> )                                       |
| v141_deck_selection.py       | Deck Selection Theorem (R5): the established integer deck $G = T_A \Sigma$ ( <b>v97/v98</b> ) pairs the $Q_+ = 1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; $i^{Q+}$ excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ ( <b>v92</b> ); Euler degree = $Q_+ \circ \rho$ <b>[E]/[C]</b>                                                                                                                                                              |
| v142_frame_integrality.py    | Frame Integrality ( $R4'$ ): pairing triples of <i>integer</i> covectors against $(1, a, \sigma)$ form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ , $11 = \ \text{Pl}(K)\ _1$ ); with $(x, y) = ( \mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \pmod{11}$ , primitivity forces even $t$ ; Cramer reductions for the two line pairings; $R4'$ : three pairings $\rightarrow$ two <b>[E]/[O]</b> ( <i>since v145: the values are atom identities — residue = one lift map</i> )                                                                                                       |
| v143_graded_frobenius.py     | Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k = C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$ ; glue average = carrier projector (index $ \mathbb{Z}_4  = 4$ ); each sector one $W(D_5) \times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ ( <b>v125</b> ); residual = the one conformal-net statement <b>[E]/[C]</b>                                                                                                                                                                                                      |
| v144_detratio_family.py      | Det-ratio cancellation within the SdS family (R1, the <b>v131</b> step): $e_2$ -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0) = -\frac{2}{3}$ , <b>v132</b> ) — no first-order term in the split; deficit coefficient $\frac{32}{27} =  \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply <b>[E]/[C]</b> ( <i>the bend since shown <math>\det'</math>-clean, v147</i> )                                                                                        |
| v145_pairing_atoms.py        | $R4'$ values are <i>atom identities</i> : with the lift $n = w_0(\sigma) +  \mu_4 e_3$ ( $w_0$ = the $A_3$ exponent duality), $n \cdot 1 =  \mu_4  -  \mathbb{Z}_2  =  \mathbb{Z}_2 $ , $n \cdot a =  \mu_4 a_3 = 8$ via the new closure $ \mu_4  + g_{\text{car}} = N_{\text{fam}}^2$ ( $4 + 5 = 9$ , $\sigma \perp w_0(a)$ ), $n \cdot \sigma = \ \sigma\ ^2 - N_{\text{fam}}^2 +  \mu_4 g_{\text{car}} = 121$ ; residue = derive the <i>one</i> lift map <b>[E]/[C]</b>                                                                                                                                                   |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v146_moebius_d4.py             | Möbius $D_4$ realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — $\iota^*$ swaps $\chi_1 \leftrightarrow \chi_3$ , fixes $\chi_2$ with $+1$ , $\det = -1$ ( $= T_A$ in the cusp basis); $\delta\iota = \Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification [E]/[C]                                                                                                                                                                       |
| v147_clock_gaussian.py         | The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born <sup>2</sup> $\Gamma$ -ratio $(1-\alpha)^{p_2}$ , rate $= -p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ <i>det'-clean</i> ; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference [E]/[C]                                                                                                                                                                                       |
| v148_fock_census.py            | NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ( $2^8=256$ ) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection ( $128 = \text{one } SO(16) \text{ chirality}$ ), $248 = 120_{\text{NS}} + 128_{\text{R}}$ ; R2 is irreducibly a twisted-sector net statement [E]/[C]                                                                                        |
| v149_cusp_normal.py            | Both dual normals are <i>anchor data</i> (R4'): in the cusp frame $n$ pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per $Q_+$ line ( $d = \frac{3}{2}a - 2 \cdot 1$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the $w_0$ -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$ , sum $14 = \dim G_2$ ; residue = <i>one</i> assignment [E]/[C]                                                                                                                                                                                                                                          |
| v150_replica_eh_model.py       | The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and $t$ -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$ ); the gap makes the $\zeta$ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$ ; linearisation $= (\ln m / 12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] ( <i>Calderón transfer since answered — the DtN kernel is conically clean</i> , v151) |
| v151_bfk_split.py              | BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ( $C_N = C_D$ , derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}} = C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]                           |
| v152_norm_is_anchor.py         | R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio $m/\mu$ is physical; $k=c_3/2 \Leftrightarrow m/\mu = e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, $\pi$ }; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]                                                                                                                                                                                                          |
| v153_no_unit_theorem.py        | No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$ , a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu = e^{3/4}$ (a ratio); $v_{\text{geo}}$ is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, $\pi$ } [E]/[O]                                                                                                                                                                              |
| v154_simple_current_theorem.py | Simple-Current Extension Theorem (the central $G_{\text{net}}$ theorem, assembled): $A = (D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L = \langle (1, 1) \rangle$ has index $ L =4$ , $c=5+3=8$ , $\mu(B) = \mu(A)/ L ^2 = 16/16=1 \Rightarrow$ holomorphic $\Rightarrow B = (E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]                                                                                                                                                                                     |
| v155_quasifree_boundary.py     | Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $PTP$ (a contraction, hence a valid covariance), so the $G_{\text{net}}$ analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]                                                                                                                                                                                           |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v156_seam_net_construction.py    | Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$ ; the $E_8$ currents are $248=120_{\text{NS}}+128_{\text{R}}$ ; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v157_rigid_fixed_point.py        | Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\hbar=0 \Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $ \mathbb{Z}_2 $ that halves $8\pi$ in $c_3$ IS the Ramond projection giving $\mu=1$ ; (iv) holomorphic $c=8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v158_fixed_point_stable.py       | The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$ , bilinear current $h=1$ , quartic $h=2$ ; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v159_pyrate_gauge_crosscheck.py  | PyR@TE cross-check of the $F_{\text{transfer}}$ gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1=\frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1=g_{\text{car}}2^{g_{\text{car}}-2}+1=41$ , the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$ , and the external RGE generator PyR@TE 3 (arXiv:2007.12700) which writes $\beta_{g_1}=(\frac{41}{10})g_1^3$ verbatim; the $M=41$ split $10b_1=40_{(\text{ferm})}+1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in <i>verification/pyrate</i> . The transfer functor’s gauge inputs are not free knobs [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v160_seam_gaussianity_from_pf.py | Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> $\kappa_{2n}$ is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap}>0$ — the value $(2/3)^6$ only sets the rate. <b>The load [O]:</b> v56’s gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ , while the seam correlation cone has $\binom{16}{4}=1820$ , $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement ‘the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum’ [E]/[O]                                                                                                                                                                        |
| v161_one_particle_reduction.py   | The cone reduces to one particle (discharging v160’s scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$ , the second quantisation of a one-particle contraction $t$ ; $\Gamma(t)$ acts on the degree- $m$ correlator sector as the exterior power $\Lambda^m(t)$ , so (1) $\text{spec}(\Gamma(t) _{\text{deg } m}) =$ products of $m$ one-particle eigenvalues [E]; (2) <b>the cone gap equals the one-particle gap</b> — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) [E]; (4) quasi-freeness <i>forced</i> not assumed — the $120=\dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. <b>Residual [O]:</b> v160’s three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 $K_\Sigma$ , gap $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O] |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v162_seam_transport_identification.py | The final identification, forced as far as it goes — the <b>v161</b> one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, <b>v115</b> ), transfer eigenvalue $(1-\alpha)^{p_2}$ , $p_2=6= R_+(A_3) $ , all carrier data <b>[E]</b> ; (2) $\mu_4$ -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} =  \mu_4  \text{diag}(X)$ , commutant of $U$ = the diagonal algebra — so the gapped $\mu_4$ -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ <b>[E]</b> ; (3) the cusp grading <i>is</i> the $A_3$ grading on the 16 Majoranas ( <b>v137</b> ) <b>[E]</b> ; (4) the fixed space = the rank-8 Calderón polarization ( <b>v113/v156</b> ) <b>[E]</b> . <b>Closure [C]/[O]</b> : with <b>v160+v161</b> , (A) closes given only $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, <b>HOR.CLOCK</b> ) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) <b>[E]/[O]</b>                                                                                                                                                                                                         |
| v163_rg_stability_flavor.py           | $F_{\text{transfer}}$ scale-tagging: the lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ , $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are <i>RG-stable</i> — PMNS running is $y_\tau^2$ -suppressed ( $\sim 1.8 \times 10^{-5} \ll \text{precision}$ ), so seam value = measured value; the quark mass-ratio readouts ( $m_t/m_b \sim 41$ ) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries $y_{\text{max}}^2$ (PyR@TE SM RGEs) <b>[E]/[C]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v164_qcd_scale.py                     | the QCD confinement scale is parameter-free from the carrier $b_3 = -7$ : running $\alpha_s(M_Z)$ down (two-loop, $n_f$ thresholds) gives $\alpha_s(m_b) \sim 0.22$ , $\alpha_s(m_c) \sim 0.38$ and confinement ( $\alpha_s \sim 1$ ) at $\sim 0.5$ GeV — so the QCD-scale <i>numerator</i> of $m_p/m_e$ is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ( $m_p = O(1)\Lambda$ is lattice) <b>[E]/[O]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v165_premise_a_closure.py             | Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics <b>[C]</b> — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the $\mu_4$ extension is a Longo Q-system of index 4 ( <b>v125</b> ), $\mu(B) = 16/16 = 1$ , $248 = 120 + 128$ , character $E_4/\eta^8 = j^{1/3}$ ( <b>v156</b> ); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $=  k $ ( <b>v156/v157</b> ); (2) R1 (seam clock) reduces to <b>GATE.QGEO [C]</b> — $\mu_4$ -equivariance forces diagonal-in-cusp, the flat $\mu_4$ -equivariant generator is <i>unique</i> $= A_0^*$ ( <b>v115/v116</b> ) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary’ = the $\mu_4$ -punctured sphere $\mathbb{P}^1 \setminus \mu_4$ = <b>GATE.QGEO</b> (cohomology established, <b>v137</b> ); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> by the No-Unit Theorem ( <b>v153</b> ). <b>Final accounting</b> : (A) = A2 assembly <b>[C]</b> + R1= <b>GATE.QGEO</b> + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof <b>[E]/[O]</b> |
| v166_higgs_free_seam.py               | the Higgs quartic from the <i>free seam</i> (premise A / <b>v158</b> ): a Gaussian seam UV forces $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured $(m_H, m_t)$ gives $\lambda(\bar{M}_{\text{Pl}}) \sim 0.002$ , $\beta_\lambda(\bar{M}_{\text{Pl}}) \sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim 0.022$ to $\sim 0.002$ ). The double condition predicts $m_H \sim 129\text{--}134$ GeV; at the scalaron scale it gives $\sim 107$ GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) <b>[E]/[C]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v167_inventory_<br>post_closure.py       | The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123's FIVE classes to: <b>Tier 0</b> — the irreducible core $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> (No-Unit, v153), nothing to close; <b>Tier 1</b> — ONE hinge (seam = flavour = horizon): $A_2$ net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P} \setminus \mu_4$ , cohomology $b_1=N_{\text{fam}}=3$ ); <b>Tier 2</b> — folded ( $\rightarrow v_{\text{geo}}, \rightarrow G_{\text{net}}$ ); <b>Tier 3</b> — $F_{\text{transfer}}$ , one functor with four typed conditional interfaces ( $\eta_B$ carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C] |
| v168_qgeo_<br>rigidity.py                | The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z \rightarrow iz, z \rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful $D_4$ are the $\mu_4$ square up to Möbius; $H^1(\mathbb{P} \setminus \mu_4)$ has rank $4-1=3=N_{\text{fam}}$ ; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ( $k=1, 2, 3$ ) are $\mu_4$ -eigenforms with $z \rightarrow iz$ eigenvalue $i^k$ = the three nontrivial characters $\chi_1, \chi_2, \chi_3$ , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$ . QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P} \setminus \mu_4$ stays the realisation premise) [E]/[C]                                                                                                                                                                                     |
| v169_etaB_<br>boltzmann_<br>interface.py | The $\eta_B$ leptogenesis transfer <i>operationalised</i> (Tier-3 $F_{\text{transfer}}$ , not a new class): type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$ . Davidson–Ibarra $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3 / v^2 \sim 8.5 \times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1) \sim 0.03\text{--}0.19$ . <b>Kill test:</b> over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, $\eta_B$ spans $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$ which <i>brackets</i> the observed $6.1 \times 10^{-10}$ (canonical point gives $6.0 \times 10^{-10}$ , not tuned) $\Rightarrow$ the route is viable; $M_1$ /washout are scenario inputs, so $\eta_B$ stays [C]. If excluded, the route falls, not the theory [C]                                                                                            |
| v170_diamond_slice_<br>compression.py    | E8 Slice Compression Lemma: the seven $E_8$ audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ( $p_n=2+2^n$ ) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$ . The $K_4$ edge products (12, 18, 30, 24, 40, 60) are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$ ; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$ ; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2 \Delta$ . The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]                                                                                                                                                                                                                                                                                                                                                 |
| v171_os_moment_<br>cluster.py            | Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$ , $\xi=0.41105$ , Källen–Lehmann masses $6 \log \frac{3}{2}, 6 \log 3$ ; gap safety $\Delta_{\text{eff}}=6 \log \frac{3}{2} - \frac{31}{4\pi^2} > 0$ with $31=1+h^\vee(E_8)$ , $c((E_8)_1)=248/31=8$ [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v172_trace_anomaly_<br>seed.py           | Trace-anomaly origin of the seed prefactor $\frac{4}{3}$ : over the seam $S^2$ ( $\chi=2$ ) the $(E_8)_1$ ( $c=8$ ) Weyl anomaly is $c\chi/6=\frac{8}{3}$ , halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$ , the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$ . Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v173_pfaffian_cp_<br>car.py              | Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$ , $\gamma_5$ -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ( $\arg \in \{0, \pi\}$ ), and reflection positivity ( $Z>0$ ) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v174_seam_fock_readings.py      | Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ( $2^{15}/16=2^{11}$ ), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$ , with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$ ) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig $g$ -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v175_net_existence_full_cone.py | The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) <b>full-cone RP for all <math>m</math></b> — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction $t$ acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ( $\dim 2^{16}=65536$ ); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$ , verified on all $2^{16}$ products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 $K_\Sigma$ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ , so full-cone RP reduces <i>for all <math>m</math></i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) <b>A2 net existence</b> assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$ , $E_8$ Cartan even unimodular ( $\det 1$ ); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O] |
| v176_seam_collar_realisation.py | The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful $D_4$ marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with $H^1$ grading $(1, 2, 3)$ , the Calderón contraction is the rank-8 $K_\Sigma$ polarisation and the index-4 extension is $(E_8)_1$ . The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius $D_4$ order 8, $b_1=N_{\text{fam}}=3$ ), Calderón DtN symbol $ k $ (v156), $H^1$ character = generation grading $(1, 2, 3)=A_3$ exponents= $\text{Spec}(Q_+)$ (v137/v168), $\mu_4$ -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all $m$ (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar’s boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$ ), left honestly [O][E][O]                                                                                                                                                 |
| v177_seam_marking_kernel.py     | The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$ , a non-fixed orbit scales to $\mu_4$ , the reflection $z \mapsto 1/z$ gives $\sigma \rho \sigma = \rho^{-1}$ ( $D_4$ , order 8) — a genus-0 four-marked faithful- $D_4$ boundary <i>is</i> $(\mathbb{P}, \mu_4)$ ; [E] COHOM: $\rho^* \omega_k = i^k \omega_k$ so the $H^1$ grading is $(1, 2, 3)=A_3$ exponents= $\text{Spec}(Q_+)$ ; [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$ , $\omega_2$ fixed); [E] NET (v154/v175): index-4 $\mu_4$ extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$ . The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked $D_4$ boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the $\mu_4$ -equivariant free gapped contraction as an <i>operator</i> ); the whole structural residual is now exactly these two doors [E][O]                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v178_marks_kernel_attempt.py  | An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock $\rho$ ( $\sim z \mapsto iz$ , fixed points $\{0, \infty\}$ ) and reflection $\tau$ forces the marks to be a generic $\rho$ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$ , $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful $D_4$ order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free $H^1$ the $\mu_4$ characters $(1, 2, 3)$ are distinct, so by Schur a $\mu_4$ -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $ , Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O] |
| v179_conformal_realisation.py | The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 is P1 — $c_3=1/( \mathbb{Z}_2  2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$ , the same $\chi=2$ that gives the ‘8’ (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$ , order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$ , a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to ‘the carrier clock acts conformally on the genus-0 double’. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$ , the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]                                                                                                                                             |
| v180_clock_is_mobius.py       | Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: ‘the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric’. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$ , so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of $S^2$ is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically $z \mapsto iz$ ; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$ ; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$ , multiplier $i$ , free orbit $\mu_4$ ). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O]                                                                              |

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| Script                                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v181_clock_<br>is_conformal_<br>symmetry.py | <p>The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form <math>z \mapsto iz</math>. The clock is the Coxeter element of <math>W(A_3)=S_4</math> (v117), order <math>h(A_3)=4= \mu_4 </math>, the <math>\mu_4</math> deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i>, not a missing theorem: ‘the seam’s conformal structure <i>is</i> the <math>\mu_4</math>-deck structure of the carrier clock’ (the carrier <math>\mu_4</math> glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam’s conformal geometry, not reducible further without relabeling. The chain <math>\text{REALIZE}(\text{v175}) \rightarrow \text{theorem}(\text{v176}) \rightarrow \text{MARKS} + \text{KERNEL}(\text{v177}) \rightarrow \text{finite cores}(\text{v178}) \rightarrow \text{CONF.01}(\text{v179}) \rightarrow \text{ISO.01}(\text{v180}) \rightarrow \text{SYM.01}(\text{v181})</math> has reached bedrock; we stop here honestly [E][O]</p> |
| v182_reviewer_<br>residual_map.py           | <p>The four external-review concerns reduce to the two already-named residuals + one data-only item. <math>A</math> (mapping <math>2D \rightarrow 4D</math>) [E]/[C]: the Plücker readout <math>11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5</math> (QBL count, v174) and the holographic reduction give <math>A = \text{QGEO.SYM.01} + F_{\text{transfer}}</math>. <math>B</math> (UV gravity) [E]: ‘gravity = geometric boundary readout’ <i>is</i> QGEO.SYM.01. <math>C</math> (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor <math>a=(1, 1, 2)</math> in <math>\geq 6</math> independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). <math>D</math> (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by <math>\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})</math> on the deck, <math>t \sim 2.84</math> only locates the point, and the missing RG generator <i>is</i> <math>F_{\text{transfer}}</math>, so <math>D = \text{QGEO.SYM.01} + F_{\text{transfer}}</math>. NET: three reduce onto <math>\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}</math>, one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C]</p>                                                                                                                                                                                  |
| v183_koide_f_<br>corner_transfer.py         | <p>The Koide source <math>\rightarrow</math> pole factor <math>\frac{53}{54}</math> has an <i>operator</i> origin (external-review proposal, validated): <math>u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}</math>.<br/> (1) <math>\mathbf{1}^T R a = 27</math> is the <math>E_6 \times A_2</math> cubic family block<br/> <math>(248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}})</math>; (2) <math>F=R+Q</math> is the <i>missing</i> Sheet-Diamond corner <math>M(1, 1)</math> with <math>\det B_F = 52 = \dim F_4</math> (v80) and anchor response <math>a^T(R+Q)\mathbf{1} = 53 = 52+1</math>; (3) <math>\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}</math> exactly (<math>54=2 \cdot 27</math>);<br/> (4) negative control: only <math>F</math> gives 53 (<math>R/Q/K/L \rightarrow 32/21/35/56</math>). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source <math>\rightarrow</math> pole reading the <math>F</math> corner stays an <math>F_{\text{transfer}}</math> conjecture [E][C]</p>                                                                                                                                                                                                                                                                                                                                                    |
| v184_etaB_anchored_<br>boltzmann.py         | <p>Honest test of the external-review <math>\eta_B</math> proposal (<math>M_1=M_R\phi_0^4</math>, <math>\tilde{m}_1=m_3/A_\Lambda</math>) — firewall verdict: <i>sharper scenario</i>, not a closure. The washout anchors plausibly (<math>\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5 \text{ meV}</math>, <math>A_\Lambda=10= E(K_5) </math>, in the strong-washout window) [C]; but <math>M_1</math> does <i>not</i> — <math>M_R</math> is the seesaw scale <math>\sim v^2/m_3 \approx 6 \times 10^{14} \text{ GeV}</math> and the nearest clean TFPT powers of <math>\tilde{M}_{\text{P1}}</math> both miss it (<math>c_3</math>-power off <math>\sim 75\%</math>, <math>\phi_0</math>-power off <math>\sim 40\%</math>), so <math>M_1=M_R\phi_0^4</math> merely relocates the free input <math>M_1 \rightarrow M_R</math> (the reviewer’s <math>6.0 \times 10^{-10}</math> uses <math>M_R=1.3 \times 10^{15}</math>; the seesaw value gives <math>3.4 \times 10^{-10}</math>, the tell that <math>M_1</math> is unpinned). <math>\eta_B</math> stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — <math>\eta_B</math> stays [C], a sharper scenario [C]</p>                                                                                                                                                                                                                                                                               |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v185_axion_relic_solver.py     | Axion relic, honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this $f_a$ (prefactor $\sim 0.026/\theta^2$ , $\theta \sim O(1)$ gives $\sim 21\%$ of $\Omega_{\text{DM}}$ , need $\theta_{\text{eff}}^2 \sim 4.7$ ), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128 = (16, 4) + (16, 4)$ ) [C][O]                                             |
| v187_ftransfer_laws.py         | The $F_{\text{transfer}}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, $\eta_B$ , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3 = -7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]                                                                                                                                                                                                    |
| v188_frontier_wording_guard.py | The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” ( $M_R$ is the seesaw scale, not a compiler power), “relic scale $f_a, m_a$ open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q = 0.664 \dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]                                                                                                                 |
| v189_riemann_roch_carrier.py   | The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$ . [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8$ , v1), $N_{\text{fam}} = 3$ . [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$ , but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C] |
| v190_nariai_entropy_bound.py   | The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2 + 1)/\Phi_3(x)$ ( $x = r_b/r_c$ ; $\Phi_3 = x^2 + x + 1$ the $N_{\text{fam}} = 3$ cyclotomic, already in <code>tfpt_horizon_readouts</code> ) satisfies $S_{\text{tot}} \geq \frac{2}{3} S_{\text{dS}}$ with equality exactly at the Nariai merge $x = 1$ , via the perfect square $3(x^2 + 1) - 2\Phi_3(x) = (x - 1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x > 0$ is $2/3 =  Z_2 /N_{\text{fam}}$ ) [E]                                                                                                                                                                                                                                                                                                                                                                             |
| v191_universal_branch_line.py  | The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ( $m = 0 \rightarrow 7/2$ ). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 =  Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]                                                                               |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v192_qgeo_energy_clock.py         | The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ( $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda =  k $ . But ‘ $\rho$ preserves $\Lambda$ ’ is plausibly equivalent to ‘ $\rho$ is the conformal deck’, so it is a sharper restatement, not a theorem, unless $\rho$ is independently defined and the invariance independently proven (open). Bedrock stays [O]                                                   |
| v193_qgeo_energy_commutator.py    | QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, $\rho$ the carrier clock, $\Lambda_\Sigma$ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] $\rho$ is carrier-defined (Coxeter of $W(A_3)=S_4$ , order $4= \mu_4 $ , v117), not imported from the seam; (2) [O] the non-circularity hinge — $\Lambda_\Sigma$ must be the <i>raw</i> RP-seam DtN, not the $\mu_4$ normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the (1, 2, 3) grading on $H^1$ (QGEO.COHO.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by $5/3$ ) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O] |
| v194_raw_seam_dtn_rp.py           | QGEO.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i> , not a closure. (1) [E] finite core: $\rho: z \mapsto iz$ on $H^1=\langle w_1, w_2, w_3 \rangle$ , $w_k=z^{k-1}dz/(z^4-1)$ , acts as $\rho^* w_k=i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma]=0$ on $H^1$ (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so $\Lambda_\Sigma$ is RP-definable without assuming $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] residual relocates: full- $L^2$ commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGEO.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]                      |
| v195_marks_lefschetz_character.py | QGEO.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing $D_4$ marks, they are <i>forced</i> to be a free $\mu_4$ orbit. [E]: $\rho: z \mapsto iz$ (order 4) fixes only $\{0, \infty\}$ ; $H^1=\chi_1+\chi_2+\chi_3$ (the $A_3$ exponents (1, 2, 3), v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; rank $H_1=n-1=3 \Rightarrow n=4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks $=\mu_4$ (cross-ratio 2); $\text{Tr}(\rho H^1)=i+i^2+i^3=-1$ , $L(\rho)=1+1=2=\#\text{fixed}$ . [O] residual relocates from “the seam is $\mathbb{P} \setminus \mu_4$ (geometric)” to “ $H^1$ carries the $A_3$ -exponent rep” — less circular [E][O]                                                                                                                                                                        |
| v196_seam_energy_functional.py    | QGEO.VARI.01 — the $E_{\text{fail}}$ variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: $E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2$ , with $E_{\text{fail}}=0 \Leftrightarrow$ the QGEO.SYM.01 conditions. [E] on the finite $H^1$ block ( $\rho=\text{diag}(i, -1, -i)$ , $\Lambda$ diagonal, $\Theta=\text{conj}$ ) all three vanish $\Rightarrow E_{\text{fail}}=0$ ; [E] a $\mu_4$ -breaking $\Lambda$ or non-order-4 $\rho$ gives $E_{\text{fail}}>0$ ( $\mu_4$ a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]                                                                                  |
| v197_rr_carrier_clifford_d5.py    | ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from $g_{\text{car}}$ to $D_5$ itself (review point 7): the 5-dim carrier mode space $C_{\text{car}}:=H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10)=D_5$ half-spinor by its even Clifford algebra. [E] $\dim \Lambda^{\text{even}}(\mathbb{C}^5)=\binom{5}{0}+\binom{5}{2}+\binom{5}{4}=1+10+5=16=2^{g_{\text{car}}-1}=\dim S^+$ ; the $\mathfrak{so}(10)=D_5$ half-spinor $2^4=16=S^+=\Lambda^{\text{even}}(C_{\text{car}})$ ; closes the chain $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ from one four-point divisor (cycle side rank $H_1=3=N_{\text{fam}}(A_3)$ ; function side $h^0=5=g_{\text{car}}$ ; Clifford spinor $16=D_5$ half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined $g_{\text{car}}=5$ [E][C]       |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v198_modular_commutator_reduction.py | QGE0.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual ( $[\rho, \Lambda_\Sigma]=0$ on full $L^2$ , framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $ k =\sqrt{-d^2}$ is $\text{diag}( n )$ and the clock $\rho:z\mapsto iz$ is $\text{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho,  k ]=0$ exactly on <i>all</i> of $L^2$ ; [E] Tomita–Takesaki: $[\rho, \Lambda_\Sigma]=0$ follows from $\omega\circ\rho=\omega$ (the modular $\Delta$ is intrinsic to $(M, \Omega)$ ; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite $H^1$ block is already $\mu_4$ -invariant (v177). [O] residual: the full raw seam state is $\mu_4$ -invariant ( $\omega\circ\rho=\omega$ ) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational QGE0.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O] |
| v199_seam_state_invariance.py        | QGE0.STATE.01 (work package A) — attacking $\omega\circ\rho=\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ( $\rho$ = Coxeter of $W(A_3)=S_4$ , $\rho^4=1$ , an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega\circ\rho=\omega \Leftrightarrow [\rho, H_1]=0$ ( $C_\Sigma=\frac{1}{2}(1+\text{sgn } H_1)$ ); [E] character criterion: $[\rho, H_1]=0 \Leftrightarrow H_1$ block-diagonal in the four $\mu_4$ -character classes $\{n=r \bmod 4\}$ ; principal symbol $ k =\text{diag}( n )$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on $H^1$ (v177); not a closure [E][O]                                                                                                                                                                                                                                                                                                                                                   |
| v200_seam_variational_scan.py        | QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}}=0$ at the $\mu_4$ config on the fixed block; here the clock angle is free and $E_{\text{fail}}$ is minimised over it. [C] scanning $E_{\text{fail}}(\theta)=\ \rho(\theta)^4-1\ ^2+\text{faithfulness}$ over $\theta\in(0, \pi)$ gives the unique minimum at $\theta=\pi/2$ ( $E_{\text{fail}}=0$ : order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the $\mu_4$ normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite $H^1$ grading, not the full operator-valued minimisation (the v199 residual) [C][O]                                                                                                                                                                                                                                                                                                                                                                                                              |
| v201_seam_subprincipal_marks.py      | QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda= k +M_f$ , $M_f$ = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n M_f n'\rangle=f_{n-n'}$ . [E] $M_f$ is $\mu_4$ -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ( $ k $ already commutes, v198). [E] a $\mu_4$ -mark sum $f(\theta)=\sum_{j=0}^3 g(\theta-2\pi j/4)$ has $f_m=4g_m[m\equiv 0 \bmod 4]$ for <i>any</i> profile $g$ — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks= $\mu_4$ orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it. [O] new residual: $\omega\circ\rho=\omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the $\mu_4$ marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O]                   |

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| Script                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v202_rare_kaon.py       | <p>The rare kaon sector <math>K \rightarrow \pi \nu \bar{\nu}</math> from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix (<math>s_{12}=\lambda_C</math>, <math>s_{23}=\varphi_0/(1+\lambda_C)= V_{cb} </math>, <math>s_{13}=\lambda_C^3/3</math>, <math>\delta=\pi/3+3\lambda_C^2=1.19823</math> rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives <math>\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu})=9.45 \times 10^{-11}</math> and <math>\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu})=3.33 \times 10^{-11}</math>. [E] CKM point + exact apex <math>(\bar{\rho}, \bar{\eta})=(0.1374, 0.3509)</math>, <math>\lambda_t=V_{ts}^* V_{td}</math>; [C] the branching ratios (<i>not</i> parameter-free — <math>X_t, P_c, \kappa_{\pm}</math> are external EW/QCD input); [E] <math>\text{BR}(K^+) \text{BR}(K^+)=9.45 \times 10^{-11}</math> lands ON the NA62 2016-2024 combination <math>(9.6^{+1.9}_{-1.8}) \times 10^{-11}</math> (La Thuile 2026), <math>+0.08\sigma</math> (the 2016-2022 obs <math>(13.0^{+3.3}_{-3.0}) \times 10^{-11}</math> was <math>\sim 1.2\sigma</math> above) — a strong consistency hit, but an <math>F_{\text{transfer}}</math> bridge, not a unique TFPT-vs-SM discriminator (SM <math>(8.6 \pm 0.4) \times 10^{-11}</math> also inside the NA62 error); [E] Grossman–Nir ratio <math>0.352 \ll 4.3</math>; [X] dated kill test: a stable NA62 <math>\text{BR}(K^+)</math> outside <math>[7, 12] \times 10^{-11}</math>, or a KOTO-II <math>\text{BR}(K_L)</math> off the GN point, breaks the flavor bridge (core untouched) [C][X]</p> |
| v203_eht_achromatic.py  | <p>The EHT achromatic polarization-rotation intercept <math>\beta_{\text{BH}}</math> around a horizon-scale black hole — the surviving core of the old UFE/BH notes (<i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: <math>\beta_{\text{BH}}(r)=Q_e^{\text{eff}} Q_m^{\text{eff}}/(256\pi^4 r^2)=16c_3^4(Q_e Q_m/r^2)</math>, <math>16c_3^4=1/(256\pi^4)</math> exactly. [E] <math>\alpha</math>-kernel cross-link: <math>16c_3^4=\delta_{\text{top}}/3</math>, <math>\delta_{\text{top}}=48c_3^4=3/(256\pi^4)</math> — the <i>same</i> top-form coefficient that fixes the <math>\alpha</math>-kernel precision-zone correction (one row from <math>\beta_{\text{rad}}=\varphi_0/4\pi</math>). [C] TFPT fixes the <math>1/r^2</math> profile, achromaticity, and sign-flip under <math>E \cdot B</math> reversal; the amplitude <math>Q_e Q_m</math> is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial <math>1/r^2</math>, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v204_muon_seam_g2.py    | <p>The muon anomalous magnetic moment <math>a_\mu=(g_\mu-2)/2</math> as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon-<math>g-2</math> note). The carrier fixes the second-order defect <math>\delta_2=B\gamma \delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2</math> (<math>\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4</math>, v2/v23; <math>B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}</math>, v51); the seam-loop projection <math>a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9) \approx 2.879 \times 10^{-9}</math>. [E] the value is an exact compiler number (<math>\delta_2=4! \text{Tr}_{S^+}(X^2)c_3^8</math>, <math>\text{Tr}=120=5!</math>); [C] the identification of <math>\delta_2/(2\pi)</math> as a magnetic vertex (<math>1/2\pi=4c_3</math>) is a physical bridge; [E] <math>0.81\sigma</math> vs dispersive <math>\Delta a_\mu=(2.49 \pm 0.48) \times 10^{-9}</math>; [C] HONEST: lattice/CMD-3 HVP shrinks it (<math>\sim 1.5 \times 10^{-9}</math>), the fixed value then <math>\sim 1.5\sigma</math> high; [X] dated kill test: a converged <math>\Delta a_\mu</math> outside <math>2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}</math> excludes the mechanism [E][C][X]</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v205_xi_threequarter.py | <p><math>\xi=c_3/\varphi_{\text{tree}}=3/4</math>: an <i>independent</i> route to the gravitational <math>3/4</math> (archive integration of the old Eliminating-<math>K</math> note). The torsion-compression ansatz <math>\kappa^2=\xi\varphi_0/c_3^2</math> with the Einstein limit <math>\kappa^2=8\pi G</math> fixes <math>\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0</math> (since <math>8\pi c_3^2=c_3</math>), and at tree level <math>\varphi_{\text{tree}}=1/(6\pi)</math> this is exactly <math>3/4</math>. [E] tree identity <math>\xi=c_3/\varphi_{\text{tree}}=3/4</math>; [E] Einstein-limit reduction <math>8\pi c_3^2=c_3 \Rightarrow G</math> is an output; [E] over-determination <math>3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)</math> (v152 <math>k=c_3/2</math>) — the torsion-compression <math>\xi</math> and the gapped EH replica are two independent appearances of the same gravitational <math>3/4</math>; [E] with the retained seed <math>\xi=c_3/\varphi_0=0.748303</math> (<math>-0.226\%</math>); [C] the <math>\kappa^2=\xi\varphi_0/c_3^2</math> ansatz is the bridge, the modern derivation is v152/v68, <math>1/G</math> stays the one anchor (v60/v153) [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v206_h0_lambda_branch.py  | The cosmological-constant Letter branch: the seam number $\delta_\Sigma = m_1 e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1 = \Omega_{\text{adm}} = 48$ , the Planck-convention split, and the quantitative $H_0 = 66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative $H_0$ -vs- $\Lambda$ row). [E] $\delta_\Sigma^{\text{lead}} = 48 e^{-2\pi \cdot 137.036/3} \approx 1.085 \times 10^{-123}$ ; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4 = (8\pi)^2 = 631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda = M_{\text{Pl}}^4 \delta_\Sigma \approx 2.4 \times 10^{-47} \text{ GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10} \delta_\Sigma  \approx 122.96$ , consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}} = 66.51$ , $H_0^{\text{disc}} = 67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; $H_0$ uses imported $\Omega_b, \omega_{\text{DM}}$ , a [C] readout [E][C] |
| v207_asymptotic_safety.py | Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close $G_{\text{metric}}$ . [E] the axion coupling is fixed and small $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.025330$ exactly ( $g_{a\gamma\gamma} = -4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g = 2 + \eta_R$ is $+2 > 0$ as $g \rightarrow 0$ and $< 0$ at large $g$ (torsion $K^2$ damping) $\Rightarrow$ a root $g_* > 0$ ; [E] $g_* = 2/a_1 > 0$ finite, Reuter-type $O(1)$ , no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]                                                                                                                             |
| v208_bh_thermodynamics.py | Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H$ : $\Delta^{it} = e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta = 2\pi$ and $T_H = \kappa/(2\pi)$ ; the $2\pi$ is the seam unit $1/(4c_3)$ ( $1/2\pi = 4c_3$ , v58), reproducing $T_H = c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R) = R + R^2/(6M_s^2)$ ( $M_s = c_3^{7/2} \bar{M}_{\text{Pl}}$ , v28), $S_W = (f_R A)/(4G) = (A/4G)(1 + R_h/3M_s^2)$ — leading $A/4G$ is the $c_3$ area law ( $1/4 = 1/ \mu_4 $ ), the correction an exact scalaron consequence; [E] SdS $T_H = (1/4\pi r_h)(1 - \Lambda r_h^2)$ , $1/4\pi = 2c_3$ ; [E] area law $S_{BH} = M^2/(2c_3) = 4\pi M^2 = A/4G$ [E][C]                                                                                                                                                        |
| v209_bh_regular_core.py   | The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2 = (2/3)^6 = 64/729$ with gap $\Delta = 6 \log(3/2) > 0$ (v56) — the core is a fixed point, $\rho \rightarrow \infty$ replaced by $\varphi \rightarrow \varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3 - 3t + 2 = (t-1)^2(t+2)$ , roots $(1, 1, -2)$ = the traceless projection of $a = (1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2 = (2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH} = A/4G$ , $1/4 = 1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v210_mark_local_dtn.py        | A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega \circ \rho = \omega$ , not just $[\rho, \Lambda] = 0$ ); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four $\mu_4$ marks $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN $\Lambda =  D_\theta  + M_f$ . [E] every off-(mod 4) Fourier coefficient of the real $f$ vanishes to $< 10^{-16}$ (exact 4-fold cancellation), although $g$ has all modes. [C] $\ [\rho, \Lambda]\  < 10^{-15}$ . [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ satisfies $\rho C \rho^\dagger = C$ to $< 10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$ ; convergence stable for $N \in \{16, 32, 64\}$ . [O] certifies ‘mark-local DtN $\Rightarrow \omega \circ \rho = \omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O] |
| v211_axion_spine_angle.py     | Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ ( $108^\circ$ ): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$ ). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$ , no fit); [E] the v185 finite-T solver ( $\theta=1 \rightarrow 0.03$ ) reaches $\Omega_{\text{DM}}$ at $\theta \approx 106^\circ$ , so $108^\circ$ is on target (harmonic $0.03\theta^2 = 0.107$ ); [C] <i>robust</i> : $108^\circ$ is $62^\circ$ below the $170^\circ$ hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i = \pi(1 - \varphi_{\text{seam}}) \approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes the branch [E][C][O][X]                                                                                                                                                                                                                                                                                                                                                                  |
| v212_leptogenesis_decouple.py | Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda = 10 =  E(K_5) $ — $M_1 = M_{\text{scal}} \varphi_0^2 / A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3 / A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1 = M_R \varphi_0^4$ (which only relocated the free input to the un-pinned seesaw $M_R$ ). [E] shared decouple $A_\Lambda = 10$ ; [E] $M_1 = 8.65 \times 10^9$ GeV inside the thermal window $[3 \times 10^9, 3 \times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5 \times 10^{-7}$ , $\eta_B \approx 1.2 \times 10^{-9}$ brackets observed $6.1 \times 10^{-10}$ ; [C] HONEST: $M_1$ uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]                                                                                                                                                                                                                                                                                                                                                       |
| v213_ftransfer_functor.py     | The <b>F_transfer</b> <i>functor contract</i> (CONTRACT.F.01): the four frontier transfers are ONE typed functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]; consolidates v183 ( $F_{\text{pole}}$ ), v212 ( $F_{\text{Boltzmann}}$ ), v211 ( $F_{\text{relic}}$ ), v164/FR.MPME ( $F_{\text{QCD}}$ ), the structural complement of the v187 typing guard. [E] axiom 1 $\mu_4$ -deck equivariance ( $\lambda_2 = (2/3)^6 = 64/729$ the deck transfer eigenvalue); [E] axiom 2 Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ , $\text{Pl}(F) = (1, 22, 30)$ , $\ \text{Pl}(K)\ _1 = 11$ ); [E] axiom 3 positivity/stochasticity ( $\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f \in (0, 1)$ ); [E] axiom 4 external modules explicit ( $b_3 = -7$ a carrier output, $ b_3  = \Omega_{\text{adm}} - 10b_1 = 7$ ); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure [E][C]                                                                                                                                                                     |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v214_seam_pillowcase.py    | <p>The <i>pillowcase</i> reduction of QGEO.SYM.01 (QGEO.PILLOW.01): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the <math>\mu_4</math> marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. [E] <math>\chi_{\text{orb}}(S^2(2,2,2,2))=2-4(1-\frac{1}{2})=0 \Rightarrow</math> flat (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits <math>\pi</math> sum to <math>4\pi=2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}}=0</math>, flat away from the marks). [E] NEW LINK: cross-ratio(<math>\mu_4</math>)=2 (v168) <math>\Rightarrow j(\lambda)=256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)</math> with <math>j(2)=1728</math> (all six harmonic cross-ratios <math>\{2, -1, \frac{1}{2}\}</math> give 1728) <math>\Rightarrow</math> the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. [E] <math>j=1728 \Leftrightarrow \tau=i</math> (square torus) <math>\Leftrightarrow</math> CM by <math>\mathbb{Z}[i] \Leftrightarrow \text{Aut}=\mathbb{Z}/4</math> (<math>j=0</math> gives <math>\mathbb{Z}/6</math>; generic <math>\mathbb{Z}/2</math>). [E] explicit CM <math>(x,y) \mapsto (-x, iy)</math> on <math>y^2=x^3-x</math> has order <math>4=z \mapsto iz</math>. [E] negative control <math>\lambda=3 \Rightarrow j=21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut}=\mathbb{Z}/2</math>, no clock. [E] UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. [O] does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s [E]</p> |
| v215_seam_deck_killtest.py | <p>The bedrock QGEO.SYM.01 (<math>\omega \circ \rho = \omega</math>) recast as a <i>falsifiable</i> kill-test (QGEO.KILL.01), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): <math>\omega \circ \rho = \omega \Leftrightarrow [\rho, C]=0</math>; the principal symbol <math> k </math> commutes exactly, so the residual is the sub-principal <math>M_f</math> with <math>[\rho, M_f]=0 \Leftrightarrow f</math> <math>\mathbb{Z}_4</math>-invariant; measure form = the raw Gaussian bulk form <math>A</math> is <math>\mu_4</math>-deck-invariant. Four levers (carrier-side [E], seam-side [O]): <b>K1</b> exactly 4 marks (order-4 clock free orbit <math>\mu_4 + b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 =  \mu_4 </math>); <b>K2</b> square config (cross-ratio <math>2 \Rightarrow j=1728</math>, <math>\text{Aut}=\mathbb{Z}/4</math>; order 4 selects <math>\mathbb{Z}/4</math> not the <math>j=0</math> <math>\mathbb{Z}/6</math>; generic <math>\Rightarrow \mathbb{Z}/2</math>); <b>K3</b> mod-4 sub-principal support (<math>\mu_4</math>-mark sum <math>\Rightarrow f_m=0</math> for <math>m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f]=0</math>); <b>K4</b> transfer gap <math>(2/3)^6=64/729</math> (cusp weights <math>\{0, \frac{1}{3}, \frac{2}{3}\}</math>). Freeze-bound (freeze_file.csv seam_deck_symmetry). [O] the decisive kill is <i>deferred</i> (QGEO.REALIZE.01): the raw <math>\mathbb{P} \setminus \mu_4</math> collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure</p>                                                                                                                                                                                                     |
| v216_marks_gauss_bonnet.py | <p>The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = <math>2\chi</math>: <math>\mathbb{Z}_2</math> branch points (deficit <math>\pi</math>) on a flat sphere (<math>\chi=2</math>) give <math>n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 =  \mu_4  = e_1(a) = N_{\text{fam}}+1</math> (the same <math>\chi=2</math> as the 8 in <math>c_3=1/(8\pi)</math>). [E] the Euclidean sphere 2-orbifolds (<math>\chi_{\text{orb}}=0</math>) are exactly <math>(2,3,6), (2,4,4), (3,3,3), (2,2,2,2)</math>. [E] all-order-2 (the <math> \mathbb{Z}_2 </math> sheet branch) <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math>; [E] <math>N_{\text{fam}}=3</math> (rank <math>H^1=\# \text{marks}-1</math>) <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math>, picking the 4-mark square over the 3-mark hexagonal <math>(3,3,3)</math>; [E] a free order-4 deck <math>\Rightarrow</math> uniquely <math>(2,2,2,2)</math> — three independent selectors converge. [O] residual: the square modulus (cross-ratio <math>2 \Rightarrow j=1728</math>, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v217_marks_emergence_scan.py  | The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks $n$ and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\  \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$ , only $n-1=3=N_{\text{fam}}$ ; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$ ); [C] gap: $\mu_4$ -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$ ), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ , gap $(2/3)^6 = 64/729$ ; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square $\sim 0.4$ ); [C] negative controls $n=3, 5, 6$ . [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only |
| v218_diamond_axis_geometry.py | Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: $C$ center, $U$ winding, $V$ sheet), $F$ the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $ ), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$ ; $\det B(C+xU)=2 \det(C+xU)$ , anchor-block sheet curvature $6= R^+(A_3) $ . [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$ , steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2 \text{Disc}(q)$ for $Q, K, C, F$ ; squares $(1, 3, 4, 6)$ , kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$ , $105=N_{\text{fam}}g_{\text{car}} \cdot 7$ . [C] heuristic: $F$ the transfer-completion corner, a search principle, not a hard filter; the $G_2/F_4$ pair-sum labels stay audit-only (cf. v94/v95) [E][C]                                                              |
| v219_icosahedral_mckay.py     | McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$ . $E_8$ is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ( $2T \rightarrow \hat{E}_6$ , $2O \rightarrow \hat{E}_7$ , $2I \rightarrow \hat{E}_8$ ). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ , containing the $2, 3, 5$ axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$ , sum of squares $120= R^+(E_8) $ ); the natural-rep McKay adjacency <i>is</i> the affine $E_8$ graph with Kac marks = the degrees. A backward certificate of the closed $E_8$ [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v220_cp_hexagonal_modulus.py  | CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$ , $\text{Aut}=\mathbb{Z}/4$ (the $\mu_4$ clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$ , $\rho=e^{i\pi/3}$ , $\text{Aut}=\mathbb{Z}/6$ , CM by $\mathbb{Z}[\omega]$ , $\arg \rho=\pi/3$ . [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$ ; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v221_seam_qecc.py             | The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$ ). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$ ; the gapped transport $T$ on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$ ) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$ ; on the trace-zero subspace $\ T\delta\  \leq (2/3)^6 \ \delta\ $ hence $\ T^n \delta\  \leq (2/3)^{6n} \ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v222_cm_norm_duality.py       | CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$ , $j=1728$ ): $N(g_{\text{car}}+i \mu_4 )=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$ . [E] HEX (Eisenstein $\mathbb{Z}[\omega]$ , $j=0$ ): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=$ scalaron. The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21 \neq 41$ ). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v223_coxeter_totative_clock.py        | The $\mu_4$ clock is the order-4 character of the $E_8$ Coxeter cycle. [E]<br>$(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ = the $E_8$ exponents = live Coxeter phases;<br>$\varphi(30)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5$ . [E] $(\mathbb{Z}/30)^\times \cong \mathbb{Z}/4 \times \mathbb{Z}/2$ with order-4 generator 7<br>$(7^2=19, 7^4=1 \bmod 30)$ : a literal $\mu_4$ clock $\langle 7 \rangle = \{1, 7, 13, 19\}$ on the exponents; the<br>conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes.<br>Only $E_8$ has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$ ; the seam-clock identification is [C][E][C]               |
| v224_diamond_fttransfer_path.py       | $F_{\text{transfer}}$ as one path on the Sheet Diamond, not four external interfaces. [E]<br>family $M(s, t) = R + Q \text{diag}(s, t, t)$ : the corners<br>$K=M(1, -1), C=M(1, 0), F=M(1, 1)$ lie on the sheet axis ( $s=1$ ); the sheet axis is<br>curved $\det M(1, t) = 4t^2 + 14t + 14$ (2nd diff $8=\text{rank } E_8$ ), the winding axis<br>$\det M(s, 0) = 6s + 8$ flat (slope $6= R^+(A_3) $ ); Plücker steps $(1, 8, 10), (1, 8, 16)$ . [C]<br>$F_{\text{transfer}}$ is the $K \rightarrow C \rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius<br>projection (v37/v183) [E][C]                                                                       |
| v225_dual_normal_frame.py             | The dual normal frame $(d, n)$ : one boundary compass for flavor and horizon, with<br>CP as orientation. [E] $d = a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}(1, 1, -2)$ (the<br>Nariai/horizon normal; $d \cdot \mathbf{1} = 0, d \cdot a = 1$ ); $n = (5, -9, 6)$ with<br>$n^\top R = (\det R, 0, 0) = (8, 0, 0)$ , $n^\top L = (20, 0, 0)$ (the first-generation/torsion selector).<br>[E] oriented volume $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$ . [C] CP as orientation:<br>$\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$ for $\rho = e^{i\pi/3}$ — CP sits where the magnitude<br>bijection (Target D) fails [E][C]   |
| v227_degree_exponent_channel_split.py | $248=120+128$ as a magnitude/phase channel typing (replaces the over-strong<br>$S^-$ -dark-matter reading). [E] $E_8$ exponents sum $120= R^+(E_8) $ ( <i>magnitude</i><br>channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank} \cdot \dim S^+$<br>( <i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$ ;<br>$\sum \deg - \sum \exp = \text{rank} = 8$ . [C] ratios+products (Target D) live in the 120 channel,<br>the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector<br>reading of 128 stays Frontier [E][C][O]                                                                                  |
| v228_rr_index_gate.py                 | P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ;<br>bedrock language for P2, not a proof from nothing). [E] $\deg D = 4 =  \mu_4 $ on<br>$\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D)) = 5 = g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1 \setminus 4) = 3 = N_{\text{fam}}$ ;<br>$h^0 + \text{rank } H_1 = 8 = \text{rank } E_8$ , $h^0 - \text{rank } H_1 = 2 =  \mathbb{Z}_2 $ ; $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$ .<br>The mode-space reading is [C], conditional on the raw seam producing the<br>degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]                                      |
| v229_lepton_frobenius_algebra.py      | The charged leptons as an étale Frobenius algebra over the $\mu_6$ resolvent. [E] exact<br>coefficients $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure $c_e c_\tau = \frac{8}{3} =  \mathbb{Z}_2  c_\mu$ , product<br>$\frac{32}{9} = 2g_{\text{car}} / N_{\text{fam}}^2$ . [E] $A = \mathbb{Q}[t]/(m)$ , $m = \prod(t-c)$ : the trace-form Gram is<br>nondegenerate ( $\det = \text{disc}(m) \neq 0$ , $A$ étale) — a commutative Frobenius algebra;<br>the $C_6$ shift has characteristic polynomial $t^6 - 1$ (spectrum $\mu_6 = \mu_3$ family $\times \mu_2$<br>sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C] |
| v230_center_budget_norms.py           | The Sheet-Diamond center budget $(7, 11, 13)$ is the three <i>local norms</i> of the theory.<br>[E] $C = R + Q \text{diag}(1, 0, 0)$ , row sums $(7, 11, 13)$ : $7 = N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex =<br>scalaron), $13 = N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = $\Delta_Q$ ), $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 1+4+6$ (QBL<br>boundary Fock count on the 4 $\mu_4$ corners = $\dim S^+ - g_{\text{car}} = 16 - 5$ ). The two CM<br>rings (v222) and the seam boundary count meet in one operator center; the<br>reading is [C][E][C]                                                                                                              |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v231_cp_mu6_phases.py             | Both CP phases are $\mu_6$ powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] $\rho$ is a primitive 6th root ( $\rho^6=1$ , $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] $\delta_{\text{CKM}}$ leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i> ); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the $C_6/\mu_6$ monodromy has charpoly $t^6-1$ (spectrum $\mu_6$ carries $\rho$ ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$ , $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG $\mu_3$ has no $-1$ (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]                                                                                                                                                          |
| v232_e8_kleinian_seam.py          | The seam as the $E_8$ <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine $E_8$ ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> $E_8$ Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$ 's of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$ 's = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \# \text{ nontrivial } 2I \text{ irreps}$ — a fourth reading of the 8 in $c_3=1/(8\pi)$ . [E] the negated intersection form is the $E_8$ Cartan (even, $\det=1$ , positive definite; each $\mathbb{P}^1$ a $(-2)$ -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$ : the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $ . [C] seam reading: the raw RP seam ( $\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes $E_8$ ); bedrock [O][E][C][O]                                                                                                                                                             |
| v233_cp_triality_phase.py         | The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$ : the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$ , the sheet $\mu_2=\{1, -1\}$ ; $\rho=\omega^2 \cdot (-1)$ . [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$ , so the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase $\omega^2$ (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] $\omega$ is the $\mu_3$ factor of the order-30 $(2, 3, 5)$ Milnor/Coxeter monodromy that builds $E_8$ (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]                                                                                                                                                                                                                                                                                                                                                    |
| v234_seam_holomorphy_selection.py | The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing $E_8$ . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic ( $\mu$ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$ ; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) =  \Gamma^{\text{ab}}  =  H_1(S^3/\Gamma)  = \#(1\text{-dim irreps})$ : $A_n \rightarrow n+1$ , $D_n \rightarrow 4$ , $E_6 \rightarrow 3$ , $E_7 \rightarrow 2$ , $E_8 \rightarrow 1$ — only $E_8$ gives 1 ( $2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = $E_8$ . [E] the three faces (v83/v154, v232, v219) are the SAME $E_8$ -selector, so Target A, $G_{\text{net}}$ , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$ ; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O] |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v235_seam_chern_simons.py         | The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer $K$ -matrix with $\# \text{anyons} =  \det K $ , edge $c = \text{signature}(K)$ , holomorphic $\Leftrightarrow  \det K =1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ( $\det 16$ , 16 anyons) $\rightarrow SO(16)_1 = D_8$ ( $\det 4$ ) $\rightarrow (E_8)_1$ ( $\det 1$ , holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$ . [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev $E_8$ quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$ ). [E] NEG: $D_8$ is free+bosonic but has 4 anyons; an odd $K$ is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]                                        |
| v236_brieskorn_capstone.py        | The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ . [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$ , $m$ the $E_8$ exponents (charpoly $\Phi_{30}$ , $\deg \varphi(30)=8=\mu$ ) — so it <i>is</i> the order-30 $E_8$ Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the $\mu_3$ triality (CP, v233) as $h^{10}$ ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the $\mu_4$ clock ( $\times 7$ , v223). [E] Milnor lattice = $E_8$ , link = Poincaré sphere. [C] one singularity generates atoms + 8 + $E_8$ + order-30 + both clocks + seam; a unification, not a closure [E][C]                                                                                                                  |
| v237_seam_sre_closure.py          | The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the $K$ -matrix ground-state degeneracy on a genus- $g$ surface is $ \det K ^g$ (carrier $\rightarrow 16$ , $D_8 \rightarrow 4$ , $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$ . [E] among rank-8 even positive-definite $K$ ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only $E_8$ has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev $E_8$ ), edge holomorphic; $D_8/\text{carrier}$ are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$ , the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk ( $D_8$ ) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev $E_8$ phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O] |
| v238_modular_lindblad_dynamics.py | DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the $\mu_4$ clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ( $e^L = T$ , $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap $\Delta$ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.                                                         |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v239_kms_thermal_time.py        | DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$ , and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$ . [E] KMS at $\beta = 1$ : $\omega(A\sigma_i(B)) = \omega(BA)$ ( $\omega(X) = \text{Tr}(\rho X)$ , $\rho = e^{-\Lambda_\Sigma}/Z$ ), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$ ). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.                                                                                                                                                                                                                                                                                            |
| v240_gns_os_reconstruction.py   | DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data $(M, \omega)$ , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ( $\rho > 0$ ), $\dim n^2$ , identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$ , gap = $6\log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$ : minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.                                                                                                                                            |
| v241_dhr_sectors.py             | DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$ , $\theta(a) = e^{2\pi i q(a)}$ . [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ( $= \det K_{\text{carrier}} = \dim S^+$ ), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$ . [E] tower = v235/v237: Lagrangian $\langle(1, 1)\rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$ ; order-2 $\langle(2, 2)\rangle \rightarrow 4 = D_8$ ; carrier $16 \rightarrow D_8 \rightarrow E_8$ . [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact. |
| v242_spin_statistics_modular.py | DHR.MODULAR.01 — spin-statistics and the modular $(S, T)$ data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$ . Same MTC as v241 ( $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q = (5x^2 + 3y^2)/8$ , $\theta = e^{2\pi i q}$ ). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ( $\theta = 1$ ), 2 fermions ( $\theta = -1$ ) = $\{(0, 2), (2, 0)\}$ , 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$ . [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.                               |
| v243_haag_ruelle_braiding.py    | DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$ . [E] mass gap $\Delta = 6\log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M  = 1$ , diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: $M$ is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$ , $M(a, -b) = M(a, b)^*$ . [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$ , trivial braiding ( $\det 1/\text{SRE}$ ). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).                                                                                                              |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v244_spectral_action_contract.py    | <p>CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net <math>\rightarrow</math> 4d physics). A single Dirac operator whose <math>\text{Tr } f(D/\Lambda)</math> yields gravity AND matter. [E] carrier spinor = one generation: <math>SO(10)</math> half-spinor = <math>\dim S^+ = 2^{g_{\text{car}}-1} = 16</math> (<math>15 + \nu_R</math>), <math>N_{\text{fam}} = 3</math>. [E] SM gauge group inside the carrier: <math>SU(3) \times SU(2) \times U(1)</math> (<math>\dim 12</math>) <math>\subset SU(5)</math> (24) <math>\subset SO(10) = D_5</math> (45); colour <math>SU(3) \subset A_3 = SU(4)</math> (Pati-Salam); <math>\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8</math>. [E] one Dirac operator <math>\rightarrow</math> gravity (<math>a_2 \sim R</math>, <math>a_4 \sim R^2</math>, <math>\sqrt{36}</math>) + matter (<math>a_4 \sim F^2</math> Yang-Mills + <math> D\phi ^2</math> Higgs, Gilkey). [X] hooks: <math>g_3 = g_2 = \sqrt{5/3} g_1</math> + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam <math>\rightarrow</math> Dirac <math>D</math>; finite geometry = carrier; spectral action reproduces SM couplings) <math>\rightarrow</math> the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.</p> |
| v245_spectral_matter_content.py     | <p>QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the <math>SO(10)</math> <b>16</b>. [E] <b>16</b> = one generation: <math>SO(10)</math> <b>16</b> = <math>SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})</math>; SM dims <math>6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+</math>, <math>N_{\text{fam}} = 3</math> (15 Weyl fermions + <math>\nu_R</math>). [E] electric charges <math>Q = T_3 + Y</math> are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: <math>\sum Y = 0</math>, <math>\sum Y^3 = 0</math>, <math>[SU(2)]^2 U(1) = 0</math>, <math>[SU(3)]^2 U(1) = 0</math> (exact). [E] unification: <math>\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8</math>, i.e. <math>g_3 = g_2 = \sqrt{5/3} g_1</math>. [E] Higgs = the <math>(1, 2)_{1/2}</math> inner fluctuation (one doublet). [O] residual: the seam <math>\rightarrow</math> Dirac realisation + measured-coupling reproduction stay the open contract.</p>                                                                                                                                                                                                                                              |
| v246_unification_data_crosscheck.py | <p>QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does <math>g_1 = g_2 = g_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math> survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, <math>M_Z</math>): <math>\alpha_i^{-1} = (59.01, 29.59, 8.47)</math>, SM 1-loop <math>b = (41/10, -19/6, -7)</math> (v159). [E] the prediction: <math>\alpha_1 = \alpha_2 = \alpha_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math>. [X] tension: pairwise crossings spread <math>\sim 4</math> decades (<math>\sim 10^{13} - 10^{17}</math> GeV), residual spread at <math>2 \times 10^{16}</math> <math>\Delta \alpha^{-1} \sim 8.8</math> — <math>g_1 = g_2 = g_3</math> not realised by the plain SM. [X] <math>\sin^2 \theta_W</math> 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread <math>\sim 3.2</math> at <math>1.7 \times 10^{14}</math> GeV). [O] no rescue: SM gauge content + no new states (the <math>E_8</math> spine is arithmetic, not a threshold tower) <math>\Rightarrow</math> no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.</p>            |
| v247_e8_branching_no126.py          | <p>PS.E8BRANCH.01 — the <math>E_8</math> audit hull forbids the <b>126</b> and supplies exactly <math>\{1, 10, 16, 45\}</math> for <math>SO(10)</math>. Carrier <math>D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8</math> (via <math>SO(16)</math>); <b>248</b> = <math>(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248</math>. [E] ranks <math>5 + 3 = 8</math>; [E] dims = 248 (<math>SO(16)</math>: 120 adj + 128 spinor); [E] <math>SO(10)</math> content <math>\{1, 10, 16, 45\}</math>, the <b>126</b> ABSENT <math>\Rightarrow 126_H</math> algebraically forbidden, <math>10 + 16 + 45</math> <math>E_8</math>-supplied not fitted; [E] caveat only one 45/one 15 <math>\rightarrow</math> proton-marginal; [O] the field-content principle + NCG complement (v250).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v248_ps_algebra.py                  | <p>PS.ALGEBRA.01 — the Pati-Salam NCG algebra <math>A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})</math> and its exact SM matching. [E] unimodular group <math>SU(2)_L \times SU(2)_R \times SU(4)_c</math> (rank 5, <math>\dim 21</math>); [E] <b>16</b> = <math>(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8</math>; [E] <math>Y = T_{3R} + (B - L)/2</math> gives the exact SM hypercharges and <math>Q = T_{3L} + Y</math> the exact charges; [E] matching <math>\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}</math>, <math>\sin^2 \theta_W = 3/8</math>. [O] <math>A_F</math> uniquely from seam+sheet+<math>\mu_4</math>; [O] caveat <math>SU(4)_c \subset SO(10) = D_5</math> is not the carrier <math>A_3 = SU(4)</math> (family).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v249_ps_unification.py              | <p>PS.RGTEST.01 — carrier-native Pati-Salam/<math>SO(10)</math> two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread <math>\sim 8.8</math> @ <math>2 \times 10^{16}</math>); [C] <math>M_{PS}/M_s = 1.0 - 1.7</math> (1-loop; 2-loop <math>\sim 1.0 - 1.2</math>) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match (<math>b_{2R} &gt; 0</math>; also <math>E_8</math>-forbidden v247); [X] NEG proton decay selects content (minimal <math>M_{GUT} \sim 2 \times 10^{15}</math> excluded; <math>+(15, 1, 1)</math> safer). [O] the gauging fork, exact <math>\kappa</math>, the one-45 marginality.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v250_dirac_triple_contract.py     | CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target $(A, H, D, J, \gamma)$ almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$ , $D_F$ from $(R, L, \varphi_0, \text{Majorana})$ . [E] $H_F = 3 \times 16 = 48$ ; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + $\nu_R$ Majorana; $M_{PS} = \kappa M_s$ ) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.                                                                                                                                                                             |
| v251_first_order_sigma.py         | PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ $\sigma$ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + $\sigma$ ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: $\sigma$ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$ . [E] order-zero; [E] real-even ( $J^2 = +1$ , $\gamma^2 = 1$ , $D\gamma = -\gamma D$ , KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$ , supported exactly on the $\nu_R \nu_R^c$ blocks ( $\sigma$ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).                                                                                                              |
| v252_full_finite_triple.py        | PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ <b>16</b> -plets (dim 96). [E] dim 96; $D_F$ Hermitian+odd; reality $J^2 = +1$ , $JD = DJ$ ; grading $\gamma^2 = 1$ , $[\gamma, a] = 0$ , $J\gamma = -\gamma J$ ; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$ ); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2 / M_R$ . [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion. |
| v253_kappa_scalaron_ps.py         | PS.KAPPA.01 — pinning $\kappa$ in $M_{PS} = \kappa M_s$ , discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] $\kappa$ pinned by RG (scan PDG errors $\times$ the two $E_8$ -allowed contents) to a tight $O(1)$ interval $\sim 1.3$ (1-loop $[1.0, 1.7]$ , 2-loop $[1.0, 1.2]$ ) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ , a ratio of heat-kernel traces over the 96-dim $H_F$ (v252) — $\Lambda$ cancels, $\kappa$ scale-free and $O(1)$ . [X] the <b>126</b> -content drives $\kappa$ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments $f_2/f_0$ fixed.                                                                                                                                                              |
| v254_inner_fluctuations.py        | PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not <b>126</b> ) + the no-junk obligation + the $\sigma$ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no <b>126</b> : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$ ); [E] the $\sigma$ ( $\nu_R$ Majorana) direction is absent for the SM algebra but present for Pati–Salam — $\sigma$ is the CCvS beyond-first-order scalar; [E] $\sigma$ is an SM singlet $(1, 1, 0)$ with $B-L = 2$ ; [C] $\sigma = \text{scalaron consistency}$ .                                                                                                                                                                                  |
| v255_spectral_action_expansion.py | PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4 \Lambda^4 a_0 + 2f_2 \Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure ( $a_0$ cosmological, $a_2$ Einstein–Hilbert+Higgs-mass, $a_4$ Yang–Mills+Weyl <sup>2</sup> + $R^2$ +Higgs-quartic, $N=96$ ); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with $\sigma$ ; [E] scalaron = $R^2$ spin-0 mode (gravitational $a_4 \sim f_0 N$ ); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.                                                                                                 |

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| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v256_orientability_<br>poincare.py         | PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$ , full orientability is a degree-4 cycle on $M \times F$ ; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.                                                                                                                                                                                                                                                                                                                                                                                            |
| v257_quasiorient_<br>modpd.py              | PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$ ) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] quasi-orientability holds (left/right $A$ -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$ ); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.                                                                                                                                                                                                                                                          |
| v258_dirac_<br>covariance_<br>induction.py | PS.DIRAC.03 — $D_F$ is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance $C$ (contraction, $\text{spec} \subset (0, 1)$ ) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ( $\beta=1$ , v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ( $D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$ ); [E] unitarily equivalent to v252 (inducing $C_F$ from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$ ); [C] $C_F = P_F C_\Sigma P_F$ , so $D_F$ is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of $C_\Sigma$ . |
| v259_modular_<br>cutoff_kappa.py           | PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes $\beta$ , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$ , $f_2=\int f=1$ , $f_4=\int u f=1$ , so $f_2/f_0=f_4/f_2=1$ ; [E] seam-fixed $\beta=1$ (no new parameter beyond $c_3$ ); [E] non-vacuous neg control: a Gaussian cutoff $e^{-u^2}$ gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$ ; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$ , no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ( $O(1)$ ); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$ .                                                                                                                          |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v260_k3_kummer_unification.py    | <p>ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the <math>E_8</math> lattice as facets of one smooth object, the geometric unification of the v1 ‘glue <math>D_5 \oplus A_3 + \mu_4 \Rightarrow E_8</math>’ step (the K3/Kummer proposal). The seam square torus <math>\tau=i</math> (<math>j=1728</math>, v214) under the elliptic involution <math>z \mapsto -z</math> squares to the Kummer K3. [E] the <math>E_8</math> Cartan is even, <math>\det=1</math>, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice <math>L_{K3}=U^3 \oplus E_8(-1)^2</math> has rank <math>22=b_2</math>, <math>\det=-1</math> (unimodular), is even, signature <math>(3, 19)</math>, with <math>E_8(-1)^2</math> an orthogonal summand; [E] seam = <math>T^2/(z \mapsto -z)</math>: <math>4= (\mathbb{Z}/2)^2 </math> fixed points, pillowcase <math>S^2(2, 2, 2, 2)</math>, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes <math> A[2] =2^4=16 \rightarrow 16</math> exceptional <math>\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16</math> (v197), <math>16=4 \times 4</math>; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally <math>E_8(-1)^2</math> (distinct sublattices of <math>H^2(K3)</math>); [C] two faces of <math>E_8</math> in K3 (local du Val <math>\mathbb{C}^2/2I</math>, v232/v236; global <math>H^2</math> summand); [C] unification (lattice [E], carrier<math>\leftrightarrow</math>node identification [C]).</p> |
| v261_modular_spectral_closure.py | <p>QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object<br/> TFPT<sub>QFT</sub> = <math>(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})</math> reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones <math>[B:A]=4</math> (v89) = <math> \mu_4 =4</math> (v92) = seam marks <math>2\chi=4</math> (v216) = pillowcase/Kummer fixed points <math> (\mathbb{Z}/2)^2 =4</math> (v260); [E] one carrier-16: <math>\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16</math>, <math>H_F=2N_{\text{fam}} \cdot 16=96</math>; [E] one gap <math>\Delta = -\log((2/3)^6) = 6\log(3/2) &gt; 0</math> = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov <math>\log((1-C)C^{-1})=H</math>, v258; KMS cutoff <math>f_2/f_0=1</math>, v259; K3 lattice signature <math>(3, 19)/\det -1/\text{even}</math>, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.</p>                                                                               |
| v262_fqcd_mp_me.py               | <p>FR.MPME.02 — the <math>F_{\text{QCD}}</math> transfer pipeline for <math>m_p/m_e</math>, implemented as a real solver (the one <math>F_{\text{transfer}}</math> pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] <math>b_3 = -7 = -(11-2n_f/3)</math> at <math>n_f=6</math> (the only TFPT number in the run); [E] a two-loop <math>\alpha_s(M_Z)=0.1179</math> run with thresholds gives <math>\alpha_s(2\text{ GeV}) \sim 0.30</math> (real ODE solve); [C] <math>\Lambda_{\text{MS}}^{(3)} \sim 0.33\text{ GeV} \times</math> the lattice bridge <math>C_p = m_p/\Lambda</math> (external <math>O(1)</math>) over the closed <math>m_e</math> gives <math>m_p/m_e \sim 1824</math> vs the measured 1836.15 within the <math>\sim 7\%</math> external budget; [X] firewall: <math>m_p/m_e</math> is not an admissible compiler integer (<math>SU(9)</math> absent from <math>D_5 \oplus A_3</math>). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v263_mnu_from_df.py              | <p>FLAV.PMNS.02 — the light neutrino matrix <math>M_\nu</math> as the type-I seesaw of <math>D_F</math>, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from <math>D_F</math>’s blocks. [E] <math>M_\nu = -m_D M_R^{-1} m_D^\top</math> symmetric + seesaw-suppressed (the light masses an output of <math>D_F</math>); [E] a <math>\mu\tau</math>-symmetric <math>(m_D, M_R)</math> gives a <math>\mu\tau</math>-symmetric <math>M_\nu</math> (the v9 texture is in the seesaw image); [C] diagonalising gives <math>\theta_{23}=45^\circ</math> and <math>\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675</math> under the seam misalignment; [O] <math>\theta_{13}=0</math> in the leading texture (physical <math>\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}</math> a separate readout), the CP magnitude <math>\delta_{\text{PMNS}}=240^\circ</math> is the <math>\mu_6/\text{triality}</math> phase (v231/v233), and <math>M_R</math> is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v264_raw_seam_<br>marklocal.py          | QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit $\pi$ each, sum $4\pi=2\pi\chi(S^2)$ ), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the $\mu_4$ orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] $M_f$ then commutes with the clock $\rho=\text{diag}(i^n)$ ; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the $\mu_4$ orbit) breaks it. A reduction/sharpening, not a closure.                                                                                                                                                                                                                                                                                                                                                         |
| v265_qft4d_fork_<br>freeze.py           | QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread $\sim 8.8$ at $2 \times 10^{16}$ , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch ( $E_8$ content $\{1, 10, 16, 45\}$ no <b>126</b> v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake $\tau_p$ window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v266_ps_threshold_<br>proton.py         | PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ : the carrier-native PS UV branch survives proton decay <i>only</i> with the $E_8$ -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$ , $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$ ; [X] minimal KILLED ( $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K $2.4 \times 10^{34}$ , even high-band); [C] $+(15, 1, 1)$ survives now ( $\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> ( $\tau_p$ at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.                                                                                                                                                                                                                                                                                                                                     |
| v267_qgeo_rigidity_<br>minimal_axiom.py | QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier $\mu_4$ clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$ ) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of $a$ ); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$ ) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$ ; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$ , only $\mathbb{Z}_2$ ; hexagonal $j=0$ has $\mathbb{Z}_6$ , order $6 \neq 4$ ); [C] arithmetic rigidity ( $\{1, i, -1, -i\}$ over $\mathbb{Q}$ , $j=1728$ CM by $\mathbb{Z}[i]$ , Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$ ). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom. |
| v268_theta13_<br>carrier_trace.py       | FLAV.TH13.01 — the reactor angle $\theta_{13}$ <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$ ), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG $\sim 0.0222$ ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$ , so $\theta_{13}$ is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ $\theta_{13}$ from $M_\nu$ ’ route); [O] CP magnitude + absolute $\nu$ scale remain open.                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v269_spert_paqft_skeleton.py   | QFT4D.SPERT.01 — the perturbative 4D S-matrix ( $S_{\text{pert}}$ ) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ( $S_{\text{top}}$ 2d DHR braiding v243 / $S_{\text{pert}}$ 4d EG of the spectral action / $S_{\text{phys}}$ LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action $a_4$ operator has mass dimension $\leq 4$ in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ( $\Delta=6 \log(3/2)>0$ ) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.                      |
| v270_pmns_jarlskog_assembly.py | FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\text{PMNS}}$ (the CP <i>strength</i> ), assembled from the four TFPT channels ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$ , $\sin^2 \theta_{23}=\frac{1}{2}$ , $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$ , $\delta=4\pi/3$ ) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM $J$ (v88); [C] data: $J_{\text{max}}$ vs NuFIT 0.0332 ( $\sim 3\%$ ), $J$ vs best fit $-0.026$ , the negative sign reproduced; [O] $\delta$ is the $\mu_6$ /triality phase (assembled input, <i>not</i> from $M_\nu/D_F$ ) and the absolute $\nu$ -mass scale ( $\Sigma m_\nu$ , $M_R$ ) stays open. Assembles the PMNS matrix + CP strength; does not derive $\delta$ from $D_F$ nor fix the scale.                                                   |
| v271_eg_oneloop_quartic.py     | QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\text{pert}}$ skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=L_{\text{int}}$ : needs no extension, the first extension is $T_2$ ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$ , the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$ ; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ( $C(\omega+d,d)=1$ , renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action $a_4$ geometric quartic; [C] the $\phi^4$ one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the $s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$ ; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number. |
| v272_nu_mass_scale.py          | FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a $\Sigma m_\nu$ number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like $v_{\text{geo}}$ (v153), while the ratios ( $m_2/m_3$ , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$ , no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ( $< 0.072$ – $0.12$ eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number.              |
| v273_eg_gauge_running.py       | QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $\text{sd}=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop $b_i$ from the explicit carrier/SM content: $b_3=-7$ , $b_2=-19/6$ , $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact $b_i$ mirrored in Wolfram.                                                                                                                                                                                                                                                                                                                                                                               |

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| Script                                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v274_scale_<br>overdetermination.py         | SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\text{Pl}}^d$ ( $\rho_\Lambda/M_{\text{Pl}}^4$ , $M_{\text{scal}}/M_{\text{Pl}}=c_3^{7/2}$ , the spectrum); [C] two independent routes to $M_{\text{Pl}}$ agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\text{Pl}}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit ( $G$ ), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.                                                                                                                                                                                                                        |
| v275_qgamb_<br>roadmap.py                   | QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2)-2\cdot 248\,c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ( $c=248/31=8$ ; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.                                                                                                                                                                                                                                                                                                                                                          |
| v276_qgeo_<br>flat_closes_<br>commutator.py | QGEO.SYM.03 — the flat $\tau=i$ pillowcase premise <i>closes</i> the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to <i>one</i> geometric postulate (it does <i>not</i> prove the premise). For a quasi-free seam state $\text{QGEO.SYM.01} = (\omega \circ \rho = \omega) \Leftrightarrow [\rho, C]=0 \Leftrightarrow [\rho, H]=0$ . [E] isometry $\Rightarrow$ commutator: if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \rightarrow iz)$ is a flat- $\tau=i$ isometry then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ <i>exactly</i> (all orders), verified on the discretised square torus ( $\rho^4=I$ , $[\rho, \Delta]=0$ , $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full $H$ (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ( $j=1728$ ; hexagonal $j=0$ is order-6, excluded; v214/v267); [O] QGEO.SYM.01 collapses to the single statement ‘the raw seam is the flat $\tau=i$ pillowcase quasi-free state’ — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form. |
| v277_seam_calderon_<br>e8_match.py          | QGAMB.TIERB.01 — the seam-Calderón $\rightarrow (E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to <i>one</i> bit (it does <i>not</i> close it). [E] $c=8$ (Sugawara, $248/31=g_{\text{car}}+N_{\text{fam}}$ ), but $c=8$ alone does not pin the net; [E] the discriminator is holomorphy — $\det \text{Cartan}(E_8)=1$ (1 primary, holomorphic) vs the $c=8$ counterexample $\det \text{Cartan}(D_8=SO(16))=4$ (4 primaries, non-holomorphic), so the single discriminator is $ \det K =1$ ; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248=\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic $\Leftrightarrow \det \text{Cartan } 1 \Leftrightarrow \text{unique even unimodular rank-8 (v83)} \Leftrightarrow \text{homology-sphere link} \Leftrightarrow  \det K =1$ (v235); [C] seam-side match via v276 (flat $\tau=i$ ) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, <i>not</i> closed.                                                     |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v278_lsz_bridge_unitarity.py    | <p>QFT4D.SPERT.04 — the <math>S_{\text{pert}} \rightarrow S_{\text{phys}}</math> LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (<math>S_{\text{phys}}</math> = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: <math>S_{\text{phys}}</math> = amputate+on-shell-residue of the connected time-ordered products, and the EG <math>T</math>-products of <math>S_{\text{pert}}</math> are those correlators (OS Wick-rotation, v240)</p> <p><math>\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}</math>; [C] the gap <math>\Delta=6\log(3/2)&gt;0</math> gives Haag–Ruelle asymptotic states so LSZ is valid; [E] <i>one-loop unitarity</i> (optical theorem): the bubble discontinuity <math>\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}</math> <i>exactly</i> = the two-body phase space, so <math>2\text{Im } M = \sum \int d\Pi  M ^2</math> (cutting rule) holds <math>\Rightarrow</math> matter+gauge unitary; [E] threshold: <math>\text{Im } B=0</math> below <math>s=4m^2</math>, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary <math>\rightarrow</math> QG.AMB.01; [C] finite field-strength <math>Z</math> from the EG scheme.</p> |
| v279_qgeo_obligation_lemma.py   | <p>QGeo.OBLIG.01 — the QGeo.SYM.01 bedrock written as <i>one</i> precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does <i>not</i> close it). Lemma: given the premise ‘the raw seam carries the flat <math>\tau=i</math> pillowcase metric <math>g_{\text{flat}}</math> (Trojanov-unique, cone angles <math>\pi</math>), so <math>\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}</math>’, then <math>\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow</math> mark-locality <math>\rightarrow E_8</math>. [E] the reduction DAG from ‘free RP seam’ has 8 closed nodes and <i>exactly one</i> open leaf (‘the raw seam state is the flat state’); [E] the geometric side is fixed (flat DtN diagonal, <math>\rho</math> commutes to all orders, Lean FORM.QGeo.03), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Trojanov + OS); [E] the all-orders step is a Lean theorem (SeamDeckClosure.flat_all_orders_clock, standard axioms, no sorry); [O] one constructive-QFT statement closes QGeo.SYM.01.</p>                                                                                                                                                                                                       |
| v280_pillowcase_steklov.py      | <p>QGeo.STEKLOV.01 — a direct numerical experiment on the flat <math>\tau=i</math> pillowcase orbifold (the seam geometry): the self-investigable test of the QGeo.SYM.01 obligation (does <i>not</i> prove the premise). [E] pillowcase = the <math>Z_2</math>-even sector of the flat torus, 4 cone points (2-torsion, deficit <math>\pi</math> each, sum <math>4\pi=2\pi \cdot 2</math>); [E] the order-4 deck <math>\rho (z \rightarrow iz)</math> descends (<math>[\rho, Z_2]=[\rho, P_{\text{even}}]=0</math>) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact:</p> <p><math>[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega</math> (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.</p>                                                                                                                                                                                                                                                                                                                                                                              |
| v281_holomorphy_modular_data.py | <p>QGAMB.MODULAR.01 — the modular-data / anyon-condensation layer of the QG.AMB.01 Tier-B holomorphy bit. [E] <math>\# \text{anyons} =  \det \text{Gram} : E_8 \rightarrow 1</math> (holomorphic), <math>D_8=SO(16) \rightarrow 4</math>, carrier <math>D_5 \oplus A_3 \rightarrow 16</math>; [E] the anyon-condensation tower <math>D_5 \oplus A_3 (16) \rightarrow D_8 (4) \rightarrow E_8 (1)</math>, each step condensing a Lagrangian <math>Z_2</math> boson, the Kitaev <math>E_8</math> endpoint (v92/v235); [E] Gauss–Milgram recovers <math>c=8</math>; [E] <math>E_8</math> is the unique holomorphic <math>c=8</math> (<math>\det 1</math>) — <math>SO(16)</math> has the same <math>c</math> but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to <math>\det K=1</math>? Reduced to one bit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v282_e8_tau_i_unification.py    | <p>QGAMB.UNIFY.01 — the <math>E_8</math>-at-<math>\tau=i</math> unification: the candidate <i>simplification</i> reducing the two open obligations to <i>one</i> object (the raw seam is the <math>(E_8)_1</math> holomorphic CFT at <math>\tau=i</math>). <math>\chi_{E_8}(\tau)=\Theta_{E_8}/\eta^8=j^{1/3}</math>. [E] <math>\tau=i</math> is the order-4 CM point (<math>j=1728</math>, the QGeo.SYM.01 flat geometry); [E] <math>\chi_{E_8}=j^{1/3}</math> (<math>\chi^3=qj</math> as <math>q</math>-series), so <math>\chi_{E_8}(i)=1728^{1/3}=12</math> (the <math>(E_8)_1</math> character at the SAME <math>\tau=i</math>, the QG.AMB.01 holomorphy); [E] one object, two faces; [C] obligation count <math>2 \rightarrow 1</math>: prove ‘the raw seam is <math>(E_8)_1</math> at <math>\tau=i</math>’ and <i>both</i> follow; [O] the unified premise stays open but is now one object.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v283_live_killtest_scorecard.py              | PRED.LIVE.01 — a live kill-test scorecard of the recent round’s computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2 \theta_{12}=0.30675$ ( $+0.31\sigma$ ), $\sin^2 \theta_{13}=0.02311$ ( $+1.45\sigma$ ), $\theta_{23}$ maximal; [C] leptonic CP $J_{\text{PMNS}}=-0.0297$ , $\delta=240^\circ$ vs $\sim 232(30)$ ; [X] $\Sigma m_\nu$ NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p=1.55\times 10^{35}$ yr for $+(15, 1, 1)$ (Hyper-K decisive); [E] closed readouts $\alpha^{-1}$ , $\Omega_b$ , $\beta$ ; all rows consistent $< 2\sigma$ , two sharp near-term kill tests. A consolidation, not new physics.                                                                                                                                                                                                                                                                                                                       |
| v284_route_i_rp_uniqueness.py                | QGEO.ROUTEI.01 — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does <i>not</i> prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip DtN( $k$ )/ $k \rightarrow 1$ , principal symbol $ \xi $ with no curvature term), L4 covariance $C = (1 + e^H)^{-1}$ from the DtN (v258), L5 $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat ( $\tau=i$ ), so $L_{\text{open}}$ reduces to ‘the raw seam is the constant-curvature geometric state’; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step). |
| v285_route_ii_seam_condensation.py           | QGAMB.ROUTEII.01 — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma <i>coincides</i> with Route (i)’s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow  \det K =1 \Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic $c=8 \Leftrightarrow (E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower $16 \rightarrow 4 \rightarrow 1 =$ the unique $\mu_4$ -Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share <i>one</i> open lemma — both say ‘the raw seam is the $(E_8)_1$ -at- $\tau=i$ object’ (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$ . $M_{\text{open}} = \text{QGEO.SYM.01}$ .                                                                     |
| v286_seam_equivalence_contract.py            | SEAM.EQUIV.01 — the Seam Equivalence contract: concentrates the whole open structure on <i>one</i> named arrow (does <i>not</i> prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau=i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] <b>import firewall</b> : no Route-i module imports any Route-ii module or vice versa, so the routes converge only through SEAM.EQUIV.01 (kills the ‘ $E_8$ into the geometry and back’ circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary’.                                                                                                      |
| v287_free_rp_bulk_to_holomorphic_boundary.py | SEAM.EQUIV.A01 — Route A (AQFT) theorem-dependency checker for the open Gral arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary net’: types the 5-lemma chain and isolates the <i>one</i> missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality $\Rightarrow$ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk $\Rightarrow$ single-sector boundary ( $\mu$ -index 1) — the one missing standard theorem (the AQFT form of $\det K=1$ ); [E] L4 $\mu=1+c=8 \Rightarrow$ holomorphic (v234); [E] L5 holomorphic $c=8 \Rightarrow (E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces SEAM.EQUIV.01 to one cited theorem.                                                                                                                                                                                                                                                                      |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v288_full_l2_subprincipal_z4.py       | SEAM.EQUIV.B01 — Route B (DtN): the full- $L^2$ lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the ‘probably provable’ step and lifting v201/v284 from the finite block to $L^2$ . [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod{4}$ ; [E] full- $L^2$ block-diagonality: $\Lambda =  n  + M_f$ commutes with $\rho = \text{diag}(i^n)$ , $\ [\rho, \Lambda]\  \sim 2 \times 10^{-15}$ on $-50..50$ ; [E] negative control: a generic curvature gives $\ [\rho, \Lambda]\  = 4.44 > 0$ ; [E] Lean consistency (geom_sum_fourth_root + markLocal_blockDiagonal); [O] the sharper residual — why is the raw seam sub-principal term mark-local?                                                                |
| v289_marklocal_raw.py                 | MARKLOCAL.RAW.01 — decompose the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $=  D_\theta  + M_f + \text{cone sources}$ (b-calculus/Melrose, imported); [O] L2 <b>Flat-Away</b> — the one open analytic lemma: $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$ (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one $\mu_4$ orbit (v195/v216); [E] L4 Fourier-support: $\mu_4$ -orbit curvature $\Rightarrow$ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: $\Rightarrow [\rho, \Lambda] = 0$ on full $L^2$ (v288). 4/5 discharged; Route B reduces to exactly L2.              |
| v290_z4_smooth_curvature_adversary.py | SEAM.ADVERSARY.01 — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is <i>not</i> mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$ -symmetric off-mark curvature $f = \varepsilon \cos(4\theta)$ : [E] it is $\mathbb{Z}_4$ -symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$ ) but <i>not</i> mark-local ( $f(\pi/4) = -\varepsilon \neq 0$ ); [E] it <i>passes</i> the v288 commutator $\ [\rho, \Lambda]\  = 0$ (same as mark-local) — so $[\rho, \Lambda] = 0$ is necessary but <i>not</i> sufficient; [E] but it shifts the DtN spectrum off the flat ladder ( $\max  \Delta\lambda  = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input. |
| v291_flataway_contract.py             | FLATAWAY.RP.01 — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four $\mu_4$ marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] <i>spectral-rigidity</i> : scanning $f = \varepsilon \cos(4\theta)$ , the spectrum deviation from the flat ladder is 0 only at $\varepsilon=0$ — flat is the unique spectral match; [E] <i>heat-kernel</i> : the heat-trace deviation is 0 only at $\varepsilon=0$ ; [C] <i>RP-energy</i> : $\text{RP} + \text{gap}$ is a Dirichlet-energy minimisation in the Troyanov conformal class $\Rightarrow$ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B.                                        |
| v292_flataway_heat_reduction.py       | FLATAWAY.HEAT.01 — the Heat-Kernel route made precise: reduces Flat-Away to <i>one</i> named spectral invariant. [E] the heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ( $c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$ ; positive on all combinations), so $\Delta \text{Tr} = 0 \Leftrightarrow f = 0$ ; leading invariant $\ f\ _{L^2}$ (Parseval). [O] reduction: Flat-Away $\Leftarrow$ ‘the seam fixes the $a_2$ heat coefficient to its flat value’ $\Rightarrow \int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.                                                                                                                                                                                                         |
| v293_flataway_spectral_hessian.py     | FLATAWAY.SPEC.01 — the Spectral-Rigidity route upgraded from one direction (v291) to the <i>full</i> smooth- $\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $ n $ is split by mode $4k$ at $ n =2k$ by $\pm\varepsilon/2$ ; [E] $S = \sum (\Delta\lambda)^2 \sim 0.50\varepsilon^2$ ; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$ , so the flat seam is a <i>strict</i> isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non- $\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam’s Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation $\Rightarrow$ Flat-Away.                                                                                                                                                                             |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v294_flataway_rp_energy.py       | FLATAWAY.ENERGY.01 — the RP-energy / Troyanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: $S^2$ with 4 cone angles $\pi$ , $\sum(2\pi - \theta) = 4\pi = 2\pi\chi$ ; [C] Troyanov/Euclidean-cone uniqueness (Troyanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f] = \int f^2$ , Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: ‘RP+gap realises the energy-minimiser’, converging with v292/v293 on one selection principle; closing any one closes Route B.                                                                                                                                                                                                                                                                                                                                                                 |
| v295_flataway_a2_exact.py        | FLATAWAY.A2.01 — the <i>exact</i> analytic $a_2$ coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \text{Tr } e^{-t\Lambda(\varepsilon)} _0 = -t\hat{f}(0) \sum e^{-t n } = 0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x) = e^{-tx}$ strictly convex $\Rightarrow \Lambda \mapsto \text{Tr } e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon=0$ is the global minimum and $\Delta \text{Tr} \geq 0$ for all $\varepsilon$ , $= 0$ iff $f = 0$ ; [E] exact $a_2$ kernel via the divided difference of $e^{-tx}$ (Duhamel), matched to numerics to rel.err $< 10^{-6}$ ; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1) \geq 0$ ; [O] reduction now analytic: Flat-Away $\Leftarrow$ ‘the seam fixes $a_2$ ’. Lean-mirrored (FORM.FLATAWAY.01).                                                                                                         |
| v296_flataway_a2_closed.py       | FLATAWAY.A2.CLOSED.01 — the exact $a_2$ coefficient in <i>closed form</i> . [E] the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ), giving $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ ; [E] a finite $(4k-1)$ -term middle (incl. the degenerate diagonal $\Phi(2k, 2k) = (t^2/2)e^{-2kt}$ ); [E] closed form $W_k(t) = \frac{t(1-e^{-4kt})}{8k(1-e^{-t})} + \frac{1}{2} \sum_{m=1}^{4k-1} \Phi(m, 4k-m)$<br>$(W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8)$ , reproducing v295 to machine precision; [E] manifestly positive; [E] small- $t$ : $W_k = t/2 + O(t^3)$ (mode-independent leading $1/2$ , the $t^2$ term cancels).                                                                                                                                                                                                                                                                                                                        |
| v297_route_a_literature_stack.py | SEAM.EQUIV.A.LIT.01 — Route A’s ‘one standard import’ as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c=8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into ‘invertible bulk $\Rightarrow (E_8)_1$ ’, all legs established; [O] the one open hypothesis is the input of LIT-A, ‘the raw quasi-free seam bulk is invertible (SRE)’ = Route B’s Flat-Away. [E] non-circular with Route B (the v286 firewall holds).                                                                                                                                                                                                                                                                                  |
| v298_e8_network_attractor.py     | <i>E8 as a network attractor</i> — a computational-substrate (Wolfram-program) reading of the $(2, 3, 5)$ /McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the $E_8$ Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the <i>unique</i> attractor of the minimal lazy graph update $m \leftarrow (A + 2I)/4 \cdot m$ on the affine- $E_8/(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$ , top eigenvalue 2). [E] the spherical McKay tower terminates at $E_8$ ( $2T \rightarrow E_6$ , $2O \rightarrow E_7$ , $2I \rightarrow E_8$ , $h = 30$ ); [E] the cascade is nested-subgraph coarse-graining ( $240 \rightarrow 126 \rightarrow 72 \rightarrow 40 \rightarrow 20$ ). [C] Wolfram reading: $E_8$ as the universal attractor of a minimal $(2, 3, 5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph <i>rewrite</i> whose attractor is the full TFPT structure stays open; $\varphi_0 = (4/3)c_3 = 1/(6\pi)$ is the analytic seed, not an edge fraction. |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v299_hypergraph_0d_autonomous.py | The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth <i>reaches</i> $E_8$ as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary $\hat{E}_8$ ) — <i>not</i> ‘P2 derived’. [E] icosahedron = 3-uniform hypergraph ( $ \text{Aut}  = 120 =  I_h $ ; the McKay group is the $SU(2)$ binary lift $2I$ , a different order-120 group); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine $E_8$ McKay graph; [E] Smith: $\rho \leq 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \leq 2 \Leftrightarrow$ a positive witness $m$ with $Am \leq 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \leq 2m$ ) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ under the $(2, 3, \cdot)$ seed; [E] $A_n$ ( $\rho = 2 \cos \frac{\pi}{n+1}$ ) and $D_n$ ( $\rho = 2 \cos \frac{\pi}{2n-2}$ ) controls are $< 2$ for all $n$ , never reaching $E_8$ ; [O] the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).                                                     |
| v300_flataway_rigid.py           | FLATAWAY.RIGID.01 — the attack on Flat-Away. [E] the flat fingerprint: $\text{spec}( D_\theta )$ is the integer ladder, every level $n \geq 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy $2 \times 2$ $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ ), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \rightarrow 1+1$ ) and changes the spectral multiset, so preserving the flat ladder <i>with</i> its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 ‘deviation grows’ scan; [E] with the v295 convexity, flat is isolated analytically <i>and</i> discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer $q$ -coefficients (248 at level 1; $E_8$ even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries <i>zero</i> new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact.                                 |
| v301_route_a_invertible.py       | SEAM.EQUIV.A.INV.01 — discharge Route A’s last open hypothesis (‘the raw quasi-free seam bulk is invertible/SRE’) via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of SEAM.EQUIV.01. [E] carrier = 16 Majoranas, $c = 8$ ( $g_{\text{car}} + N_{\text{fam}} = 5 + 3 = 8 = 16/2$ , v148); [E] $SO(16)_1 \supset (E_8)_1$ : $\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248 = $\dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$ -classified by the chiral Majorana number and is <i>always</i> invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $ \det K  = \#\text{anyons}$ is 1 for $E_8$ (invertible) vs 4 for the gauged $D_8 = SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A’s ‘invertible bulk’ now follows from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input ‘is the quasi-free bulk gapped?’; [O] with v300 the whole open content of SEAM.EQUIV.01 reduces to the single spectral statement ‘the quasi-free seam bulk is gapped’. |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v302_seam_gap.py           | <p>SEAM.EQUIV.GAP.01 — the last lever: after v300/v301 the whole open content of SEAM.EQUIV.01 was ‘the quasi-free seam bulk is gapped’; that gap is <i>not</i> a free input but the established Recovery gap of the seam transfer operator. [E] <math>\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}</math> (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: <math>\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 &gt; 0</math> (multiplicative gap <math>1 - (2/3)^6 = 665/729</math>; <math>6 = 2N_{\text{fam}}</math>, <math>(2/3)^6</math> the <math>\mu_4</math>-deck transfer eigenvalue forced by the carrier); [E] the positivity margin <math>\Delta - 31/(4\pi^2) = 1.6476 &gt; 0</math> (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap <math>\Delta &gt; 0</math> is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap <math>6 \ln(3/2) &gt; 0</math>, and SEAM.EQUIV.01’s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v303_ftransfer_dynamics.py | <p>FR.DYNAMICS.01 — the discrete→dynamic lens on F_transfer: the main-branch mechanism (a gapped, positivity-preserving relaxation to a <i>unique</i> attractor; the local-averaging update → the <math>E_8</math> marks, gap <math>6 \ln(3/2)</math>) is the shared <i>shape</i> of all four instances. [E] F_pole (Koide) <i>is</i> the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier <math>F' = (2/3)^6 = \lambda_2</math>, the seam-clock subleading eigenvalue (v82/v54); <math> F'  &lt; 1 \Rightarrow</math> a unique attractor at the seam rate; [C] F_Boltzmann (<math>\eta_B</math>) is a gapped positive contraction (washout <math>\kappa_f \in (0, 1)</math>, H-theorem; compiler fixes <math>A_\Lambda = 10</math>, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed <math>\theta_i = 3\pi/5</math>, rate cosmological); [E] F_QCD (<math>m_p/m_e</math>) is the 1-loop RG flow with the carrier <math>b_3 = -7</math> (asymptotic freedom <math>\Rightarrow</math> the Gaussian UV attractor, <math>\Lambda_{\text{QCD}}</math> generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so F_transfer is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v304_idg_nonlocal_ghost.py | <p>QGAMB.IDG.01 — the KMS spectral-action cutoff suggests a <i>nonlocal</i> (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a <i>truncation artefact</i> — a [C] narrowing of the red-team rt_F perturbative-gravity attack, <i>not</i> a closure of QG.AMB.01 (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator <math>1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]</math> has a healthy massless graviton (residue <math>+1/M^2</math> at <math>p^2=0</math>) and a massive spin-2 mode with <i>negative</i> residue <math>-1/M^2</math> at <math>p^2 = -M^2</math> (the Stelle ghost, reproducing v278); [E] for an IDG operator <math>p^2 a(p^2)</math> with <math>a(z) = e^{z/M^2}</math> entire and nowhere zero, the dressed propagator <math>1/(p^2 a)</math> keeps its <i>only</i> pole at <math>p^2=0</math> — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation <math>a(z) = 1 + z/M^2</math> (the <math>R+R^2</math> order) has a real zero at <math>z = -M^2</math> = the ghost pole, so ‘entire vs polynomial’ is the discriminator; [C] the cutoff is the <math>\beta=1</math> seam KMS weight <math>f(u) = e^{-u}</math> (v259), not a choice — truncating <math>\text{Tr} e^{-D^2/\Lambda^2}</math> at <math>a_4</math> is what makes the local <math>R+R^2</math> and the ghost; [C] <b>conditional narrowing</b>: <i>if</i> the resummed form factor is entire (Biswas–Mazumdar–Siegel class) <i>then</i> the perturbative graviton is ghost-free and rt_F’s attack is a truncation artefact; [O] <b>open</b>: (i) the trace regulator <math>f</math> is not the kinetic form factor <math>a(\square)</math> — entire analyticity of the resummation is unproven; (ii) a ghost-free <i>perturbative</i> graviton is not the nonperturbative ambient measure — QGAMB.IDG.01 stays [C]/[O], not a QG.AMB.01 closure.</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v305_witness_independence.py  | Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ( $e=(4,5,2)$ , $p_n=2+2^n$ ; v23/v228), $\pi$ the only transcendental primitive (ARCH.CORE.01) — 13 atoms / 2 free primitives = $6.5 \times$ compression made explicit (the recurring $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ and $(2/3)^6$ are <i>one</i> ratio); $E_8/g_{\text{car}}$ /anchor are overdetermined ( $\geq 3$ structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); $\alpha^{-1}$ 's '137' is foreign to the anchor family = an independent witness [E] |
| v306_seed_crossval.py         | Leave-one-out cross-validation of the shared seed $\varphi_0$ : back-solving $\varphi_0$ independently from the five seed-grammar observables ( $\sin^2 \theta_{12}$ , $\sin^2 \theta_{13}$ , $\beta$ , $\Omega_b$ , $\lambda_C$ ) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within $3\sigma$ out of sample (most-tensioned = $\sin^2 \theta_{13}$ , $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the $\chi^2$ by $> 100 \times$ [E]                                                                                                     |
| v307_data_watchdog.py         | Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live $n_\sigma$ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items $\alpha^{-1}$ ( $1.9\sigma$ ), $s_{23}$ ( $1.8\sigma$ ) and the most-tensioned core $\sin^2 \theta_{13}$ ( $2.0\sigma$ ); the prediction-of-record $\sin^2 \theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) [E]       |
| v308_seam_equiv_chain.py      | TOE attack 1 — the SEAM.EQUIV.01 keystone as a composition certificate: the v297 stack as one well-typed chain $\text{gap}>0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$ with exact discriminators $ \det \text{Cartan}(E_8) =1$ vs $ \det \text{Cartan}(D_8) =4$ , carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ , $c=g_{\text{car}}+N_{\text{fam}}=8$ ; the input gap is the derived recovery gap $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure [E][O]                                                                                                                    |
| v309_modular_bedrock.py       | TOE attack 2 — the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) $\Rightarrow \mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of QGEO.SYM.01 [E][O]                                                                                                                                                                                     |
| v310_carrier_sm_anomaly.py    | TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5 \rightarrow SU(5) \subset SO(10)$ as $10+5+1 = \text{one generation}$ ; gauge-anomaly-free ( $\sum Y=0$ , $\sum Y^3=0$ , $[SU(2)]^2 U(1)=[SU(3)]^2 U(1)=0$ , exact) and reproduces $(b_1, b_2, b_3)=(\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts $b_1$ , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ $\beta$ sub-question; the full EG $S$ -matrix + Yukawa stay open [E][C]                                                                                                                                                                 |
| v311_gap_decoupled_measure.py | TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ , correlation length $1/(6 \ln(3/2))$ , finite susceptibility $729/665$ ; the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=EH/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); QG.AMB.01 stays [O] (gap-decoupled, inert) [E][O]                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v312_hypergraph_substrate.py  | TOE attack 5 — the hypergraph substrate sharpened: the affine- $E_8$ Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly $2\cos(\pi/5)=$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is <i>not</i> any eigenvalue and $\varphi_0$ is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ (gap $6\ln(3/2)$ ) and the analytic seed $\varphi_0$ . The open question is precisely delimited, not closed [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v313_golden_atoms_spectral.py | The golden ratio is the $g_{\text{car}}=5$ signature: the $(2,3,5)$ atoms <i>are</i> the affine- $E_8$ network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , the golden minimal polynomial $x^2-x-1$ divides it ( $\phi=2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k \in \{2,3,5\}=\{ \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}}\}$ (giving $0, 1, \phi, 1/\phi$ ), so the golden ratio is the $g_{\text{car}}=5$ (5-fold) signature — a new spectral witness for $g_{\text{car}}$ (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2+1/3+1/5=31/30$ ties the v63/v76 margin numerator $31=2^{g_{\text{car}}}-1=248/8=1+h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: $\phi$ is structural, not a readout. [C] direction: $\phi$ is the $E_8$ -quasicrystal (Elser-Sloane) ratio, a candidate substrate [E][C]                                                                                                                                                                            |
| v314_rate_translation.py      | Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine- $E_8$ adjacency spectrum splits into the rational part $\{0, \pm 1, \pm 2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi, \pm 1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$ ; the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2\cos(\pi/2)=0= \mathbb{Z}_2 $ , $2\cos(\pi/3)=1=N_{\text{fam}}$ build the recovery rate $(2/3)^6=( \mathbb{Z}_2 /N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$ ), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$ ) has <i>no</i> rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ( $30=g_{\text{car}}(2N_{\text{fam}})=5\cdot 6$ ); the $\phi$ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C]                                                                                                                                                                                                                       |
| v315_coxeter_coupling.py      | Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5,6)=1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: $E_8$ exponents = totatives(30) = totatives(5) $\times$ totatives(6), $\phi(30)=4\cdot 2=8=$ rank $E_8$ ; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}]=8=$ rank $E_8$ , $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) (\ni \sqrt{5})$ and the family $\mathbb{Q}(\zeta_3)$ ; [E] GALOIS = $\mu_4 \times \mathbb{Z}_2$ : $(\mathbb{Z}/30)^\times = (\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2$ (order 8, max order 4), $\mu_4=(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2=(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}=\phi+1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$ . [C] $30=5\cdot 6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C] |
| v316_galois_readout.py        | Does the Galois $\mu_4 \times \mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho=\zeta_6=\zeta_{30}^5$ is the order-6= $2N_{\text{fam}}$ FAMILY factor $c^5$ (the carrier is $\zeta_5=\zeta_{30}^6$ ); [C] $\delta_{\text{CKM,lead}}=\pi/3=\arg \zeta_6$ , $\delta_{\text{PMNS}}=4\pi/3=\arg \zeta_6^4$ (power $4= \mu_4 $ ), $\zeta_6^4=-\zeta_6$ the sheet flip ( $\mu_6=\mu_3 \times \mu_2$ ); [E] the Galois $\mathbb{Z}_2=\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q})=\{\zeta_6 \mapsto \zeta_6^5=\bar{\zeta}_6\}$ IS CP conjugation ( $\delta \mapsto -\delta$ ); [E] the carrier factor $\zeta_5$ (golden $\sqrt{5}$ ) carries no phase (static); [E] the magnitudes ( $\sin^2 \theta_{12}=1/3-\varphi_0/2$ , transcendental $\varphi_0$ ) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C]                                                                                                                                                                   |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v317_galois_family.py       | Are the 3 generations a $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ ( $\text{Spec}(Q_+) = \{1, 2, 3\} = A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$ ; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive $\mu_3$ orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$ ). [E] the Galois $\mathbb{Z}_2$ ( $\zeta_3 \mapsto \zeta_3^2$ , = CP conjugation v316) fixes $w=0$ (rational) and swaps $w=1/3 \leftrightarrow 2/3 \Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C]                                                                                                                                                                                                                                                                                                                                                                                                   |
| v318_arithmetic_capstone.py | The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$ , and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$ : $[\mathbb{Q}(\zeta_{30}) : \mathbb{Q}] = \varphi(30) = 8 = \text{rank } E_8$ , $30 =  \mathbb{Z}_2  N_{\text{fam}} g_{\text{car}} = h(E_8)$ , Galois $(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$ ; [E] rigidity: forced by the anchor atoms $\{2, 3, 5\} = e(a)$ ; [E] magnitude reduction: $\varphi_0 = (4/3)c_3 + 48c_3^4$ is a pure function of $\pi$ (no free parameters), so the magnitudes are $\{a, \pi\}$ -determined, not free; [E] free inventory: 0 dimensionless free parameters, $\pi$ the unique transcendental, $v_{\text{geo}}$ the one dimensional unit. [O] residual: the raw-seam realisation stays open (QGEO.SYM.01/SEAM.EQUIV.01) [E][O]                                                                                                                                                                                                                                                                                                                          |
| v319_translation_clock.py   | The translation clock: the discrete $\leftrightarrow$ dynamic bridge is the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , $\text{gcd}(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ( $\mathbb{Z}/6 = 2N_{\text{fam}}$ , ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$ , rate $(2/3)^6$ ; order-6 generator $c^5$ ( $\zeta_6 = \zeta_{30}^5$ ). [E] LAW VS LIVE: the resummed clock rate( $n$ ) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from $n=1$ ; the $E_8$ live phases (totatives of 30) start at 1, so 0, 1, 2, 3, 4, 5 is law-inclusive, 1, 2, 3, 4, 5 live-only. [E] STATIC HAND ( $\mathbb{Z}/5 = g_{\text{car}}$ , ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$ , field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$ , $ \mu_4 =4$ planes, v55); the “it is one clock” reading is [C]. |
| v320_galois_cp_relation.py  | A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ . [E] both phases are powers of one hexagonal unit $\rho = \zeta_6$ (v316): $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ ( $60^\circ$ ), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ ( $240^\circ$ ), $\rho^4 = -\rho$ , so $\delta_{\text{PMNS}} - \delta_{\text{CKM}}^{\text{lead}} = \pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: $\delta_{\text{PMNS}}$ robustly away from $240^\circ$ ( $> 3\sigma$ ; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X]                                                                                                                                                                                                                                                                                                                                                                      |
| v321_killtest_board.py      | The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a freeze_file.csv kill row: #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD $r$ (kill $r > 0.01$ ), #3 DESI/Euclid $w = -1$ , #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ( $\sim 2\sigma$ now), #8 LiteBIRD $\beta$ . A forward decision ledger complementing v307; no new physics [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v322_cp_lock_sharpened.py    | The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$ ; $\delta_{\text{CKM}}^{\text{lead}} = 60^\circ$ (node 1), $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (node 4). [E] the node selector is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} =  \mu_4  \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3) = 4\pi/3 = \delta_{\text{CKM}}^{\text{lead}} + \pi$ ( $\rho^4 = -\rho$ ). [C] sub-leading budget: $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2$ ( $\sim 68.7^\circ$ ); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7^\circ \Rightarrow$ band $240 \pm \sim 9^\circ$ . [C] current tension: vs NuFIT 6.0 NO ( $\delta_{CP} = 212_{-41}^{+26^\circ}$ ) the leading $240^\circ$ sits at $+1.08\sigma$ , band edge $\sim 231^\circ$ at $+0.74\sigma$ . [X] kill: the nearest wrong node is $60^\circ$ away $\gg$ budget; DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic $\delta_{CP}$ hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X]                                                                                                                         |
| v323_bw_geometric_modular.py | The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow <i>geometric</i> , so the $\mu_4$ deck invariance is not an independent axiom. [E] the clock $\rho = \text{diag}(i^n) = \exp(i(\pi/2)L)$ is a geometric rotation ( $L = \text{diag}(n)$ , $\rho^4 = I$ , $\mu_4 \subset U(1)_{\text{rot}}$ ). [E] for a rotation-covariant $C = f(L)$ the modular Hamiltonian $K = \log((1 - C)/C) = g(L)$ , so $[K, L] = 0$ (the modular flow commutes with all rotations) and $[\rho, K] = [\rho, C] = 0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is <i>not</i> geometric ( $[K, L] \neq 0$ ); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ QGEO.SYM.01 ( $\omega \circ \rho = \omega$ ) is downstream of SEAM.EQUIV.01 + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]                                                                                                                                                                                                                      |
| v324_hypergraph_fiber.py     | The minimal hypergraph substrate that works: a $(2, 3, 5)$ -network <i>fibered</i> by a 3-fold cusp channel reproduces both the $E_8$ skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}} = (A + 2I)/4$ fixes the Kac marks (eigenvalue $1 = E_8$ skeleton), subleading $(\varphi + 2)/4$ (golden); $(2/3)^6$ is <i>not</i> in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}} = \text{diag}((1 - w)^{2N_{\text{fam}}})$ , $w \in \{0, 1/3, 2/3\}$ ( $N_{\text{fam}} = 3$ , v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$ . [E] the fibered substrate $T = T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ( $w = 0$ ), skeleton $\times$ democratic cusp) <i>and</i> $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ( $T_{\text{net}} \otimes I_3$ ) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ . [E] minimality: the smallest fiber is the 3-node cusp ( $= N_{\text{fam}}$ ) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of $2/3$ from a pure rewrite [E][C] |
| v325_pillowcase_keystone.py  | The seam keystone as ONE citable theorem: the raw seam state <i>is</i> the flat $\tau = i$ pillowcase state, with <b>exactly one</b> open lemma. IF (raw RP-collar seam state = flat $\tau = i$ pillowcase / rotation-invariant state) THEN 4 square marks + $\mu_4$ deck + $\omega \circ \rho = \omega$ + holomorphic $c = 8 \Rightarrow (E_8)_1$ , with the carrier recovery gap as input. [E] pieces all exact: $n = 2\chi(S^2) = 4$ (v216); cross-ratio $2 \Rightarrow j = 1728$ (v214/v267); flat orbifold $S^2(2, 2, 2, 2)$ , $\chi_{\text{orb}} = 0$ (Trojanov); DtN sub-principal at $m \equiv 0 \pmod 4$ (v201/v280); $\det \text{Cartan}(E_8) = 1$ vs $D_8 = 4$ (v83/v143/v308); gap $6 \ln \frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau = i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (QGEO.SYM.01, SEAM.EQUIV.01; v323), machine-pinned in Lean (FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01); an assembly/reduction certificate (like v176/v261), not a closure [E][O]                                                                                                                                                                                                                                                                  |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v326_ftransfer_suite.py    | The $F_{\text{transfer}}$ solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] $F_{\text{pole}}$ (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}(\eta_B)$ : washout contraction to the balance $6.1 \times 10^{-11}$ , monotone (H-theorem), external thermal rate. [C] $F_{\text{relic}}$ (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$ ), cosmological rate. [E] $F_{\text{QCD}}(m_p/m_e)$ : 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, $\Lambda_{\text{QCD}}$ generated. [C] $F_{\text{RareKaon}}$ : $\text{BR}(K^+)=9.45 \times 10^{-11}$ on NA62 2016-2024 (v202). [O] only $F_{\text{pole}}$ has the seam rate $(2/3)^6$ ; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O] |
| v327_hypergraph_rewrite.py | Deriving the cusp weight $2/3$ from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ( $w=0$ ) and $ \mathbb{Z}_2 =2$ surviving — supplies the cusp fiber. [E] $M$ has spectrum $\{1, 2/3, 0\}$ ; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}= \mathbb{Z}_2 /N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order- $6=2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): $2/3$ is NOT a root of the affine- $E_8$ charpoly ( $p(2/3)=-3520/19683 \neq 0$ ), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2, 3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]                                                                                                                                                 |
| v328_theta13_pressure.py   | The $\theta_{13}$ pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO ( $0.02195 \pm 0.00058$ ) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] $\theta_{13}$ is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets $\theta_{12}$ — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is $\sim 5\%$ below the axiom, PMNS $\theta_{13}$ RG running is $< 1\%$ (NO), and $5/6$ is exact — the prediction stays frozen (REG.FREEZE.01). [X] pre-registered kill: a stable $\sin^2 \theta_{13}$ away from 0.023108 at $> 3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]                                                                                                                                                                |
| v329_os_gap_reduction.py   | Closing the OS step: the bulk gap <i>is</i> the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy = $6 \ln \frac{3}{2}$ , independent of system size — exactly “OS bulk gap = transfer gap” (the v308 open input), discharged at the many-body level. [E] $H \Rightarrow \omega \circ \rho = \omega$ : under $H$ the covariance is rotation-covariant $C=f(L)$ , $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ : $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise $H$ (Lean-pinned FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01). [O] residual: the necessity of $H$ on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]                                                                                     |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v330_qgamb_admissible_measure.py | Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer $T$ (v221/v311; spectrum $\{1, (2/3)^6, (1/3)^6\}$ ) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \rightarrow \lambda_2=(2/3)^6$ ( $\xi=1/\Delta$ , $\Delta=6 \ln \frac{3}{2}$ ). [E] tightness $\Rightarrow$ projective limit: $\chi=\sum_n C(n)=729/665$ finite, so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu=\lim \mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{OS}=-\log T \geq 0$ , unique vacuum, gap $6 \ln \frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi \rightarrow \infty$ , tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v275) the physical readouts do not need it, but QG.AMB.01 stays open. A genuine PARTIAL construction, NOT a closure [E][O]                                                                                                 |
| v331_necessity_of_H.py           | The <i>necessity</i> of $H$ reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ( $\tau=i$ ), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329’s sufficiency). [E] square $\Rightarrow [\rho, \Lambda]=0$ : marks at $\{0, \pi/2, \pi, 3\pi/2\}$ make the curvature $\mathbb{Z}_4$ -symmetric (Fourier support $0 \bmod 4$ ), $M_f$ couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $ k +M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho, \Lambda] \neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio( $1, i, -1, -i$ ) $=2 \Rightarrow j=1728$ , the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ . [E] NEG: a $\mathbb{Z}_2$ mark set commutes with $\rho^2$ but not $\rho$ . [O] residual: the dynamical <i>selection</i> of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]                                                                                                                                                                        |
| v332_qgamb_metric_sector.py      | The metric sector of QG.AMB.01 characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330’s metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\text{conf}}(D)=- (D-1)(D-2)/4$ is negative for $D \geq 3$ ( $c_{\text{conf}}(4)=-3/2$ ), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so $e^{-S}$ diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76). [C] IDG conditional route: the entire form factor $e^{z/M^2}$ (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial $(R+R^2)$ truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full QG.AMB.01 needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O] |
| v333_cm_selection.py             | The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S=\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\text{SL}(2, \mathbb{Z})$ has $S^2=-I$ , $S^4=I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\text{PSL}(2, \mathbb{Z})$ (the other, $\rho$ , is order 3). [E] deck order $\rightarrow$ CM point: order-4 $\rightarrow$ cross-ratio 2 $\rightarrow j=1728 \rightarrow \tau=i$ (square); order-3 $\rightarrow j=0 \rightarrow \tau=\rho$ (hexagon); $ \mu_4 =4$ selects $\tau=i$ . [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2$ = the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects $\rho$ , not the square. [O] residual: only “the deck acts <i>geometrically</i> (= QGEO.SYM.01) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]                                                      |

*continued on next page*

| Script                                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>v334_conformal_resolution.py</code> | The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a <b>CONDITIONALLY</b> convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the <code>v332</code> follow-up). <b>[E]</b> the obstruction: $c_{\text{conf}}(4) = -3/2 < 0$ , the unrotated conformal Gaussian diverges. <b>[E]</b> GHP rotation: $\phi \rightarrow i\phi$ flips the sign, $\int e^{- c \phi^2} = \sqrt{\pi/ c }$ <b>FINITE</b> (GHP 1978). <b>[E]</b> IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, <code>v304</code> ), the dressed per-mode action positive post-GHP and UV-suppressed. <b>[E]</b> NEG: a polynomial truncation crosses zero (the Stelle ghost, <code>v304/v278</code> ). <b>[C]</b> conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). <b>[O]</b> residual: whether the GHP-rotated IDG-dressed measure is the <b>CORRECT</b> nonperturbative <b>QG.AMB</b> measure (the contour’s nonperturbative validity); gap-decoupled ( <code>v76</code> ), blocks no test, <b>NOT</b> a closure <b>[E][C][O]</b> |

**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The **[E]** (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

## TFPT verification suite: the journey of ~300 machine-checked scripts

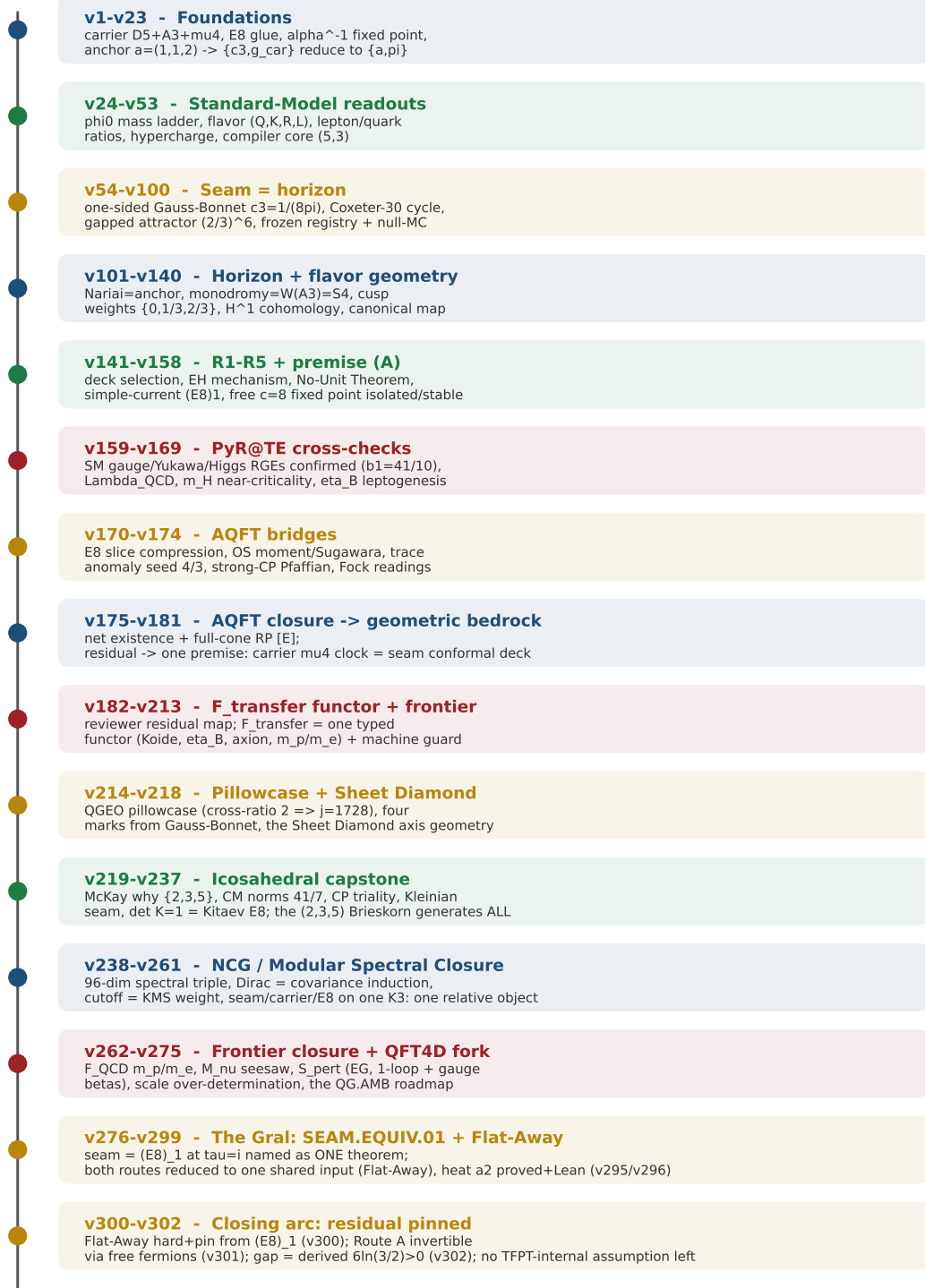


Figure 1: The development timeline of the verification suite — the eleven phases of the journey and what each did, mathematically and physically: from the foundations (carrier,  $E_8$  glue,  $\alpha^{-1}$ , the anchor  $a = (1, 1, 2)$  to which  $\{c_3, g_{\text{car}}\}$  reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flux geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the  $F_{\text{transfer}}$  functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, to the *icosahedral capstone* — where the single  $(2, 3, 5)$  Brieskorn singularity is shown to generate the whole skeleton and the closing step becomes the physical question “is the seam short-range-entangled ( $\det K = 1$ )?”. The master script  $\rightarrow$  check map follows.

## Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers vN refer to `verification/vN_*.py`.

**2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)**

- **v333\_cm\_selection.py** (QGE0.CMSELECT.01, [E]/[O]). The CM-selection mechanism for the last keystone residual:  $\tau=i$  (the square) is the *unique* order-4 modulus —  $S=\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $S^4=I$  and fixes  $i$ , and an order-4 deck selects  $j=1728$  (vs the order-3 hexagon  $j=0$ ) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal  $\pi/2$  spacing). So given the  $\mu_4$  deck,  $\tau=i$  is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGE0.SYM.01). NOT a closure.
- **v334\_conformal\_resolution.py** (QGAMB.CONFORMAL.01, [E]/[C]/[O]). The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation  $\phi \rightarrow i\phi$  flips the wrong-sign kinetic term, making the divergent conformal Gaussian  $\int e^{+|c|\phi^2}$  into the convergent  $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$ ; the entire IDG form factor  $e^{p^2/M^2}$  dresses the propagator pole-free (the polynomial  $R+R^2$  ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The Conway–Sloane lattice side: the  $E_8$  Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ( $|\det|=1$ ) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness:  $E_8$  is the only even unimodular lattice in  $\dim < 16$ ) is the cited classification residual.

**2026-06-22 (endgame round: the necessity of  $H$  reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem,  $|\det E_8|=1$  and the  $\mu_4$  commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)**

- **v331\_necessity\_of\_H.py** (QGE0.NECESS.01, [E]/[O]). The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam  $\Lambda=|k|+M_f$  commutes with the  $\mu_4$  clock *iff* the four marks sit at the square ( $\tau=i$ ) — moving a mark off the square breaks  $\mathbb{Z}_4$  and  $[\rho, \Lambda] \neq 0$ . The square has cross-ratio 2 ( $j=1728$ ), the unique 4-config with an order-4 automorphism (v214/v267), so  $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$  and “necessity of  $H$ ” sharpens to: the raw dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.
- **v332\_qgamb\_metric\_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient  $-(D-1)(D-2)/4$  ( $=-\frac{3}{2}$  at  $D=4$ ), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial  $R+R^2$  ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator  $|\det \text{Cartan}(E_8)|=1$  ( $E_8$  has one anyon, vs  $D_8$ ’s four) is now proved principally



and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the  $8 \times 8$  `PRODUCT` gives `IsUnit (det E8)` via  $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$  — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).

- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers  $\mathbb{Z}[i]$ :  $i^4=1$ ,  $i^2=-1$ , and (on concrete  $6 \times 6$  matrices) an offset-4 coupling commutes with the  $\mu_4$  clock while an offset-2 coupling does not — all by kernel `decide`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)**

- **v329\_os\_gap\_reduction.py** (QGEO.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization  $d\Gamma(-\log T)$ , whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy =  $6 \ln \frac{3}{2}$ , *independent of system size*. Together with  $H \Rightarrow \omega \circ \rho = \omega$  and  $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ , this exhibits the rotation-invariant flat-pillowcase premise  $H$  as the *single sufficient premise* of both bedrock IDs; residual = the necessity of  $H$  on the raw collar.
- **v330\_qgamb\_admissible\_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ( $C(n)/C(n-1) \rightarrow (2/3)^6$ ), with finite susceptibility  $\chi = 729/665$  so  $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$  converges  $\Rightarrow$  uniform tightness (Prokhorov)  $\Rightarrow$  the projective limit exists; OS reconstruction gives  $H_{\text{OS}} \geq 0$  with gap  $6 \ln \frac{3}{2}$ . The full  $\text{QG}_{\text{amb}}$  (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom):  $|\det \text{Cartan}(A_3)|=4=|\mu_4|$  ( $3 \times 3$ , `det_fin_three`) and  $|\det \text{Cartan}(D_5)|=4$  ( $5 \times 5$ , `decide`), giving the carrier  $|\det \text{Gram}|=4 \cdot 4=16$  (the v281 anyon count). The rank-8  $E_8/D_8$  case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the  $F_{\text{transfer}}$  solver suite, the cusp weight derived from a minimal rewrite, and the  $\theta_{13}$  pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)**

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum`  $\rightarrow$  `GeometricModularFlow`  $\rightarrow$  `Mu4ModularSymmetry`  $\rightarrow$  `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator  $(1 \mid 4) \wedge \neg(4 \mid 1)$  (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325\_pillowcase\_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat  $\tau=i$  pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks,  $\mu_4$  deck,  $\omega \circ \rho = \omega$ , holomorphic  $c=8 \Rightarrow (E_8)_1$ ,

with the recovery gap as input); all pieces machine-checked (incl. the pillowcase  $\chi_{\text{orb}}=0$ ), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.

- **v326\_ftransfer\_suite.py** (FR.SUITE.01, [E]/[C]). The four  $F_{\text{transfer}}$  readouts productised into one runnable harness, each a gapped relaxation to a unique attractor:  $F_{\text{pole}}$  (Koide, seam rate  $(2/3)^6$ ),  $F_{\text{Boltzmann}}$  ( $\eta_B$ , H-theorem),  $F_{\text{relic}}$  (axion adiabatic invariant),  $F_{\text{QCD}}$  ( $m_p/m_e$ , RG to the UV fixed point), plus the  $F_{\text{RareKaon}}$  bridge. Only  $F_{\text{pole}}$  carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327\_hypergraph\_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight  $2/3$  is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum  $\{1, 2/3, 0\}$ , so  $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}=|\mathbb{Z}_2|/N_{\text{fam}}$  emerges from the rule arity and  $(2/3)^6=64/729$  is the recovery gap; with a PROOF that  $2/3$  is not a root of the affine- $E_8$  charpoly ( $p(2/3)=-3520/19683 \neq 0$ , sharpening v312). v324's injected datum is reduced to the anchor arity  $\{2, 3\}$ .
- **v328\_theta13\_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified:  $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}=0.023108$  is  $+2.0\sigma$  above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG  $< 1\%$ , exponent  $5/6$  exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon  $F_{\text{transfer}}$  bridge row.
- **Wolfram mirror** 285  $\rightarrow$  288. v325 ( $\chi_{\text{orb}}=0$ ) and v327 (rewrite spectrum  $\{0, 2/3, 1\}$ ,  $(2/3)^6=64/729$ , and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

## 2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ” framing was replaced by the current  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  with the two structural items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs**. Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected**. Version label 5.0 $\rightarrow$ v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886 $\rightarrow$ 3357 jobs (FORM.SEAMEQUIV.01); README Python suite v213/ 212 $\rightarrow$ v324/ 322 modules; Wolfram extension 269/269 / 249/249 $\rightarrow$ 285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

**2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)**

- **Bug fix —  $\delta_{\text{CKM}}$  display value.** The predictions table (`tex-artefacts/predictions.tex`) showed  $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ , but that formula evaluates to  $68.65^\circ$  (the canonical frozen `DELTA_CKM_RAD = 1.198232` rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to  $68.65^\circ$ . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202\_rare\_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  ( $\sim 1.2\sigma$  above) to the full 2016–2024 combination  $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$  (La Thuile 2026), on which the prediction  $9.45 \times 10^{-11}$  lands at  $+0.08\sigma$ . Framed honestly: a strong *consistency* hit for the closed CKM point, but an  $F_{\text{transfer}}$  bridge — *not* a unique TFPT-vs-SM discriminator (the SM  $(8.6 \pm 0.4) \times 10^{-11}$  is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new  $F_{\text{transfer}}$  bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ( $\pi/3$ ) component of the quark phase”: the lock is to the structural  $\pi/3$ , not the full measured  $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$ , so  $\delta_{\text{PMNS}} = 240^\circ$  (not  $248.7^\circ$ ). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305\_witness\_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule:  $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$  is the *static* carrier ( $\mathbb{Z}/5$ ) field, every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6$  hand), so a dynamic rate explained by  $\phi$  is a cross-field false friend (`v314/v315`).
- **v203\_eht\_achromatic.py — the  $\mu_4$  Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at  $m \equiv 0 \pmod{4}$  (the  $\mu_4$  fingerprint of the four marks, `v201`); a  $\mu_4$ -orbit profile has  $\sim 0$  off-mode power while a  $\mathbb{Z}_2$  control leaks at  $m = 2$ .
- **Seed-hyperplane figure (Fig. ??).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed  $\varphi_0$ , the error-weighted mean matching the axiom to 0.04%, with  $\sin^2 \theta_{13}$  the  $\sim 2\sigma$  pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

**2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)**

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks,  $275 \rightarrow 285$ , *all passing*): `v313` (the affine- $E_8$  characteristic polynomial with the golden factor  $x^2 - x - 1$ ; the  $(2, 3, 5)$  atoms as  $2 \cos(\pi/k)$ ; the  $31/30$  selection), `v314` (the rational kernel vs  $\mathbb{Q}(\sqrt{5})$  dynamic-rate split), `v315` ( $30=5 \cdot 6$ ,  $\varphi(30)=8$ , the Galois group  $\mu_4 \times \mathbb{Z}_2$ , the  $\sqrt{5}$  Gauss sum), `v316` ( $\rho=\zeta_6=\zeta_{30}^5$ ,  $\rho^4=-\rho$ ,  $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$ ), `v317` ( $N_{\text{fam}}=3$  as the  $\mu_3$  orbit, the  $1+2$  Galois split), `v318` ( $\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$  a pure function of  $\pi$ ) and `v320` (the CP lock  $\delta_{\text{PMNS}}=240^\circ$ ). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.

- **v322\_cp\_lock\_sharpened.py** (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]). The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes  $\arg \rho^k = k \cdot 60^\circ$ , and *which* one is selected is the deck order:  $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$  (node 4); [C] the sub-leading correction is bounded by the quark analogue  $3\lambda^2 \approx 8.7^\circ$ , so the prediction is the band  $240^\circ \pm \sim 9^\circ$ ; [C] against NuFIT 6.0 (NO,  $\delta_{CP} = 212_{-41}^{+26^\circ}$ ) the leading value is  $+1.08\sigma$ , the band edge  $+0.74\sigma$  – compatible today; [X] the nearest wrong node is  $60^\circ$  away, far beyond the budget, so DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate cleanly. Cited in `origin_theory`; ledger GALOIS.CP.SHARPEN.01.
- **v323\_bw\_geometric\_modular.py** (QGEO.BW.01, [E]/[C]/[O]). The keystone pushed one honest step via Bisognano–Wichmann. [E] the  $\mu_4$  clock is a geometric rotation  $\rho = \text{diag}(i^n) = \exp(i\frac{\pi}{2}L)$ ,  $\mu_4 \subset U(1)_{\text{rot}}$ ; [E] for a rotation-covariant covariance  $C = f(L)$  the modular Hamiltonian  $K = \log((1-C)/C) = g(L)$ , so  $[K, L] = 0$  (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric,  $[K, L] \neq 0$ ); [C] given the seam is the  $(E_8)_1$  chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so QGEO.SYM.01 ( $\omega \circ \rho = \omega$ ) is *downstream* of SEAM.EQUIV.01 + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger QGEO.BW.01.
- **v324\_hypergraph\_fiber.py** (HYP.FIBER.01, [E]/[C]). The constructive v312 follow-up: the *minimal* substrate that carries both the  $E_8$  skeleton and the recovery gap is a *product*. [E] the carrier network  $T_{\text{net}} = (A + 2I)/4$  (attractor = Kac marks = skeleton) fibred by the 3-node family cusp  $T_{\text{cusp}} = \text{diag}((1-w)^{2N_{\text{fam}}})$ ,  $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$  (spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ); the fibred  $T_{\text{net}} \otimes T_{\text{cusp}}$  has top eigenvalue 1 (eigenvector marks  $\otimes (w=0)$ ) and  $(2/3)^6$  as a genuine eigenvalue (marks  $\otimes (w=\frac{1}{3})$ ) – the  $N_{\text{fam}}=3$  cusp is the minimal fibre and the substrate = carrier  $\times$  family = the  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \times 6$  of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of  $2/3$  from a pure rewrite. Cited in `origin_theory`; ledger HYP.FIBER.01.

## 2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321\_killtest\_board.py** (KILL.BOARD.01, [E]). The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO  $\sin^2 \theta_{12} = 0.306747$  (fastest), #2 CMB-S4/LiteBIRD  $r$  (kill  $r > 0.01$ ), #3 DESI/Euclid  $w$  (kill  $w \neq -1$ ), #4 DUNE/Hyper-K  $\delta_{\text{PMNS}} = 240^\circ$  (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering  $+m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  now), #8 LiteBIRD  $\beta$ . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

## 2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes  $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$  assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Muger / Kawahigashi–Longo–Muger holomorphy; Conway–Sloane  $E_8$  uniqueness) as named *axioms* and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a *decide*-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the  $|\det \text{Cartan}|$  that separates the holomorphic  $E_8$  from the same- $c$  rival  $D_8 = SO(16)$ ). Built end-to-end (`lake build`, 3357 jobs) on Lean



v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

## 2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ : v320)

- **v320\_galois\_cp\_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit  $\rho = \zeta_6$  of the family Galois factor:  $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$  ( $60^\circ$ ),  $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$  ( $240^\circ$ ), and  $\rho^4 = -\rho$  locks them as  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  (the  $\mathbb{Z}_2$  sheet flip); [C] this upgrades the previously assigned  $\delta_{\text{PMNS}} = 240^\circ$  to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a  $\delta_{\text{PMNS}}$  robustly incompatible with  $240^\circ$  ( $> 3\sigma$  at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation;  $240^\circ$  is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

## 2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0.5) or live-only (1..5): v319)

- **v319\_translation\_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314);  $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  and  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (two coprime hands, v315). [E] the *dynamic* hand is  $\mathbb{Z}/6 = 2N_{\text{fam}}$  (ticks  $0, \dots, 5$ ): the recovery exponent is exactly  $6 = 2N_{\text{fam}}$ , rate  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ , order-6 generator  $c^5$  ( $\zeta_6 = \zeta_{30}^5$ , v316). [E] position 0 is the *conserved law* — the resummed clock rate( $n$ ) =  $-p_2 \ln(1 - n/N_{\text{fam}})$  (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from  $n=1$ , and the  $E_8$  live phases (totatives of 30) start at 1 ( $\{1, 7, \dots, 29\}$ ), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is  $\mathbb{Z}/5 = g_{\text{car}}$  (ticks  $1, \dots, 5$ ): the golden field  $\mathbb{Q}(\sqrt{5})$  ( $\varphi = 2 \cos(\pi/5)$ ), field-obstructed (no rational dynamic image, v314) — static-only;  $\zeta_5 = \zeta_{30}^6$  carries no phase. [E] one full turn =  $\text{lcm}(5, 6) = 30 = h(E_8)$ , rank  $E_8 = 8 = \varphi(30)$  live phases pairing into  $|\mu_4| = 4$  invariant planes (v55). [C] verdict: the discrete $\rightarrow$ dynamic translation *is* the order-30 clock = (static 5-ring)  $\times$  (dynamic 6-ring), the dynamic hand carrying the rate (exponent  $6 = 2N_{\text{fam}}$ ), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row TRANSLATE.CLOCK.01.

## 2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$ , and the complete input is $\{a, \pi, v_{\text{geo}}\}$ — zero dimensionless free parameters: v318)

- **v318\_arithmetic\_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field  $\mathbb{Q}(\zeta_{30})$  with Galois group  $\mu_4 \times \mathbb{Z}_2$  (degree  $\varphi(30) = 8 = \text{rank } E_8$ ,  $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$ ), forced by the anchor atoms  $\{2, 3, 5\} = e(a)$ ; [E] the magnitude seed  $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$  ( $c_3 = 1/(8\pi)$ ) is a pure function of  $\pi$  (no free parameters, anchor-derived coefficients), so the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ ) are  $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters,  $\pi$  the



unique transcendental primitive (ARCH.CORE.01),  $v_{\text{geo}}$  the unique dimensionful input (No-Unit Theorem) — complete input  $\{a, \pi, v_{\text{geo}}\}$ . [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in `origin_theory`; ledger row ARITH.CAPSTONE.01.

**2026-06-21 (are the 3 generations a  $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)**

- **v317\_galois\_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$  (with  $\text{Spec}(Q_+) = \{1, 2, 3\}$ , the  $A_3$  exponents) are the cube roots of unity  $\{1, \zeta_3, \zeta_3^2\}$  under  $w \mapsto e^{2\pi i w}$ , and the cyclic family symmetry  $w \mapsto w + \frac{1}{3}$  is a *transitive*  $\mu_3$  orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues  $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$  attach to the weights (attractor = eigenvalue 1 at  $w=0$ ); [E] the family Galois  $\mathbb{Z}_2$  ( $\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$ , = the v316 CP conjugation) refines the orbit to 1+2 — fixing  $w=0$  (rational) and swapping  $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$ ; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a  $\mu_3$  orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row GALOIS.FAMILY.01.

**2026-06-21 (does the Galois  $\mu_4 \times \mathbb{Z}_2$  organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois  $\mathbb{Z}_2$  = CP conjugation: v316)**

- **v316\_galois\_readout.py** (GALOIS.READOUT.01, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling  $\mu_4 \times \mathbb{Z}_2$  reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit  $\rho = \zeta_6 = \zeta_{30}^5$  is exactly the order-6 =  $2N_{\text{fam}}$  factor  $c^5$  of v315 (the carrier being  $\zeta_5 = \zeta_{30}^6$ ); [C] the leading CP phases are its cyclotomic arguments  $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$  and  $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$  (power 4 =  $|\mu_4|$ ), with  $\zeta_6^4 = -\zeta_6$  the  $\mathbb{Z}_2$  sheet flip and  $\mu_6 = \mu_3 \times \mu_2$  (inherits v231/v233); [E] the family Galois  $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$  ( $\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$ ) is CP conjugation  $\delta \mapsto -\delta$ ; [E] the carrier factor  $\zeta_5$  (golden  $\sqrt{5}$ ) carries no phase ( $\mu_4$  static), and the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ , transcendental  $\varphi_0$ ) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in `origin_theory`; ledger row GALOIS.READOUT.01.

**2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois =  $\mu_4 \times \mathbb{Z}_2$ : v315)**

- **v315\_coxeter\_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  couple the carrier (5-fold, golden,  $\mathbb{Q}(\sqrt{5})$ ) and family ( $6 = 2 \cdot 3$ , rational) sectors, or just factorize? [E] the cyclic group *factorizes*:  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (direct product, group-decoupled, in line with the v314 field obstruction); [E] the  $E_8$  exponents = totatives(30) split as totatives(5)  $\times$  totatives(6),  $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$ ; [E] but the *field* couples:  $\mathbb{Q}(\zeta_{30})$  is the compositum of the carrier  $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$  and the family  $\mathbb{Q}(\zeta_3)$ , degree 8 = rank  $E_8$ ; [E] the Galois group is exactly  $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$  (order 8, max element order 4), so the  $\mu_4$  seam clock *is* the Galois group of the carrier's golden field and  $\mathbb{Z}_2$  that of the family field (refining v223); [E]  $\sqrt{5} = \phi + 1/\phi$  is the quadratic Gauss sum in  $\mathbb{Q}(\zeta_5)$ . [C] verdict:  $30 = 5 \cdot 6$  does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row COX.COUPLE.01.

## 2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$ : v314)

- **v314\_rate\_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- $E_8$  network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate  $(2/3)^6$ ) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part  $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$  plus the golden part  $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$ , while the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles  $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$  and  $2 \cos(\pi/3)=1 = N_{\text{fam}}$  build  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$ , same field, same atoms), but [E] the carrier 5-fold (golden,  $\mathbb{Q}(\sqrt{5})$ ) has NO rational dynamic image – a field obstruction ( $2/3$  is neither an adjacency nor a lazy-update eigenvalue; heat-kernel  $t = \Delta/(2 - \phi) \approx 6.37$  not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , and the  $\phi$ -quasicrystal  $E_8 = H_4 + \phi H_4$  ( $240 = 2 \cdot 120$ ) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row RATE.TRANSLATE.01.

## 2026-06-21 (the golden ratio made precise: the $(2, 3, 5)$ atoms are the affine- $E_8$ network spectral angles, $\phi = 2 \cos(\pi/5)$ the $g_{\text{car}}=5$ signature: v313)

- **v313\_golden\_atoms\_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- $E_8$  adjacency characteristic polynomial factors as  $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , so the golden minimal polynomial  $x^2-x-1$  divides it and  $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$  is an *exact* eigenvalue; [E] the non-trivial eigenvalues are  $2 \cos(\pi/k)$  for exactly  $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$  ( $2 \cos(\pi/2)=0$ ,  $2 \cos(\pi/3)=1$ ,  $2 \cos(\pi/5)=\phi$ ,  $2 \cos(2\pi/5)=1/\phi$ ), so the golden ratio is *specifically* the  $g_{\text{car}}=5$  (5-fold) signature and  $g_{\text{car}}=5$  gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the  $(2, 3, 5)$  spherical excess  $1/2+1/3+1/5 = 31/30 > 1$  (the condition selecting the finite  $2I \Rightarrow E_8$ ) ties the v63/v76 gap-margin numerator  $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$  to the Coxeter  $30 = h(E_8)$  — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null:  $\phi$  is a structural lattice/network quantity, not a phenomenological readout. [C] direction:  $\phi$  is the  $E_8$ -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in `origin_theory`; ledger row GOLD.ATOMS.01.

## 2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **v308\_seam\_equiv\_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic  $c=8$  ( $E_8$ )<sub>1</sub> net” assembled into ONE well-typed logical chain  $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$ , with the exact discriminators  $|\det \text{Cartan}(E_8)|=1$  vs  $|\det \text{Cartan}(D_8=SO(16))|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$  and  $c=g_{\text{car}}+N_{\text{fam}}=8$  all machine-checked; the input gap is the derived recovery gap  $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309\_modular\_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock  $\omega \circ \rho = \omega$  via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so  $\mu_4$  is derived; the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  round-trips to the seam operator and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity

of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.

- **v310\_carrier\_sm\_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  is exactly one anomaly-free Standard-Model generation ( $SU(5) \subset SO(10)$ :  $10+\bar{5}+1$ ), with  $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$  (exact) and the one-loop  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts  $b_1$ , so  $N_{\text{fam}}=3$  pins the RG. Closes the spectrum/anomaly/ $\beta$  sub-question; the interacting Epstein–Glaser  $S$ -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311\_gap\_decoupled\_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  gives geometric clustering at rate  $(2/3)^6$  (correlation length  $1/(6 \ln(3/2))$ , finite susceptibility  $729/665$ ), and the v76 margin  $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$  decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312\_hypergraph\_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- $E_8$  Kac marks are the Perron eigenvector ( $A_{m=2m}$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5) =$  the golden ratio, but the recovery rate  $(2/3)^6$  is *not* any eigenvalue and  $\varphi_0$  is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  and the analytic seed  $\varphi_0$ . The open question is precisely delimited, not closed. Cited in `introduction`; ledger row HYP.INJECT.01.

## 2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305\_witness\_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h(E_8)=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the single anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive (13 atoms from 2 free primitives,  $6.5 \times$  compression made explicit), so the recurring  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio; the  $E_8$  spine is overdetermined ( $\geq 3$  structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by v100, never multiplied); the negative control shows  $\alpha^{-1}$ ’s “137” is foreign to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, v3). Cited in the `tfpt_5_redteam` body; ledger row WITNESS.ECON.01. Python-only.
- **v306\_seed\_crossval.py** (SEED.CROSSVAL.01, [E]). A leave-one-out cross-validation turning the “many observables, one seed  $\varphi_0$ ” worry into a falsifiable out-of-sample test: back-solving  $\varphi_0$  from each of the five seed-grammar observables ( $\sin^2 \theta_{12}$ ,  $\sin^2 \theta_{13}$ ,  $\beta$ ,  $\Omega_b$ ,  $\lambda_C$ ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  out of sample (most-tensioned =  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the  $\chi^2$  by  $> 100\times$ . Cited in `tfpt_5_redteam`; ledger row SEED.CROSSVAL.01. Python-only.
- **v307\_data\_watchdog.py** (DATA.WATCHDOG.01, [E]). The operational decision pipeline over the frozen registry (v84): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live  $n_\sigma$  against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are  $\alpha^{-1}$  ( $1.9\sigma$ ),  $s_{23}$  ( $1.8\sigma$ ) and the most-tensioned core prediction  $\sin^2 \theta_{13}$  ( $2.0\sigma$ , the honest

current pressure point);  $\sin^2 \theta_{12} = 0.306747$  is compatible ( $0.02\sigma$ , the JUNO falsifier), the  $r \in [0.0033, 0.0048]$  band is below the BK18 limit and the kill, and  $w=-1$  is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.

- **Target-A holomorphy kill test (`freeze_file.csv`).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ( $|\det K| > 1$ , the same- $c$  rival  $SO(16)_1$ ) would break holomorphy and kill “seam net =  $(E_8)_1$ ” — since holomorphic  $c=8 \Leftrightarrow E_8$  is machine-proven (`v83/v143`) and  $\det K=1 \Leftrightarrow$  invertible/SRE (`v237/v301/v302`).

## 2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of $c$ ” framing of `SEAM.EQUIV.01: v304`)

- **`v304_idg_nonlocal_ghost.py` (`QGAMB.IDG.01`, [\[C\]](#)/[\[O\]](#)).** A conditional, parameter-free narrowing of the red-team `rt_F/v278` Stelle-ghost attack on the perturbative gravity sector. [\[E\]](#) the local  $R+R^2/\text{Weyl}^2$  four-derivative propagator  $1/(p^2(p^2+M^2))$  carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [\[E\]](#) for an *entire* graviton form factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [\[C\]](#) the spectral-action cutoff is the seam KMS weight  $f(u) = e^{-u}$  (`v259`), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [\[O\]](#) It does *not* close `QG.AMB.01`: the trace regulator is not yet shown to resum to an entire  $a(\square)$ , and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row `QGAMB.IDG.01`.
- **Double-pushout figure (`tfpt_1_architecture_e8`).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order- $|\mu_4|=4$  glue is *one* operation in two categories: the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  (`v1`) and the conformal-net / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$  (`v92/v281/v234/v277/v260`). Expository only; no new claim.
- **Framing (introduction, `tfpt_research_contracts`).** `SEAM.EQUIV.01` is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of  $c$  plays in relativity (cf. `v267`). Prose sharpening; no status change.

## 2026-06-19 (the discrete→dynamic lens on `F_transfer`: one shape, one seam-rate, `v303`)

- **`v303_ftransfer_dynamics.py`** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update  $\rightarrow$  the  $E_8$  marks, gap  $6 \ln(3/2)$ ) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [\[E\]](#) `F_pole` (Koide) *is* the seam dynamics (Möbius multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock eigenvalue  $\Rightarrow$  a unique attractor at the seam rate, `v82/v54`); [\[C\]](#) `F_Boltzmann` ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0, 1)$ , H-theorem); [\[C\]](#) `F_relic` (axion) freezes at an adiabatic fixed point; [\[E\]](#) `F_QCD` ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3 = -7$  (asymptotic freedom, Gaussian UV attractor). [\[O\]](#) Verdict: one shape underlies all four; `F_pole` has the seam rate  $(2/3)^6$  *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory’s simplicity is preserved because the external rates are honestly fenced (`v187`), not reduced to the seam. [\[E\]](#) suite `v303 5/5`, AUDIT OK.



## 2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$ , v302)

- **v302\_seam\_gap.py** closes the chain on the single statement v301 had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  (multiplicative gap  $1 - (2/3)^6 = 665/729$ ;  $(2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, SEAM.EQUIV.01’s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ) — *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite v302 5/5, AUDIT OK.

## 2026-06-19 (Route A’s last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301\_route\_a\_invertible.py** attacks the single remaining *unconditional* gap of SEAM.EQUIV.01. Since v300 collapsed Route B into Route A’s rationality, that gap is Route A’s open hypothesis (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev’s periodic table, class D). [E] the carrier is 16 Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); [E]  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) vs 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. [O] LIT-A’s ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of SEAM.EQUIV.01 reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. [E] suite v301 6/6, AUDIT OK.

## 2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$ , v300)

- **v300\_flataway\_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum (“the spectral mismatch grows with  $\varepsilon$ ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint —  $\text{spec}(|D_\theta|)$  is the integer ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (sympy  $2 \times 2$   $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , + an  $O(g^2)$  common shift), validated against the full Toeplitz operator; [E] any  $g_k \neq 0$  lifts the resonant degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every  $g_k = 0 \Rightarrow f = 0$  — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* [C] the  $(E_8)_1$  character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients (248 at level 1;  $E_8$  even unimodular  $\Rightarrow$  integer weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat



seam, i.e. the external pin *is* the  $(E_8)_1$  rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

## 2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches $E_8$ as the last finite point, v299)

- **v299\_hypergraph\_0d\_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ( $|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$ ; the McKay group is the  $SU(2)$  binary lift  $2I$ , a *different* order-120 group whose McKay graph is affine  $E_8$ ; 30 edges =  $h(E_8)$ ); [E] the  $1 \rightarrow 4$  subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the  $2I$  fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine  $E_8$  McKay graph; [E] Smith ( $\rho \leq 2 \Leftrightarrow$  affine ADE) + Collatz–Wielandt ( $\rho \leq 2 \Leftrightarrow$  a local witness  $Am \leq 2m$ ); [E]\* the autonomous rule (diffusion-generated witness + *local* gate  $Am \leq 2m$ , Perron only as audit) runs  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ ; [E] the negative controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$  (closed form), so neither family ever reaches  $E_8$ . Honest type: [O] the  $(2, 3, \cdot)$  seed is the single irreducible input = P2 — *the rule reaches  $E_8$  as the last finite point on this seed; it is not a derivation of P2.* [E] build green, AUDIT OK.

## 2026-06-18 ( $E_8$ as a network attractor — the Wolfram-program reading, v298)

- **v298\_e8\_network\_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- $E_8$  Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4m$  on the  $(2, 3, 5)$  McKay graph (converges from any positive start;  $Am = 2m$ , top eigenvalue 2); the spherical tower terminates at  $E_8$  and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ $E_8$  is the universal attractor of a minimal  $(2, 3, 5)$  network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives:  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

## 2026-06-18 ( $a_2$ closed form + Route A literature stack, v296–v297)

- **v296\_flataway\_a2\_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact  $a_2$  coefficient in *closed form*: the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ) to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle; e.g.  $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ . Manifestly positive, small- $t$  law  $W_k = t/2 + O(t^3)$  (the  $t^2$  term cancels). Reproduces v295 to machine precision.
- **v297\_route\_a\_literature\_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global*

minimum — so  $\Delta\text{Tr} \geq 0$  for all deformations,  $= 0$  iff  $f = 0$ . The exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (Duhamel), validated to rel. err  $< 10^{-6}$ .

- **Lean FORM.FLATAWAY.01** (`SeamDeckClosure.lean` Part 5): the formal  $\Delta=0 \Leftrightarrow$  flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes  $a_2$ ) remains. [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292\_flataway\_heat\_reduction.py** (`FLATAWAY.HEAT.01`): the heat route, made precise. The heat-trace deviation  $\Delta\text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a *positive-definite* quadratic form in the off-mark curvature ( $c_k(t) > 0$  for every  $\mathbb{Z}_4$  mode, positive on all combinations), leading invariant  $\|f\|_{L^2}$  — so Flat-Away reduces to the single fact “the seam fixes the  $a_2$  heat coefficient”.
- **v293\_flataway\_spectral\_hessian.py** (`FLATAWAY.SPEC.01`): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$  space. Mode  $4k$  splits the degenerate Steklov level  $|2k|$  at first order; the Hessian of the spectral mismatch over modes  $\{4, 8, 12\}$  is positive-definite (eigenvalues  $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294\_flataway\_rp\_energy.py** (`FLATAWAY.ENERGY.01`): the variational route. With the rigid mark divisor (4 cone angles  $\pi$ , Gauss-Bonnet  $4\pi$ ), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy  $\int f^2$  is strictly convex (Hessian  $2\pi I$ ) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290\_z4\_smooth\_curvature\_adversary.py** (`SEAM.ADVERSARY.01`) is the honest red-team that  $\mathbb{Z}_4$  block-diagonality is *not* mark-locality: a smooth  $\mathbb{Z}_4$  off-mark curvature  $f = \varepsilon \cos(4\theta)$  [E] passes the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , same as mark-local) yet [E] shifts the DtN spectrum ( $\max |\Delta\lambda| = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not  $[\rho, \Lambda]=0$  — which is exactly why Flat-Away cannot be read off the commutator.
- **v291\_flataway\_contract.py** (`FLATAWAY.RP.01`) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family — deviation zero only at  $\varepsilon=0$ ) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289\_marklocal\_raw.py** (`MARKLOCAL.RAW.01`) decomposes the one remaining Route-B residual (“why is the raw seam sub-principal term mark-local?”) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks  $\Rightarrow$  curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on  $4\mathbb{Z}$  (v264/v288), L5 full-operator  $[\rho, \Lambda]=0$  (v288). 4/5 discharged — Route B reduces to exactly L2.

- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- $L^2$  QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk  $\Rightarrow$  single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

**2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)**

- **v286\_seam\_equivalence\_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ $E_8$  into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain  $\text{RP} + \text{gap} + \text{chirality} + \text{Gaussian} + \text{invertible} \Rightarrow \mu=1 \Rightarrow \text{holomorphic} \Rightarrow (E_8)_1$  is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary* (the AQFT form of  $\det K=1$ ). 4/5 discharged.
- **v288\_full\_l2\_subprincipal\_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ , so  $\Lambda = |n| + M_f$  commutes with  $\rho = \text{diag}(i^n)$  on the full operator,  $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$ ; generic curvature  $\rightarrow 4.44$ ), lifting v201/v284 to  $L^2$ . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

**2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)**

- **v284\_route\_i\_rp\_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1  $\text{RP} \rightarrow \text{DtN}$  (v194), L2  $\text{marks} \rightarrow \text{pillowcase}$  (v195/v216/v214), L3  $\text{flat Steklov} = \sqrt{-\Delta_{\text{flat}}}$  (verified here), L4 covariance from DtN (v258), L5  $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov ( $\chi_{\text{orb}}=0 \Rightarrow$  unique flat  $\tau=i$ ) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.
- **v285\_route\_ii\_seam\_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow \text{holomorphic} \Leftrightarrow E_8$ ; the tower  $16 \rightarrow 4 \rightarrow 1$ ), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the  $(E_8)_1$ -at- $\tau=i$  object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$ . Build green, AUDIT OK.

**2026-06-18 (self-investigation round: pillowcase DtN, modular data, the  $E_8$ -at- $\tau=i$  unification, live scorecard, v280–v283)**

- **v280\_pillowcase\_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat  $\tau=i$  pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].

- **v281\_holomorphy\_modular\_data.py** (QGAMB.MODULAR.01):  $\#anyons = |\det \text{Gram}|$  ( $E_8 \rightarrow 1$ ,  $SO(16) \rightarrow 4$ ), the anyon-condensation tower  $16 \rightarrow 4 \rightarrow 1$ , Gauss–Milgram  $\rightarrow c=8$ ; holomorphy ( $\det K=1$ ) is the single Tier-B discriminator.
- **v282\_e8\_tau\_i\_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* —  $\chi_{E_8}=j^{1/3}$  gives  $\chi_{E_8}(i)=1728^{1/3}=12$ , so  $\tau=i$  is both the order-4 CM point (QGEO) and the  $(E_8)_1$  modulus (QG.AMB): the two obligations are two faces of one object (the seam is  $(E_8)_1$  at  $\tau=i$ ), **obligation count**  $2 \rightarrow 1$ .
- **v283\_live\_killtest\_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$  floor 0.0586 eV; proton decay  $1.55 \times 10^{35}$  yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

## 2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279\_qgeo\_obligation\_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat  $\tau=i$  pillowcase metric* (Trojanov-unique,  $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$ ), then  $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow \text{mark-locality} \rightarrow E_8$ . [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal,  $\rho$  commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutesclock` and `flat_all_orders_clock`: any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

## 2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278\_lsz\_bridge\_unitarity.py** (QFT4D.SPERT.04) connects the perturbative **S-matrix to the physical asymptotic one and verifies one-loop unitarity**. [E] LSZ bridge: the EG  $T$ -products of  $S_{\text{pert}}$  (v269) are the OS Wick-rotated correlators (v240), so *amputate+on-shell-residue* gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ; [C] the gap  $\Delta=6 \log(3/2) > 0$  gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity  $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$  *exactly* = the two-body phase space, so  $2 \text{Im } M = \sum \int d\Pi |M|^2$  (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt\_F) is non-unitary  $\rightarrow$  QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

## 2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277\_seam\_calderon\_e8\_match.py** (QGAMB.TIERB.01) reduces Tier-B to **ONE bit**. [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$ ,  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248 = \dim E_8$  weight-1 currents (240 roots). [E] the discriminator is holomorphy — the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (non-holomorphic), so the one condition is  $|\det K|=1$ . [C] the seam side matches via v276 (flat  $\tau=i$ ) + v260 (Kummer/K3). [O] residual

= the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

## 2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276\_qgeo\_flat\_closes\_commutator.py** (QGEO.SYM.03) **collapses the bedrock to ONE geometric postulate**. The state-level residual the whole bedrock rests on is the operator equation  $[\rho, H]=0$ . [E] if  $H = f(\Delta_{\text{flat}})$  and the order-4 deck  $\rho (z \mapsto iz)$  is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (verified on the discretised square torus:  $\rho^4=I$ ,  $[\rho, \Delta]=0$ ,  $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full  $H$ . [C]  $\tau=i$  is forced by order-4 ( $j=1728$ ; hexagonal  $j=0$  is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat  $\tau=i$  pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

## 2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt\_F)

- **redteam/rt\_F\_qft4d.py** (REDTEAM.F.01) **stress-tests the just-published round** (the perturbative  $S_{\text{pert}}$  layer, the PMNS Jarlskog, the  $\nu$ -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity  $R^2/\text{Weyl}^2$  sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator  $1/(p^2(p^2 + M^2))$  has a negative-residue massive spin-2 mode), so  $S_{\text{pert}}$  is unitary for the matter+gauge sector only — which is exactly why the nonperturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the  $\Lambda$ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

## 2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo**. Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser  $S_{\text{pert}}$  skeleton v269, the concrete one-loop quartic v271 and gauge  $\beta$ -coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute  $\nu$ -mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, `run_all.py` green, AUDIT OK, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

## 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272\_nu\_mass\_scale.py** (FLAV.NUSCALE.01) **types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{\text{EW}})^2/M_R$  (one UV input, like  $v_{\text{geo}}$ , v153); [C] the observed  $m_3=0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (v249) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; [O]  $M_R$  is not a clean  $\bar{M}_{\text{Pl}}$  power, so the absolute value stays open; [X] the NO floor  $\Sigma m_\nu=0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).



- **v273\_eg\_gauge\_running.py** (QFT4D.SPERT.03) **derives the SM one-loop  $b_i$  via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- **v274\_scale overdetermination.py** (SCALE.OVERDET.01) **shows the single mass anchor is over-determined.** [E]/[C] two independent routes to  $\bar{M}_{\text{Pl}}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV)  $= 2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- **v275\_qgamb\_roadmap.py** (QGAMB.ROADMAP.01) **gives the open gate QG.AMB.01 a module.** [E] Tier A gap decoupling ( $\Delta_{\text{eff}} = 1.648 > 0$ , v36/v76); [E]/[C] Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, run\_all green, AUDIT OK, build green.

## 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **v271\_eg\_oneloop\_quartic.py** (QFT4D.SPERT.02) **takes the v269  $S_{\text{pert}}$  skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $\text{sd}=2$  in  $d=4$ , the one-loop bubble has  $\text{sd}=4=d$ , so  $\omega=\text{sd}-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$ . [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270\_pmns\_jarlskog\_assembly.py** (FLAV.PMNS.03) **assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\delta = 4\pi/3$ ) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a *derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\text{max}}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6/\text{triality}$  phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py** (QFT4D.SPERT.01) **builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\text{dim} \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order

(causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py (FLAV.TH13.01) closes the  $\theta_{13}$  exponent origin.** The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py generates verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (status\_ledger.csv for claim→status→dependencies→supersedes, script\_registry.csv/script\_clusters.csv for the theme grouping, docs\_map.csv for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into **bash build.sh gen**. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks” ( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio  $2 \Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

## 2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266\_ps\_threshold\_proton.py (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the v265 fork.** The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes ps\_threshold\_proton.json): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr  $<$  Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed

(15, 1, 1) [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected (must be + (15, 1, 1))*, *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

## 2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265\_qft4d\_fork\_freeze.py (QFT4D.FORK.01)** turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (qft4d\_fork\_freeze.json), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1 = \alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to tfpt\_research\_contracts; [E] build green, AUDIT OK.

## 2026-06-18 (Three open-residual advances: the $F_{QCD}$ solver, $M_\nu$ from $D_F$ , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262\_fqcd\_mp\_me.py (FR.MPME.02)** implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z) = 0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{MS}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py (FLAV.PMNS.02)** makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric; [C] it reproduces  $\theta_{23} = 45^\circ$  and  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13} = 0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **v264\_raw\_seam\_marklocal.py (QGE0.ENERGY.03)** sharpens the bedrock. Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda] = 0 \Rightarrow \omega \circ \rho = \omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGE0.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in tfpt\_4\_frontier, tfpt\_2\_standard\_model and tfpt\_research\_contracts. [E] AUDIT OK, build green.

## 2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo  $\text{QGEO.SYM.01}$  (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same  $\text{QGEO.SYM.01}$  postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

## 2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise  $\text{QGEO.SYM.01}$  via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

## 2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian  $H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum); [E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .
- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).



- **v260\_k3\_kummer\_unification.py** (ARCH.K3.01) **unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes =  $\dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py** (QFT.MSC.01) **is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated \subsection\* entries, itemize items, and the inline markup \textbf/\emph/\texttt/\vref, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

**2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)**

- **v257\_quasiorient\_modpd.py** (PS.NCG.ORIENT.02) **corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.



## 2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254\_inner\_fluctuations.py** (PS.NCG.FLUCT.01) closes **v250 #5 + the no-junk obligation**. The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)/126$ . [E] the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E]  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . [C]  $\sigma =$  scalaron.
- **v255\_spectral\_action\_expansion.py** (PS.SPECACT.01) closes the **spectral-action expansion (v252 #11)**. The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : [E] structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); [E]  $b/a^2 = 1/3$  (top-dominated); [E]  $\sin^2 \theta_W = 3/8$ ; [E] the  $R^2$  spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; [C]  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; [O] exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py** (PS.NCG.ORIENT.01) — **honest status, no faked proof**. KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$  to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; [E] the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

## 2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and $\kappa$ pinned to $O(1)$ , v252–v253)

- **v252\_full\_finite\_triple.py** (PS.DIRAC.02) advances the **Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]**. The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ( $J^2 = +1$ ,  $JD = DJ$ ), grading ( $\gamma^2 = 1$ ,  $[\gamma, a] = 0$ ,  $J\gamma = -\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$ . [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) discharges residual 6(b) of **v249**.  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the 126-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

## 2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$ — the CCvS Pati–Salam mechanism, v251)

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) advances the **Dirac contract v250 from a contract to an exhibited mechanism**. On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2 = +1$ ,  $\gamma^2 = 1$ ,  $D\gamma = -\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order

condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

## 2026-06-17 (Carrier-native Pati–Salam program: $E_8$ forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** v247\_e8\_branching\_no126.py (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\bar{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one  $45 \Rightarrow$  proton-marginal). v248\_ps\_algebra.py (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). v249\_ps\_unification.py (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . v250\_dirac\_triple\_contract.py (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level qft.txt carries the full analysis.

## 2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **v246\_unification\_data\_crosscheck.py (QFT4D.RGTEST.01) — the first hard data confrontation of the spectral-action prediction.** TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (figures/qft\_skeleton.pdf, figures/qft\_unification.pdf) added via make\_figures.py (PDF + website PNG) and embedded in tfpt\_research\_contracts.tex; a root-level qft.txt summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

## 2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **v245\_spectral\_matter\_content.py (QFT4D.MATTER.01) discharges the computable core of the 4d contract v244.** The carrier half-spinor (the  $SO(10)$  16) is verified, in exact rational arithmetic, to be one Standard-Model generation: **16** =  $SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of CONTRACT.QFT4D.01 is now [E]; only the seam  $\rightarrow$  Dirac realisation and the run-down couplings stay [O]. Wired into run\_all.py, the ledger, the registry, cited in tfpt\_research\_contracts.tex, mirrored to the website.

## 2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the v64 mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237). `v242_spin_statistics_modular.py` (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

## 2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238\_modular\_lindblad\_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). [E] keystone: the dissipative gap  $= -\log((2/3)^6) = 6\log(3/2) =$  the OS mass gap  $\Delta$  (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

## 2026-06-17 (Experiments: black-hole-cosmology signatures, the $\eta_B$ Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior ( $w_{in}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{in} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai  $Q_{geom}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a [C] structural echo, no  $\mathcal{C} \leftrightarrow Q_{geom}$  map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES

$\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).

- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale** (experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); tfpt\_4\_frontier Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically** (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221). The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (internal\_consistency, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain** (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets). The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** experiments/evidence\_scorecard.json now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website EXPERIMENTS\_AUDIT mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone experiments/ surfaces (not in the verification suite or ledger); the tfpt\_4\_frontier/tfpt\_horizon\_readouts and papers.ts/predictions.ts edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to tfpt\_5\_redteam** (rt\_A\_e8net.py, redteam\_table.txt, redteam/README.md, infrastructure). The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level (v143) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , v148), while the  $q(A3)$  normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays [C]/[O] (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/** (scripts/export-shadow.sh, verification/make\_script\_index.py, infrastructure). export-shadow.sh ships figures/\*.pdf (so make\_manifest.py --check passes literally on the exported tree) while still excluding website/ and the compiled paper PDFs; the SHADOW\_MIRROR.md table is



corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet** (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). Sharper than `v220/v225` (which only *located* the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}}^{\text{leading}} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the `v88` misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays [\[C\]](#) (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity** (`v232_e8_kleinian_seam`, `origin_theory`, [\[E\]](#)/[\[C\]](#)). The du Val side of the McKay correspondence (`v219`) gives the open seam-realisation premise `QGEO.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves = rank  $E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; `TOP0.E8.01`, bedrock stays [\[O\]](#)).
- **CP is the universal family/triality phase, only sheet-split** (`v233_cp_triality_phase`, `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). One step past `v231` in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate** (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [\[E\]](#)/[\[O\]](#)). The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; `v232`), (iii) exactly one 1-dim irrep (`v219`). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ([\[O\]](#)):  $\text{RP} + \text{gap} + \text{chirality} \Rightarrow \text{quasi-free bulk}$  (`v160`)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).
- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (`v235_seam_chern_simons`, `tfpt_research_contracts`, [\[E\]](#)/[\[O\]](#)). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\# \text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (`v92`) *is* the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic =  $E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).



- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton (v236\_brieskorn\_capstone, origin\_theory, [E]).** Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]).** Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2, 3, 5\}$  (v219\_icosahedral\_mckay, origin\_theory, [E]).**  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1, 2, 3, 4, 5, 6, 10\}$  — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep *is* the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex) (v222\_cm\_norm\_duality, origin\_theory, [E]).** The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index *is* the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3,2) split generates (5, 6, 7, 13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle (v223\_coxeter\_totative\_clock, origin\_theory, [E]).**  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h)=\text{rank}$  and  $h=2 \cdot 3 \cdot 5$  (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120=|R^+(E_8)|$  (magnitude channel), degrees sum  $128=\text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D=4=|\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0=5=g_{\text{car}}$ ,  $\text{rank } H_1=3=N_{\text{fam}}$ ,  $h^0+H_1=8=\text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGEO.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+=16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type);

free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).

- **CP lives in the hexagonal phase fiber** (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]). The seam deck is the square modulus  $j=1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j=0$  ( $\mathbb{Z}/6$ ,  $\arg \rho = \pi/3$ ). The frozen  $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$  has leading term  $\pi/3 = \arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame** ( $d, n$ ) with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).  $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$  Nariai;  $n = (5, -9, 6)$  reads det on the first-generation column;  $\det(\mathbf{1}, d, n) = 21 = 3 \cdot 7$ ;  $\text{Im det}(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- $F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).  $M(s, t) = R + Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source  $\rightarrow$  pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra** (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).  $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$ , product  $\frac{32}{9}$ ;  $A = \mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget**  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).  $C = R + Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$  (hex),  $13 = N_{\mathbb{Z}[i]}(3 + 2i)$  (square),  $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes** (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]). Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C + xU) = 14 + 6x$  (slope  $6 = |R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C + yV) = 14 + 14y + 4y^2$  (2nd difference  $8 = \text{rank } E_8$ ); the anchor block has  $\det B(C + xU) = 2 \det(C + xU)$  and sheet curvature  $6 = |R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the decuple then the full generation (identity [E], source  $\rightarrow$  pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28 = 2 \dim G_2$ ,  $52 = \dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (v216\_marks\_gauss\_bonnet, tfpt\_research\_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch),  $\text{rank } H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck

— converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling `v217_marks_emergence_scan` ([C], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. `v216` exact (Wolfram extension  $\rightarrow 248/248$ ); `v217` numerical (Python-only). Suite  $\rightarrow 217$  modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198/v199/v201`) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ ,  $\text{Aut} = \mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via `v214/v54/v56`). Suite  $\rightarrow 215$  modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`). The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the `v201` “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (`v168`)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ ; all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. [O] It does *not* close `QGEO.SYM.01`: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values `mpmath`-numerical. Suite  $\rightarrow 214$  modules.

## 2026-06-15 (the $F_{\text{transfer}}$ transfer functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel [E] composed with an external continuous solver [C]: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier

$\lambda_2=(2/3)^6=64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53=a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F)=(1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1=11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3=-7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). **CONTRACT.F.01** is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger **CONTRACT.F.01**.

## 2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, [C]). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, **DM.AXION.SPINE.01**); [X]  $\Omega_a h^2$  outside  $\sim[0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch** (v212\_leptogenesis\_decuple, tfpt\_4\_frontier, [C]). Both Boltzmann inputs share the decuple  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184's  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (**FR.ETAB.04**; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, [O]). A new module machine-proves the COHO node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho : z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents  $= \text{Spec}(Q_+)$ , rank  $3 = N_{\text{fam}}$ . The MODULE *parity* is also proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^* \omega_1 = \omega_3$ ,  $\sigma^* \omega_2 = \omega_2$ ,  $\sigma^* \omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in AuditContract.lean. The geometric COHO/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (MobiusUniformisation.lean, ledger FORM.QGEO.02, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the clock  $\rho : z \mapsto iz$  has  $\rho^4=\text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma \rho \sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1, -1, swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in AuditContract.lean. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.



## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201/v210` (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]). A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]). The carrier's second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.
- **An independent gravitational  $3/4$**  (`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi \varphi_0^{\text{ret}}/c_3^2$ ) — the same  $3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.
- **Quantitative  $H_0$  from the  $\Lambda$  branch** (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.
- **Asymptotic Safety as a second QG route** (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]). A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger `GRAV.ASYMP.01`.
- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, [E]/[C]). Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to `v198/v199`)



and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.

- **The black hole as a seam fixed point** (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]). No singularity ( $\varphi \rightarrow \varphi_*$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

## 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure** (tfpt\_3, [O]). Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

## 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to  $c_3$**  (tfpt\_4\_frontier, [C]). Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

## 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- HOR.EHT.01 (v203, [E]/[C]/[X]). The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into tfpt\_horizon\_readouts (new EHT section), the ledger, registry, Wolfram (243/243), and the website (predictions.ts). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- FR.RAREKAON.01 (v202, [C]). The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_\pm$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+ \nu\bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0 \nu\bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$

is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62 BR( $K^+$ ) outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II BR( $K_L$ ) off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, **v187** guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The **v199** bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, **v198**); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \bmod 4]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by **v195**) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, **v177**)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* **v185**’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the **v185** docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.
- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py)**. A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, **v185** docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py [E]/[O] (claim QGE0.STATE.01, work package A)**. Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , **v117**), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements”

— a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.

- `v200_seam_variational_scan.py` [C]/[O] (claim QGEO.VARI.02, work package B). The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i =$  the  $(1, 2, 3)$  grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the `v199` residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- `v198_modular_commutator_reduction.py` [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- `v195_marks_lefschetz_character.py` [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- `v196_seam_energy_functional.py` [E]/[O] (claim QGEO.VARI.01). The failure functional  $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the `v194` Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- `v197_rr_carrier_clifford_d5.py` [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.

- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^* w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the (1, 2, 3) grading on  $H^1$  (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7)**. `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

## 2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side rank  $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by rank( $D_5$ ) = 5. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly*



*typed [C]* — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.

- `v192_qgeo_energy_clock.py [O]` (claim `QGEO.ENERGY.01`). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (`v186` skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

## 2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the `F_transfer-firewall` round (`v184–v188`): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the `P1/P2-vs-QGEO.SYM.01` two-layer clarification. The uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; `v186` skipped). `run_all.py` green, audit clean, website built.

## 2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)

- `v184_etaB_anchored_boltzmann.py [C]` (claim `FR.ETAB.03`). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_\Lambda$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_\Lambda = 10$  atom, in the strong-washout window) `[C]`; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{P1}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so  $M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays `[C]`, the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- `v185_axion_relic_solver.py [C]` (claim `FR.DM.03`). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a `[C]` scenario, not a sharp prediction; the relic contract is a kill test for the `BRANCH`, not the theory. Cited in `tfpt_4_frontier`.
- `v187_ftransfer_laws.py [E]` (claim `FR.TRANSFER.01`). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a `[C]/[O]` marker — never a primitive `[E]` compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be `[E]`, the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- `v188_frontier_wording_guard.py [E]` (claim `AUDIT.WORDING.01`). A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old `[E]`-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  `v184 [C]`; the old “open relic scale” dark-matter line  $\rightarrow$  `v185`; the old “near, not exact” Koide line  $\rightarrow$  `v183`; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording



that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (`v186` was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).

- *Review framing points* (*P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data*) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- `v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`). An external-review proposal, validated exactly: the Koide source→pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T Ra = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (`v80`). Hard negative control:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades `FR.KOIDE.03` from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly `HOR.TRISECT.01/v103` — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- `v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as `v175–v181`, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count `v174` + holographic reduction), B ( $\equiv \text{QGEO.SYM.01}$ ), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator =  $F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to `JUNO/CMB-S4`). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website `RosettaLexicon`): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , `JUNO`;  $r \approx 0.004$ , `CMB-S4`) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.

- **Review recommendation #4 — the bedrock as a postulate.** QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

## 2026-06-14 (external-review response: CSV fix, $G_{\text{net}}$ 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “ $a=(1,1,2)$ ”, “ $\text{PSL}(2,C)$ ”), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

## 2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the  $\sim 181$  scripts (foundations  $\rightarrow$  SM readouts  $\rightarrow$  seam = horizon  $\rightarrow$  horizon/flavor geometry  $\rightarrow$  R1–R5/premise-(A)  $\rightarrow$  PyR@TE cross-checks  $\rightarrow$  AQFT bridges  $\rightarrow$  AQFT closure), what each

did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.

- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, `v6`, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01/v23`, `ARCH.CORE.01/v53`) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows `AX.P1.01/AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py** [E]/[O] (claim `QGEO.SYM.01`). The isometry premise `QGEO.ISO.01` (`v180`) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (`v117`), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→...→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180\_clock\_is\_möbius.py** [E]/[O] (claim `QGEO.ISO.01`). Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, *non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, `v179`) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its

surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(\mathbb{Z}_2 \cdot 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee–Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` [E] — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168’s cross-ratio to the full uniformisation); `COHOM` [E] —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE` [E] — a *unique*  $\mu_4$ -equivariant iso



(multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); NET [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP}$  all  $m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1} v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- `v175_net_existence_full_cone.py` [E]/[O] (claim QFT.NETRP.01). The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$ :* the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of



the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every*  $m$  (not only  $m \leq 6$ , [v161](#)) to the one-particle contraction (Shale–Stinespring), discharging the [v160](#) scope premise. (2)  $A_2$  assembled + verified: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 [v156](#) check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, [v83](#)); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, [v125](#)). (3) Both then need the *same* input — the seam realisation `QGE0.REALIZE.01` (finite half proven, [v168](#)) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [\[O\]](#). **Net:** net existence and full-cone RP become [\[E\]](#); the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings [v174](#): CAR cone, Witten index, g-theorem)

- `v174_seam_fock_readings.py` [\[E\]](#) (claim `QFT.FOCK.01`). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind [v161](#). (2) *Witten index = the Plücker norm* 11: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the [v157/v158](#) rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression [v170](#), OS moment [v171](#), trace-anomaly seed [v172](#), Pfaffian CP [v173](#))

- `v170_diamond_slice_compression.py` [\[E\]](#) (claim `E8.SLICE.01`). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- `v171_os_moment_cluster.py` [\[E\]](#)/[\[C\]](#) (claim `QFT.OSMOMENT.01`). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in `tfpt_research_contracts` (G5).
- `v172_trace_anomaly_seed.py` [\[E\]](#) (claim `SEED.ANOMALY.01`). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2/\dim(D_5)$  factorisation. Cited in `tfpt_2` (seed).
- `v173_pfaffian_cp_car.py` [\[E\]](#) (claim `FLAV.STRONGCP.02`). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).

- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron 28/79), fed by TFPT’s normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166\_higgs\_free\_seam.py** [E]/[C] (claim HIGGS.FREESEAM.01). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer’s SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163\_rg\_stability\_flavor.py** [E]/[C] (claim FLAV.RGSTAB.01). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **v164\_qcd\_scale.py** [E]/[O] (claim QCD.LAMBDA.01). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ 's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module v159\_pyrate\_gauge\_crosscheck.py** [E]/[C] (claim EM.BUDGET.02). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,  $\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}}^{2g_{\text{car}}-2} + 1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5} Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.
- **PyR@TE reproducer (not vendored)**. `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

## 2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + $F_{\text{transfer}}$ + irreducible core, v167)

- **New module v167\_inventory\_post\_closure.py** [E]/[C]. The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (GATE.METRIC.16–19) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + GATE.QGEO (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1 = N_{\text{fam}} = 3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (v152) and the ambient measure into  $G_{\text{net}}$  (v76). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,

$m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).

- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

**2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)**

- **New module v165\_premise\_a\_closure.py [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (v125),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $DtN = |k|$  (v156/v157).  $R_1$  (**seam clock**) **reduces to GATE.QGEO** [C]: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , v115/v116), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary  $= \mathbb{P} \setminus \mu_4$ ” = **GATE.QGEO** (cohomology established, v137,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)**

- **New module v162\_seam\_transport\_identification.py [E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced:** a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,  $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (v115) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (v137). ( $\alpha$ ) is the rank-8 Calderón polarisation of  $\omega_k=|k|$  (v113/v156). **Closure:** with v160+v161, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)**

- **New module v161\_one\_particle\_reduction.py [E]/[O].** *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior



power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by **v113** (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and **v158** makes every non-Gaussian deformation irrelevant. **Consequence:** **v160**’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

## 2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py`** **[E]/[O]**. Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment $\leftrightarrow$ cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only  $\text{gap}>0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual:** **v56**’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of **v56**”, which the suite does *not* yet establish (ledger **GATE.METRIC.16**). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (**v160** is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

## 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection \{YYYY-MM-DD...\}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

## 2026-06-13 (premise A: the destination fixed point is provably stable, **v158**)

- **The free chiral  $c=8$  fixed point is provably stable (**v158**, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (**v157**, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all



interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).

- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, **v157**)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee–Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, **v156**)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual:* exact given (i) the free-bulk premise (DtN =  $|k|$ , v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN =  $|k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, **v155**)

- **$G_{\text{net}}$ ’s last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (**v155** is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, **v153–v154**)

- **$v_{\text{geo}}$  re-typed: the No-Unit Theorem (**v153**).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger `ANCHOR.VGEO.02`).
- **$G_{\text{net}}$  stated as the central closing theorem (**v154**).** The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (**v89/v92/v125/v143/v148**); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed (review):** the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  (`[E]/[C]`), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, **v152**)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (**v152**).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (**v68**). So  $v_{\text{geo}}$  (**v78**),  $1/G$  (**v68**) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and `SEAM.THEOREM.01`’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).

- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (rev `{git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ~270 files / ~3 MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (`v53–v56`, `v101`, `v105`, `v113`, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class `[O]`/`[E]`.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.

- **Build hardening.** `build.sh` now removes a document's `.aux/.toc/.out` on failure, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghlelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. SEAM.THEOREM.01 narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays [O] (ledger SEAM.EHMODEL.02).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot 1$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2 - 1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m / 12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, origin\_theory, contracts, next.txt, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0$  = the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 (4+5=9, \sigma \perp w_0(a))$  and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line ( $= T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p_2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).

- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day)*: the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue, 2/3 motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{\text{net}}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  *error* was in fact a correct-but-treacherous notation switch (see above).



**2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)**

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+=1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q_+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading  $= Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4' narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion  $= \mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1's det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_{br_c} = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays [C] with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its [O] typing untouched.
- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

**2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)**

- **Single-source script index.** The master script  $\rightarrow$  check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite  $\leftrightarrow$  `run_all.py`  $\leftrightarrow$  registry  $\leftrightarrow$  ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts,

and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.

- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-`[E]` artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611$ ,  $r=0.0045$ ,  $A_s$ -disfavoured) and states the band-of-record explicitly,

matching COSMO.NSTAR.01 and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. [I]/[L]  $\rightarrow$  [E], [N]/[P]  $\rightarrow$  [E]/[C]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction's inflation/ $N_*$  rows are made consistent with v86/COSMO.NSTAR.01:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_* \in [50, 60]$  with the scalaron-reheating conditional point  $N_*=51.4$  ( $n_s=0.9611$ ,  $r=0.0045$ ), replacing the stale point values  $N_*=55$  (predictions table) and  $N_* \simeq 57$  (residual card); the closure-ledger inflation row is re-typed [E]  $\rightarrow$  [C](conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger [P] typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and v5.2 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion

documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.

- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven **keyboxes** are reduced to the single headline “In one sentence” card; the explanatory **keyboxes** (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a **winbox** (a result, not a claim), and the explanatory *plausible/follows next* **winboxes** become prose. The loud boxes now carry only genuinely load-bearing callouts (one **keybox**, five **winboxes**, one **gapbox**) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only **keybox** (claim), **winbox** (result) and **gapbox** (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in **FLAV.QRATIO.01** is removed (the statement no longer says “stays [P]” — ratios are [E] closed on the derived stratum, only the absolute scale stays [O], with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); **tfpt\_research\_contracts** is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with Koide/ $\eta_B$ /axion/ $m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt\_N}$ ). Website mirrored throughout (**papers.ts**, **predictions.ts**, **faq.ts**, **glossary.ts**, **OpenGates**, **Hero**, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: **tfpt\_style.tex** (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and **tfpt\_status.tex** (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in **build.sh** and the manifest **TEX** list.

## 2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4’ selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1’s bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).



- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\text{det}'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+)$  = the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the  $R4'$  residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the GATE.QGEO residue collapses to one  $\mathbb{Z}_3$  choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).

- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank =  $c$ ) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- $U_{\text{wall}}$  **made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  (diag =  $a/|\mu_4|$ , off-weights (8, 0, 5)/144); resonance theorem ( $M_\infty = \mathbf{1} \iff a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad ( $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ ); the 121 audit lemma; the lepton words are the compiler atoms (8, 5, 3) with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and det 8 — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).

## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.